

Boundary and Interface Classes for Universal Chern Hamiltonians via Doubling and Jamming

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Abstract

Higson and Prodan’s universal Chern Hamiltonians are closed-surface models built from the simplicial boundary operator and Poincaré duality. We treat the boundary and interface classes obtained from finite surfaces with boundary by doubling and jamming. The finite-dimensional results are unconditional. For every finite triangulated oriented surface with boundary we construct an intrinsic Poincaré–Lefschetz Hamiltonian, compare it with the original-side compression of the doubled closed Hamiltonian, identify the discrepancy as boundary-collar supported, and prove Schur-complement estimates for finite- M jamming. Along families with negligible collar dimension these estimates identify the normalized bulk spectral limits of the intrinsic, compressed, and jammed models. The Roe-algebra interface class is conditional on a uniform closed doubled-bulk gap, and its winding is evaluated only after a closed-bulk Chern input and a compatible Ewert–Meyer bulk–interface pairing are supplied. The fully internal periodic example is the triangular Bloch class: its closed-bulk gap, Chern signs, and straight Toeplitz winding are proved in the paper. Additional hexagonal and square two-band symbols, robustness variants, and reproducibility details are collected in the appendices. Numerical checks document the finite-matrix behavior but are not used as proof inputs.

1 Introduction

1.1 Universal Chern Hamiltonians from simplicial data

The Haldane model is the prototypical two-dimensional class-A topological insulator [3]; for background on the tenfold-way classification and Chern phases, see [4, 5, 6, 21]. On irregular discretizations, direct magnetic-field or Peierls constructions can lead to mobility-gap or defect-state issues that complicate the extraction of clean topological gaps; see, for example, [22, 23, 24, 21]. Rigorous bulk–edge and bulk–interface correspondences in quantum Hall and related class-A systems are treated in [25, 26, 27, 30, 28, 29, 31, 32].

Higson and Prodan introduced a different route [8, 9]. For a triangulation of a *closed* oriented surface, one builds a local tight-binding Hamiltonian from two pieces of simplicial-topological data:

- the simplicial boundary operator B , and
- a Hermitian Poincaré-duality operator S constructed from cap product with the fundamental cycle.

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The resulting Hamiltonians

$$H_{\pm} = (B + B^{\dagger}) \pm S \quad (1.1)$$

are local and, in the sequential-refinement limit, define nontrivial topological bulk classes with Chern numbers ± 1 [8, 9]. In that theory, Chern numbers and bulk–boundary correspondences belong to the infinite-system refinement limit, while fine finite triangulations already display the expected gap and edge physics [8, 9]. The underlying Hilbert–Poincaré framework uses the algebraic identity

$$BS + SB^{\dagger} = 0, \quad (1.2)$$

rooted in analytic surgery theory and the work of Higson and Roe [10, 11, 12, 13].

1.2 The boundary case

If the input triangulation $\mathcal{K}$ has boundary, there is no closed fundamental 2-cycle supported on the faces of $\mathcal{K}$, so the closed-case cap-product recipe for S is not directly available. Compact manifolds with boundary instead carry a relative fundamental class and satisfy Poincaré–Lefschetz duality [15]. We use this relative class to define an intrinsic boundary Hamiltonian on $\mathcal{K}$, and then compare it with a second construction that reduces the boundary problem to the closed case by simplicial doubling and jamming the reflected side.

1.3 Doubling and jamming

Given a triangulated surface-with-boundary $\mathcal{K}$, let $\overline{\mathcal{K}}$ be an orientation-reversed copy. Gluing $\mathcal{K}$ and $\overline{\mathcal{K}}$ along $\partial\mathcal{K}$ produces the closed double $D(\mathcal{K})$. We then build the closed universal Hamiltonians on $D(\mathcal{K})$ and suppress the reflected copy by adding a large on-site potential on the copy degrees of freedom. This is the *doubling-and-jamming* construction.

The closed theory supplies a canonical operator $H_{\pm}(D(\mathcal{K}))$ on the doubled triangulation, while the large potential isolates the original side at low energies. The finite- M analysis gives error bounds for this approximation; the coarse section then constructs the associated Roe-algebra interface class under the stated geometric and spectral hypotheses.

The later construction uses four concrete ingredients: an intrinsic Poincaré–Lefschetz boundary Hamiltonian, a comparison with the doubled original-side compression, a collar-supported discrepancy statement, and quantitative finite- M jamming estimates. The finite-dimensional bundle is unconditional. The Roe K_1 -class is constructed after assuming a uniform closed doubled-bulk gap, and its numerical winding needs a closed-bulk Chern input together with a compatible bulk–interface pairing. The periodic triangular section is the fully internal benchmark: its closed gap, Chern signs, and straight Toeplitz winding are proved inside the paper. Additional two-band hexagonal and square symbols are recorded only as appendix benchmarks.

1.4 Theorem map

Let

$$H_{\pm, M} = H_{\pm}(D(\mathcal{K})) + M\mathbf{P}_{\text{copy}}$$

be the jammed Hamiltonian, written in block form as

$$H_{\pm, M} = \begin{pmatrix} A & C \\ C^{\dagger} & D + M\mathbf{1} \end{pmatrix} \quad \text{on} \quad \mathcal{H}_{\text{orig}} \oplus \mathcal{H}_{\text{copy}}.$$

The theorem map separates the unconditional finite-dimensional bundle from the conditional interface framework and from the fully internal periodic benchmark. The finite-dimensional bundle consists of the Poincaré–Lefschetz construction, the collar comparison, the large- M Schur-complement estimates, and the normalized bulk spectral equivalence under negligible collar dimension. The conditional framework imports exactly the closed-surface Higson–Prodan input and the compatible Ewert–Meyer pairing stated below. The periodic triangular class supplies an internal closed-bulk and straight-Toeplitz example.

1. The relative fundamental class of $(\mathcal{K}, \partial\mathcal{K})$ gives an intrinsic Poincaré–Lefschetz boundary Hamiltonian on $\mathcal{K}$. The simplicial double gives a second Hamiltonian by closed Higson–Prodan theory; its original-side compression agrees with the relative Hamiltonian up to a boundary-collar operator, and the jammed operator converges to this compression at low energy.
2. The finite- M comparison is made quantitative by Schur-complement estimates: compressed resolvents, full block resolvents, first-order effective operators, bounded-energy spectral localization, leakage, Morse-index stability, and contour spectral projections. Combined with the collar-rank comparison, these estimates also give a fully internal bulk spectral-measure equivalence between the intrinsic Poincaré–Lefschetz model, the doubled compression, and the finite- M jammed model along any family with negligible boundary-collar dimension and $M \rightarrow \infty$.
3. For refinement families with a uniform doubled-bulk Fermi gap, the quotient by the interface ideal gives a global Roe-algebra interface class

$$\text{Ind}_{\text{int}}(J_{\pm, M}^{\infty}; T^+, T^-) = \partial_Y(E_+(\pi(P_{\pm}))) \in K_1(C^*(Y \subset X)).$$

When the closed doubled family satisfies the Higson–Prodan closed-bulk Chern input and the chosen coarse partition is compatible with the Ewert–Meyer bulk–interface pairing, this class has winding number ± 1 . Section 6 gives the verification template for these conditions; it does not assert admissibility for a refinement scheme until the closed-bulk and pairing checks have been completed.

4. For the periodic triangular-lattice class, the closed-bulk gap and Chern input are checked directly. We write down the 6×6 Bloch symbol, prove a uniform zero gap from an exact determinant formula, compute the Chern signs from the isolated massless crossing, and evaluate primitive straight periodic interfaces by the Toeplitz extension. The FHS computations in Tables 3 and 4 remain reproducible numerical checks. Finite covers, annular families, finite-corner quotients, finite-defect variants, and tail-split component maps are appendix variants of this straight Toeplitz calculation.

Logical dependencies. The proof uses the following implications; no converse is assumed.

- (I) internal finite-dimensional results: the relative boundary operator, the collar comparison, and the large- M Schur-complement estimates;
- (G) a uniform closed doubled-bulk Fermi gap and uniform locality/norm bounds;
- (C) a closed-bulk Chern pairing and a compatible Ewert–Meyer bulk–interface pairing for the chosen coarse partition.

From (I) and (G), Theorem 6.35 constructs the global Roe-algebra interface K_1 -class. From (C), applied to that class, Theorem 6.38 identifies the compatible winding value. Thus a winding

conclusion is always an instance of

$$(I) + (G) + (C) \implies \text{Wind} = \pm 1.$$

The ingredient (I) is proved here. For Higson–Prodan admissible closed doubles, (G) and the closed-bulk part of (C) are imported from the cited closed theory; the Ewert–Meyer compatibility in (C) remains an external pairing check. In the periodic triangular section, the gap part of (G) and the closed-bulk Chern sign in (C) are proved directly. For primitive straight periodic interfaces the pairing in (C) is evaluated internally by the Toeplitz extension. The finite-cover, finite-defect, finite-corner, split, and tail-split variants are collected in Appendix A.5. For nonperiodic coarse partitions, the compatible interface pairing remains external. No separate boundary spectral gap, no componentwise edge-mode hypothesis, and no unverified refinement-class admissibility are used in the quotient-interface theorem. The finite-matrix checks most directly attached to these core results are the Poincaré–Lefschetz/compression comparison in Figure 6 and table 8, the finite- M rates in Figure 11 and table 7, the boundary transport and flux checks in Figures 7 and 10, and the periodic FHS checks in Tables 3 and 4. Table 1 records which conclusions are finite-dimensional, which are conditional Roe-algebra statements, and which periodic examples are evaluated within the paper.

Table 1: Theorem map. “Internal” means proved here from the inputs shown; “external” means that a cited or separately verified input is still required.

Category and conclusion	Inputs	Status
Finite-dimensional boundary theory: relative boundary Hamiltonian and collar comparison	Finite triangulated surface with full boundary subcomplex	Internal, Theorem 3.12
Finite-dimensional jamming theory: finite- M decoupling, leakage, contour projections, Morse index	Finite block separation hypotheses	Internal, Section 4
Internal bulk spectral measures: PL model, doubled compression, and finite- M jammed model have the same normalized bulk spectral limits	Uniform finite-dimensional bounds, $M \rightarrow \infty$, and boundary-collar dimension $o(\dim \mathcal{H}(\mathcal{K}_n))$	Internal, Theorem 4.17
Conditional Roe theory: global interface K_1 -class	Uniform locality, boundedness, and closed doubled-bulk gap	Internal after these hypotheses, Theorem 6.35
Conditional Roe theory: winding value ± 1	Closed-bulk Chern pairing and compatible coarse interface pairing	External pairing evaluation, Theorem 6.38
Internal periodic examples: triangular closed-bulk input	Periodic triangular Bloch symbol	Internal gap and Chern signs, Theorems 5.3 and 5.6
Internal periodic examples: straight Toeplitz benchmark and appendix variants	Periodic triangular input, primitive straight edge, and the stated finite-cover/robustness hypotheses	Internal Toeplitz winding for the primitive straight case in the body; finite-cover, annular, finite-defect, finite-corner, split, and tail-split variants in Appendix A.5

Table 2: Notation used for the bulk, jammed, and sidewise interface operators.

Symbol	Meaning
$H_{\pm} = (B + B^{\dagger}) \pm S$	Closed universal Chern Hamiltonian.
$H_{\pm, M} =$ $H_{\pm}(D(\mathcal{K})) + M\mathbf{P}_{\text{copy}}$	Finite doubled-and-jammed Hamiltonian.
$A = P_{\text{orig}} H_{\pm}(D(\mathcal{K})) P_{\text{orig}}$	Original-side compression of the doubled Hamiltonian.
$H_{\pm}^{\text{PL}}(\mathcal{K})$	Intrinsic Poincaré–Lefschetz boundary Hamiltonian.
$P_{\pm} = \chi_{(-\infty, 0)}(H_{\pm}^{\infty})$	Closed-bulk Fermi projection in the coarse disjoint union.
$J_{\pm, M}^{\infty}$	Direct-sum jammed family defining the interface.
T^{+}, T^{-}	Positive-side and negative-side comparison operators.
$Y, C^{*}(Y \subset X)$	Coarse interface and its Roe ideal.
$\partial_Y(E_{+}(\pi(P_{\pm})))$	Internally constructed interface K_1 -class.

2 Simplicial Hilbert space, locality, and universal operators

2.1 Triangulations and the manifold-with-boundary condition

Definition 2.1 (Finite two-dimensional simplicial complex). A finite abstract simplicial complex is a finite collection of finite vertex sets that is closed under passing to subsets. Its elements are simplices; a simplex with $p+1$ vertices has dimension p . A finite two-dimensional simplicial complex has simplices only in dimensions 0, 1, 2. A triangulation of a compact PL surface is such a complex whose geometric realization is PL homeomorphic to the surface; see, for example, [14, 15].

Definition 2.2 (Triangulated surface with boundary). A finite oriented 2-dimensional simplicial complex $\mathcal{K}$ is a *triangulated surface with boundary* if its geometric realization $|\mathcal{K}|$ is a compact PL 2-manifold with boundary and the boundary subcomplex $\partial\mathcal{K}$ is a *full subcomplex*: every simplex of $\mathcal{K}$ whose vertices all lie on $\partial\mathcal{K}$ is itself a boundary simplex. In combinatorial terms, the PL-manifold condition says that every vertex link is either a cycle (interior vertex) or a path (boundary vertex), while fullness is equivalent in dimension two to saying that no face has all three vertices on $\partial\mathcal{K}$ and no non-boundary edge has both endpoints on $\partial\mathcal{K}$. In particular, every edge is incident to exactly one or two faces, the edges incident to exactly one face form $\partial\mathcal{K}$, and $\partial\mathcal{K}$ is a disjoint union of cycles.

Remark 2.3. The vertex-link condition is essential. Edge-incidence conditions together with the statement that the boundary edges form cycles do *not* exclude pinched or singular 2-complexes, and the proof of Proposition 3.2 uses precisely the fact that interior vertex links are cycles and boundary vertex links are paths.

Remark 2.4 (Full boundary subcomplex). In an abstract simplicial complex, a simplex is uniquely determined by its vertex set. The doubling construction identifies boundary vertices and duplicates interior ones. This creates two types of collision if the fullness condition fails:

- (i) *Face collision.* If a face $\{v_0, v_1, v_2\}$ has all three vertices on $\partial\mathcal{K}$, its reflected copy has the same vertex set, so $D(\mathcal{K})$ is not an abstract simplicial complex.
- (ii) *Edge collision and valence failure.* If an interior edge $[v_0, v_1]$ has both endpoints on $\partial\mathcal{K}$, the reflected copy has the same vertex set $\{v_0, v_1\}$ and cannot be distinguished from the original. Worse, each face incident to $[v_0, v_1]$ in $\mathcal{K}$ produces a reflected face also incident to $[v_0, v_1]$, so this edge becomes incident to four or more faces in $D(\mathcal{K})$, violating the 2-manifold condition.

The simplex-count formula $|D(\mathcal{K})_1| = 2|\mathcal{K}_1| - |\partial\mathcal{K}_1|$ also fails: it counts such an edge twice, but only one copy exists.

Requiring $\partial\mathcal{K}$ to be a full subcomplex rules out both failures. This is a mild condition: it holds automatically after a single barycentric subdivision (which places a new interior vertex at the barycenter of every simplex, so no simplex retains an all-boundary vertex set; see the standard PL subdivision discussion in [14, 15]). All refinement families used in this paper satisfy the condition. If one starts from a triangulated PL surface whose boundary subcomplex is not full, the construction applies canonically to its barycentric subdivision, subject to the same closed-bulk and coarse-interface checks for the subdivided doubles.

2.2 Hilbert space of simplices

Let $\mathcal{K}_p$ be the set of p -simplices of $\mathcal{K}$. We work with the simplicial Hilbert space

$$\mathcal{H}(\mathcal{K}) = \ell^2(\mathcal{K}_0) \oplus \ell^2(\mathcal{K}_1) \oplus \ell^2(\mathcal{K}_2) = \bigoplus_{p=0}^2 \mathcal{H}_p(\mathcal{K}).$$

Edges are given arbitrary orientations to define the signs in the boundary operator, while faces are oriented coherently with the surface orientation. Changing these conventions only changes basis by a diagonal sign matrix and leaves the spectrum unchanged.

2.3 Boundary operator B

The simplicial boundary map B is the usual chain boundary, viewed as an operator on $\mathcal{H}(\mathcal{K})$ lowering degree by one:

$$B : \mathcal{H}_2 \rightarrow \mathcal{H}_1, \quad B : \mathcal{H}_1 \rightarrow \mathcal{H}_0, \quad B|_{\mathcal{H}_0} = 0.$$

On basis simplices,

$$\begin{aligned} B|[v_0, v_1]\rangle &= |v_1\rangle - |v_0\rangle, \\ B|[v_0, v_1, v_2]\rangle &= |[v_1, v_2]\rangle - |[v_0, v_2]\rangle + |[v_0, v_1]\rangle. \end{aligned}$$

Then $B^2 = 0$ and $B + B^\dagger$ is self-adjoint and local [8, 9].

2.4 Barycentric graph metric and locality

Let $X(\mathcal{K})$ be the set of barycenters of all simplices of $\mathcal{K}$. We equip $X(\mathcal{K})$ with the graph metric on the 1-skeleton of the first barycentric subdivision: two simplex barycenters are adjacent precisely when the corresponding simplices are incident by inclusion.

Definition 2.5 (Propagation / locality radius). A bounded operator T on $\mathcal{H}(\mathcal{K}) = \ell^2(X(\mathcal{K}))$ has propagation at most R if

$$\langle \sigma, T\tau \rangle = 0 \quad \text{whenever} \quad \text{dist}_{X(\mathcal{K})}(\sigma, \tau) > R.$$

We also call such an operator R -local.

Remark 2.6. The operator $B + B^\dagger$ is local with respect to the barycentric graph metric. In the Higson–Prodan construction, the operator S is again local/near-neighbor, and the closed Hamiltonians are therefore local tight-binding operators [8, 9]. We work with the barycentric graph metric throughout because the exact collar-support statements in Section 6 depend on this concrete model.

2.5 Orientation and sign conventions

The surface orientation fixes the orientation of every two-simplex used in the fundamental and relative fundamental chains. Edge orientations are arbitrary choices used to write the simplicial boundary matrix; changing an edge orientation conjugates the matrices by a diagonal sign unitary and does not change spectra, Chern numbers, or Toeplitz windings. The cap-product convention is the Higson–Prodan convention stated below: faces have coefficient $+1$ in the oriented fundamental cycle, and the incidence signs in the cap product determine the matrix entries of the duality operator S .

For periodic Bloch symbols on $\mathbb{T}^2$, the Chern number is computed with the orientation $dk_1 \wedge dk_2$ and the sign convention

$$\text{Ch}(P) = \frac{1}{2\pi i} \int_{\mathbb{T}^2} \text{tr}(P dP \wedge dP).$$

The FHS tables use the same oriented plaquette order. For straight periodic interfaces, the Toeplitz convention compresses the positive normal coordinate and leaves the tangential coordinate periodic; with this choice the Toeplitz connecting map has winding equal to the above Chern number. Reversing the surface orientation reverses the cap-product sign and interchanges the H_+ and H_- sign conventions.

2.6 Cap product and the Poincaré-duality operator S (closed case)

For a *closed* oriented triangulation $\mathcal{K}$ (no boundary), let

$$|X\rangle = \sum_{\tau \in \mathcal{K}_2} |\tau\rangle$$

be the fundamental 2-cycle, written after choosing the face orientations coherently with the surface orientation. Equivalently, each oriented face basis vector is included with coefficient $+1$; reversing a face orientation would change the sign of the corresponding basis vector. For an oriented p -simplex ξ , let δ_ξ denote the dual p -cochain in the simplex basis. We use the Higson–Prodan cap-product convention

$$\delta_\xi \frown [v_0, \dots, v_2] = \delta_\xi([v_{2-p}, \dots, v_2]) [v_0, \dots, v_{2-p}],$$

with the value understood to be zero unless the oriented suffix $[v_{2-p}, \dots, v_2]$ equals ξ . Thus the matrix coefficients are incidence signs. The cap product with $|X\rangle$ yields the degree-reversing map

$$P_p |\xi\rangle := \sum_{\tau \in \mathcal{K}_2} \delta_\xi \frown |\tau\rangle \in C_{2-p}(\mathcal{K}), \quad P = \bigoplus_{p=0}^2 P_p,$$

which is the Higson–Prodan operator P [8, 9]. One then defines a Hermitian operator S from P by

$$T|_{\mathcal{H}_p} = \frac{1}{2}(P^\dagger + (-1)^p P), \quad S|_{\mathcal{H}_p} = i^{p(p-1)+1} T|_{\mathcal{H}_p},$$

so that S is local, Hermitian, degree-reversing, and satisfies

$$BS + SB^\dagger = 0.$$

Remark 2.7. The relation $BS = -SB^\dagger$ does *not* by itself imply that $(B + B^\dagger)$ anticommutes with S . Later anticommutation statements are therefore stated either as hypotheses or as measured finite-matrix defects.

2.7 Universal Chern Hamiltonians (closed case)

Definition 2.8 (Closed universal Chern Hamiltonians). For closed oriented $\mathcal{K}$ we define

$$H_{\pm}(\mathcal{K}) = (B + B^{\dagger}) \pm S. \quad (2.1)$$

2.8 Exact and defect-tolerant gap criteria

The closed universal model has a refinement-stable topological gap in the Higson–Prodan theory [8, 9]. We use two elementary finite-dimensional estimates: one under exact anticommutation, and one allowing a controlled anticommutation defect.

Proposition 2.9 (Exact gap lower bound under exact anticommutation). *Let $\mathcal{K}$ be closed. Assume S is Hermitian and satisfies*

$$(B + B^{\dagger})S + S(B + B^{\dagger}) = 0.$$

Then

$$H_{\pm}(\mathcal{K})^2 = (B + B^{\dagger})^2 + S^2.$$

In particular, if

$$\mu_S := \text{dist}(0, \text{spec}(S)) > 0,$$

then

$$\text{spec}(H_{\pm}(\mathcal{K})) \cap (-\mu_S, \mu_S) = \emptyset.$$

Proof. See Appendix A.11.

Proposition 2.10 (Gap lower bound with anticommutation defect). *Let $\mathcal{K}$ be closed, and set*

$$Q := B + B^{\dagger}, \quad \mu_S := \text{dist}(0, \text{spec}(S)), \quad \varepsilon_{\text{ac}} := \|QS + SQ\|.$$

Assume S is Hermitian. Then, for each choice of sign,

$$H_{\pm}(\mathcal{K})^2 \succeq (\mu_S^2 - \varepsilon_{\text{ac}})\mathbf{1}.$$

Consequently, if $\varepsilon_{\text{ac}} < \mu_S^2$, then

$$\text{spec}(H_{\pm}(\mathcal{K})) \cap (-\sqrt{\mu_S^2 - \varepsilon_{\text{ac}}}, \sqrt{\mu_S^2 - \varepsilon_{\text{ac}}}) = \emptyset.$$

Proof. See Appendix A.12.

3 Surfaces with boundary via doubling and jamming

3.1 Doubling

The topological double of a manifold with boundary is standard. We use a simplicial version, stated with the fullness convention needed to keep the result an abstract simplicial complex without duplicate simplices; compare the usual PL discussion in [14, 15].

Definition 3.1 (Double). Let $\mathcal{K}$ triangulate an oriented surface with boundary in the sense of Definition 2.2. Let $\bar{\mathcal{K}}$ be a copy of $\mathcal{K}$ with the opposite surface orientation, having a fresh copy $\bar{v}$ of each interior vertex v and sharing each boundary vertex. The *double* is the abstract simplicial complex

$$D(\mathcal{K}) = \mathcal{K} \cup_{\partial\mathcal{K}} \bar{\mathcal{K}},$$

whose vertex set is $\mathcal{K}_0^{\text{int}} \sqcup \partial\mathcal{K}_0 \sqcup \bar{\mathcal{K}}_0^{\text{int}}$ and whose faces are the original faces of $\mathcal{K}$ together with the reflected faces of $\bar{\mathcal{K}}$ (each reflected face containing at least one copy-side vertex by the fullness condition in Definition 2.2).

Proposition 3.2 (The double is a closed oriented triangulated surface). *Let $\mathcal{K}$ satisfy Definition 2.2. Then $D(\mathcal{K})$ is a finite closed oriented abstract simplicial complex that triangulates a closed surface.*

Proof. See Appendix A.13.

Theorem 3.3 (Simplex count identities under doubling). *Let $\mathcal{K}$ be a triangulated surface with boundary in the sense of Definition 2.2. If $\partial\mathcal{K}_0$ and $\partial\mathcal{K}_1$ denote the boundary vertices and boundary edges, then*

$$|D(\mathcal{K})_2| = 2|\mathcal{K}_2|, \quad |D(\mathcal{K})_1| = 2|\mathcal{K}_1| - |\partial\mathcal{K}_1|, \quad |D(\mathcal{K})_0| = 2|\mathcal{K}_0| - |\partial\mathcal{K}_0|.$$

Moreover,

$$|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|.$$

Proof. See Appendix A.14.

Corollary 3.4 (Dimension of the doubled simplicial Hilbert space). *Under the hypotheses of Theorem 3.3,*

$$\dim \mathcal{H}(D(\mathcal{K})) = 2 \dim \mathcal{H}(\mathcal{K}) - 2|\partial\mathcal{K}_0|.$$

Proof. See Appendix A.15.

Proposition 3.5 (Euler characteristic of the double). *Let $\mathcal{K}$ satisfy Definition 2.2. Then*

$$\chi(D(\mathcal{K})) = 2\chi(\mathcal{K}).$$

Proof. See Appendix A.16.

Corollary 3.6 (Genus formula for the double). *Let $\mathcal{K}$ be a connected orientable surface of genus g with $b \geq 1$ boundary components. Then the closed surface $D(\mathcal{K})$ has genus*

$$g_{\text{double}} = 2g + b - 1.$$

Proof. See Appendix A.17.

3.2 Jamming and block decomposition

By Proposition 3.2, the double $D(\mathcal{K})$ is a closed oriented triangulated surface, so the closed-case operator S from Section 2.6 and hence $H_{\pm}(D(\mathcal{K}))$ are well-defined. We then suppress the copy degrees of freedom by a large on-site potential.

Let

$$\mathcal{H}(D(\mathcal{K})) = \mathcal{H}_{\text{orig}} \oplus \mathcal{H}_{\text{copy}},$$

where $\mathcal{H}_{\text{orig}}$ is spanned by simplices on the original side together with the glued boundary simplices, and $\mathcal{H}_{\text{copy}}$ is spanned by simplices strictly on the copy side. Let P_{orig} and P_{copy} be the corresponding orthogonal projections. Equivalently, $\mathcal{H}_{\text{copy}}$ is spanned by reflected-copy simplices that are not contained in the glued boundary subcomplex; reflected simplices meeting the boundary but not lying in $\partial\mathcal{K}$ are copy-side degrees of freedom.

Definition 3.7 (Jammed Hamiltonian). For $M > 0$ define

$$H_{\pm, M} = H_{\pm}(D(\mathcal{K})) + MP_{\text{copy}}. \quad (3.1)$$

Writing the unjammed operator in block form,

$$H_{\pm}(D(\mathcal{K})) = \begin{pmatrix} A & C \\ C^{\dagger} & D \end{pmatrix} \quad \text{on} \quad \mathcal{H}_{\text{orig}} \oplus \mathcal{H}_{\text{copy}},$$

we obtain

$$H_{\pm, M} = \begin{pmatrix} A & C \\ C^{\dagger} & D + M\mathbf{1} \end{pmatrix}.$$

3.3 The original-side boundary Hamiltonian

The same block decomposition also gives a self-adjoint Hamiltonian on the surface with boundary itself:

$$H_{\pm}^{\partial}(\mathcal{K}) := P_{\text{orig}} H_{\pm}(D(\mathcal{K})) P_{\text{orig}}|_{\mathcal{H}_{\text{orig}}} = A. \quad (3.2)$$

We call this the original-side boundary Hamiltonian associated with the doubled construction. It acts on the degrees of freedom of the original triangulation together with the glued boundary simplices. Since it is a compression of a self-adjoint local operator by a diagonal projection, it is self-adjoint and has finite propagation in the induced barycentric metric. The finite- M theory below shows that $H_{\pm}^{\partial}(\mathcal{K})$ is the large-jam low-energy limit of $H_{\pm, M}$, with explicit error bounds.

3.4 Relative Poincaré–Lefschetz boundary Hamiltonian

There is also an intrinsic boundary operator on $\mathcal{K}$ which does not require forming the double. Let

$$C_p(\mathcal{K}, \partial\mathcal{K}) := C_p(\mathcal{K}) / C_p(\partial\mathcal{K}).$$

Using the simplex basis, we identify $C_p(\mathcal{K}, \partial\mathcal{K})$ with the span of p -simplices not lying in $\partial\mathcal{K}$. Let

$$q_p : C_p(\mathcal{K}) \rightarrow C_p(\mathcal{K}, \partial\mathcal{K}), \quad j_p : C_p(\mathcal{K}, \partial\mathcal{K}) \rightarrow C_p(\mathcal{K})$$

be the quotient map and the canonical representative inclusion. Since $\partial\mathcal{K}$ is one-dimensional, $C_2(\mathcal{K}, \partial\mathcal{K}) = C_2(\mathcal{K})$. The relative fundamental chain is

$$[\mathcal{K}, \partial\mathcal{K}] = \sum_{\tau \in \mathcal{K}_2} |\tau\rangle \in C_2(\mathcal{K}, \partial\mathcal{K}),$$

with the faces oriented coherently with the orientation of $\mathcal{K}$; this is the simplicial representative of the Poincaré–Lefschetz fundamental class [15, Section 3.3].

Cap product with $[\mathcal{K}, \partial\mathcal{K}]$, after the same simplex-basis identification of cochains with chains used in Section 2.6, gives degree-reversing maps

$$P_p^{\text{rel}} : C_p(\mathcal{K}) \longrightarrow C_{2-p}(\mathcal{K}, \partial\mathcal{K}).$$

Explicitly, for an oriented p -simplex ξ ,

$$P_p^{\text{rel}}|\xi\rangle = q_{2-p} \left(\sum_{\tau \in \mathcal{K}_2} \delta_\xi \frown |\tau\rangle \right). \quad (3.3)$$

Define the lifted relative cap-product map

$$\hat{P}_p := j_{2-p} P_p^{\text{rel}} : C_p(\mathcal{K}) \longrightarrow C_{2-p}(\mathcal{K}), \quad \hat{P} := \bigoplus_{p=0}^2 \hat{P}_p.$$

The lifted map discards boundary representatives in the relative target, but it does not introduce any new simplex outside the support of the cap product.

A local example fixes the convention. If $\tau = [v_0, v_1, v_2]$ is an oriented face and $\xi = [v_1, v_2]$, then

$$\delta_\xi \frown \tau = [v_0, v_1].$$

If this complementary edge lies in $\partial\mathcal{K}$, the relative map kills the term because $q_1[v_0, v_1] = 0$, whereas the closed doubled cap product still sees the glued boundary edge. This is the elementary source of the collar term below. Similarly, if $\eta = [u, w]$ is a non-boundary edge with $u \in \partial\mathcal{K}_0$ and $w \notin \partial\mathcal{K}_0$, then

$$Bj[\eta] - j\partial^{\text{rel}}[\eta] = -|u\rangle$$

up to the chosen orientation of η ; the defect is a boundary vertex.

Definition 3.8 (Relative boundary Hamiltonian). Let $S_{\mathcal{K}}^{\text{PL}}$ be the Hermitian degree-reversing operator obtained from $\hat{P}$ by the same degree-dependent symmetrization used in Section 2.6:

$$T^{\text{PL}}|_{\mathcal{H}_p(\mathcal{K})} = \frac{1}{2}((\hat{P})^\dagger + (-1)^p \hat{P}), \quad S_{\mathcal{K}}^{\text{PL}}|_{\mathcal{H}_p(\mathcal{K})} = i^{p(p-1)+1} T^{\text{PL}}|_{\mathcal{H}_p(\mathcal{K})}.$$

The intrinsic Poincaré–Lefschetz boundary Hamiltonians are

$$H_{\pm}^{\text{PL}}(\mathcal{K}) := B_{\mathcal{K}} + B_{\mathcal{K}}^\dagger \pm S_{\mathcal{K}}^{\text{PL}} \quad \text{on } \mathcal{H}(\mathcal{K}). \quad (3.4)$$

Remark 3.9 (Local boundary calculation). The collar term in Theorem 3.12 comes from a simple local mechanism. Suppose $f = [v_0, v_1, w]$ is an oriented face whose edge $e = [v_0, v_1]$ lies in $\partial\mathcal{K}$. In the relative target complex, $q_1(e) = 0$, so any cap-product coefficient whose output representative is e is removed by the quotient $C_1(\mathcal{K}) \rightarrow C_1(\mathcal{K}, \partial\mathcal{K})$ before the lift j . In the closed double, the same local coefficient is paired with the reflected face across e , and the cancellation is part of the closed fundamental cycle instead. Thus the relative and doubled formulas can differ only in matrix coefficients involving e , f , or their incident simplices. All of these simplices lie in a fixed barycentric collar of the boundary. Away from that collar, no boundary representative is removed and the two coefficient formulas are identical.

Lemma 3.10 (Relative cap identity and collar defect). *Let $\partial_r^{\text{rel}} : C_r(\mathcal{K}, \partial\mathcal{K}) \rightarrow C_{r-1}(\mathcal{K}, \partial\mathcal{K})$ be the relative boundary, defined by*

$$\partial_r^{\text{rel}} q_r = q_{r-1} B_{\mathcal{K}, r},$$

and let $\delta_p := B_{\mathcal{K}, p+1}^\dagger : C_p(\mathcal{K}) \rightarrow C_{p+1}(\mathcal{K})$ be the simplicial coboundary under the simplex-basis inner product. With the cap-product sign convention used in Section 2.6,

$$\partial_{2-p}^{\text{rel}} P_p^{\text{rel}} + (-1)^p P_{p+1}^{\text{rel}} \delta_p = 0, \quad p = 0, 1, \quad (3.5)$$

where $P_3^{\text{rel}} = 0$. After lifting by the representative inclusion j ,

$$B_{\mathcal{K}, 2-p} \widehat{P}_p + (-1)^p \widehat{P}_{p+1} \delta_p = (B_{\mathcal{K}, 2-p} j_{2-p} - j_{1-p} \partial_{2-p}^{\text{rel}}) P_p^{\text{rel}}. \quad (3.6)$$

The right-hand side has matrix coefficients only when both the source and target simplices lie in a fixed barycentric collar of $\partial\mathcal{K}$. Consequently

$$B_{\mathcal{K}} S_{\mathcal{K}}^{\text{PL}} + S_{\mathcal{K}}^{\text{PL}} B_{\mathcal{K}}^\dagger$$

is supported in a fixed boundary collar.

Proof. See Appendix A.18.

Lemma 3.11 (Coefficient agreement off the boundary collar). *Let P_D be the closed cap-product operator on $D(\mathcal{K})$, and let $\widehat{P}$ be the lifted relative cap-product operator on $\mathcal{K}$. There is a radius $R_\frown$, depending only on the local cap-product rule, such that for original-side simplices $\sigma, \tau \in \mathcal{K}$,*

$$\langle \sigma, P_D \tau \rangle = \langle \sigma, \widehat{P} \tau \rangle$$

whenever σ and τ both lie outside $N_{R_\frown}(\partial\mathcal{K})$.

Proof. See Appendix A.19.

Theorem 3.12 (Relative boundary Hamiltonian and doubled compression). *Let $\mathcal{K}$ be a triangulated oriented surface with full boundary subcomplex. Under the natural identification of $\mathcal{H}(\mathcal{K})$ with $\mathcal{H}_{\text{orig}} \subset \mathcal{H}(D(\mathcal{K}))$, $H_{\pm}^{\text{PL}}(\mathcal{K})$ is self-adjoint and local. Moreover there is a self-adjoint operator $R_{\partial, \pm}$ such that*

$$H_{\pm}^{\partial}(\mathcal{K}) = H_{\pm}^{\text{PL}}(\mathcal{K}) + R_{\partial, \pm}. \quad (3.7)$$

The comparison error $R_{\partial, \pm}$, and also the relative duality defect

$$B_{\mathcal{K}} S_{\mathcal{K}}^{\text{PL}} + S_{\mathcal{K}}^{\text{PL}} B_{\mathcal{K}}^\dagger,$$

are supported in a bounded collar of the boundary: there is a radius R_{PL} , depending only on the local cap-product convention, such that

$$\langle \sigma, R_{\partial, \pm} \tau \rangle = 0 \quad \text{unless} \quad \sigma, \tau \in N_{R_{\text{PL}}}(\partial\mathcal{K}) \subset X(\mathcal{K}).$$

The same vanishing statement holds with $R_{\partial, \pm}$ replaced by $B_{\mathcal{K}} S_{\mathcal{K}}^{\text{PL}} + S_{\mathcal{K}}^{\text{PL}} B_{\mathcal{K}}^\dagger$. For uniformly regular refinement families using the same cap-product convention, the same R_{PL} works for all members of the family.

Proof. See Appendix A.20.

The finite-matrix comparison in Figures 6 and 15 and tables 8 and 11 illustrates this theorem: it measures the collar support of $R_{\partial,+}$ and its boundary-scale behavior under subdivision.

Remark 3.13. The collar-supported operator $R_{\partial,\pm}$ is not assumed to be small in norm. The theorem therefore identifies the doubled compression with a boundary-local perturbation of the intrinsic relative Hamiltonian; it does not claim that the finite spectra of $H_{\pm}^{\text{PL}}(\mathcal{K})$ and $H_{\pm}^{\partial}(\mathcal{K})$ are uniformly close unless the chosen cap-product convention makes $R_{\partial,\pm} = 0$.

Corollary 3.14 (Quotient equality of the relative and doubled boundary models). *Let $(\mathcal{K}_n)_n$ be a uniformly regular refinement family, using the coarse disjoint union and interface ideal notation of Section 6. Set*

$$H_{\pm}^{\text{PL},\infty} := \bigoplus_n H_{\pm}^{\text{PL}}(\mathcal{K}_n), \quad H_{\pm}^{\partial,\infty} := \bigoplus_n H_{\pm}^{\partial}(\mathcal{K}_n),$$

viewing both operators on the original-side Hilbert space and extending them by zero on the reflected-copy sector. Then

$$H_{\pm}^{\partial,\infty} - H_{\pm}^{\text{PL},\infty} \in C^*(Y \subset X).$$

Consequently the intrinsic Poincaré–Lefschetz boundary Hamiltonian and the doubled original-side compression have the same image in the quotient by the interface ideal. Moreover the relative Poincaré–duality defect

$$\bigoplus_n \left(B_{\mathcal{K}_n} S_{\mathcal{K}_n}^{\text{PL}} + S_{\mathcal{K}_n}^{\text{PL}} B_{\mathcal{K}_n}^{\dagger} \right)$$

belongs to $C^(Y \subset X)$, so the Poincaré–duality relation holds exactly in the quotient. For every continuous function f on a common bounded spectral interval containing the spectra,*

$$\pi\left(f(H_{\pm}^{\partial,\infty})\right) = \pi\left(f(H_{\pm}^{\text{PL},\infty})\right).$$

Proof. See Appendix A.21.

Lemma 3.15 (Spectral counts under collar-supported perturbations). *Let A and $A + R$ be self-adjoint operators on a finite-dimensional Hilbert space, and suppose $R = P_W R P_W$ for the projection onto a subspace W of dimension r . If*

$$N_T(t) := \text{rank } \chi_{(-\infty, t]}(T),$$

then

$$|N_{A+R}(t) - N_A(t)| \leq r \quad \text{for all } t \in \mathbb{R}.$$

Consequently, for every interval $I = (a, b]$,

$$|\text{rank } \chi_I(A + R) - \text{rank } \chi_I(A)| \leq 2r.$$

Applied to $A = H_{\pm}^{\text{PL}}(\mathcal{K})$ and $A + R = H_{\pm}^{\partial}(\mathcal{K})$, one may take $W = \ell^2(N_{\text{RPL}}(\partial\mathcal{K}))$.

Proof. See Appendix A.22.

Remark 3.16. The lemma gives a bulk interpretation of the collar term. Along refinement families for which the boundary-collar dimension is $o(\dim \mathcal{H}(\mathcal{K}))$, the normalized spectral distribution functions of the intrinsic relative Hamiltonian and the doubled compression agree asymptotically. The statement does not rule out boundary eigenvalues; it says that any discrepancy is carried by boundary-scale degrees of freedom.

Corollary 3.17 (Asymptotic bulk spectral equivalence). *For a refinement family $(\mathcal{K}_n)_n$, let*

$$d_n := \dim \mathcal{H}(\mathcal{K}_n), \quad r_n := \dim \ell^2(N_{R_{\text{PL}}}(\partial\mathcal{K}_n)).$$

Let $N_{\pm,n}^{\text{PL}}$ and $N_{\pm,n}^{\partial}$ be the spectral distribution functions of $H_{\pm}^{\text{PL}}(\mathcal{K}_n)$ and $H_{\pm}^{\partial}(\mathcal{K}_n)$. Then

$$\sup_{t \in \mathbb{R}} \left| \frac{N_{\pm,n}^{\partial}(t)}{d_n} - \frac{N_{\pm,n}^{\text{PL}}(t)}{d_n} \right| \leq \frac{r_n}{d_n}.$$

In particular, if $r_n/d_n \rightarrow 0$, the two normalized spectral distribution functions have the same limit points. For fixed-surface two-dimensional refinements with boundary length growing subquadratically relative to the number of simplices, this condition holds. Equivalently, the normalized traces of continuous test functions on a common bounded spectral interval have the same subsequential limits for the two boundary models.

Proof. See Appendix A.23.

4 Finite-dimensional decoupling theory

The finite-dimensional analysis is perturbation theory specialized to the block structure produced by doubling and jamming. We use standard Schur-complement and spectral projection tools, as in Kato's perturbation theory [16, Chapter II] and standard matrix analysis [17], but keep the constants and separation hypotheses explicit because they are later checked on the computed matrices. The estimates isolate the precise way in which the original-side block emerges as $M \rightarrow \infty$.

4.1 Schur complements

Lemma 4.1 (Schur complement resolvent formula). *Let*

$$T = \begin{pmatrix} A & C \\ C^\dagger & D \end{pmatrix}$$

be self-adjoint on $\mathcal{H}_1 \oplus \mathcal{H}_2$, and let $z \in \mathbb{C} \setminus \mathbb{R}$. Then $D - z$ is invertible and the $(1,1)$ block of $(T - z)^{-1}$ is

$$P_1(T - z)^{-1}P_1 = (A - z - C(D - z)^{-1}C^\dagger)^{-1}.$$

Proof. See Appendix A.24.

4.2 Compressed resolvent convergence on the original side

For convenience write

$$D_M := D + M\mathbf{1}, \quad d_M(z) := \text{dist}(z, \text{spec}(D_M)).$$

Since D_M is self-adjoint, $d_M(z) > 0$ whenever $\Im z \neq 0$.

Theorem 4.2 (Compressed resolvent convergence with explicit $O(1/M)$ rate). *Let $H_{\pm,M}$ be as in (3.1). For any $z \in \mathbb{C} \setminus \mathbb{R}$ one has*

$$\|P_{\text{orig}}(H_{\pm,M} - z)^{-1}P_{\text{orig}} - (A - z)^{-1}\| \leq \frac{\|C\|^2}{d_M(z)|\Im z|^2}. \quad (4.1)$$

In particular, if $M > \|D\| + |\Re z|$, then

$$\|P_{\text{orig}}(H_{\pm, M} - z)^{-1}P_{\text{orig}} - (A - z)^{-1}\| \leq \frac{\|C\|^2}{(M - \|D\| - |\Re z|)|\Im z|^2}. \quad (4.2)$$

Hence

$$P_{\text{orig}}(H_{\pm, M} - z)^{-1}P_{\text{orig}} \xrightarrow{M \rightarrow \infty} (A - z)^{-1}$$

in operator norm at rate $O(1/M)$.

Proof. See Appendix A.25.

Corollary 4.3 (Boundary Hamiltonian as the large-jam limit). *Let $H_{\pm}^{\partial}(\mathcal{K}) = A$ be the original-side boundary Hamiltonian from (3.2). For every $z \in \mathbb{C} \setminus \mathbb{R}$,*

$$P_{\text{orig}}(H_{\pm, M} - z)^{-1}P_{\text{orig}} \longrightarrow (H_{\pm}^{\partial}(\mathcal{K}) - z)^{-1}$$

in operator norm, with the explicit $O(M^{-1})$ bound in (4.1). Thus the doubled-and-jammed Hamiltonian recovers a self-adjoint boundary Hamiltonian on the original side in the norm-resolvent sense. By Theorem 3.12, the same limit can be written as

$$P_{\text{orig}}(H_{\pm, M} - z)^{-1}P_{\text{orig}} \longrightarrow (H_{\pm}^{\text{PL}}(\mathcal{K}) + R_{\partial, \pm} - z)^{-1}.$$

Proof. See Appendix A.26.

The $O(M^{-1})$ resolvent trend and the related M^{-2} first-order and leakage trends are reported in Figures 11 and 12 and tables 7 and 10.

Corollary 4.4 (Degenerate full block-resolvent limit). *Let $z \in \mathbb{C} \setminus \mathbb{R}$. Then*

$$(H_{\pm, M} - z)^{-1} \xrightarrow{M \rightarrow \infty} (A - z)^{-1} \oplus 0$$

in operator norm. More precisely,

$$\begin{aligned} \|(H_{\pm, M} - z)^{-1} - ((A - z)^{-1} \oplus 0)\| &\leq \frac{\|C\|^2}{d_M(z)|\Im z|^2} + \frac{2\|C\|}{d_M(z)|\Im z|} \\ &\quad + \frac{1}{d_M(z)} + \frac{\|C\|^2}{d_M(z)^2|\Im z|}. \end{aligned}$$

In particular, the full resolvents converge in norm to the degenerate block limit $(A - z)^{-1} \oplus 0$ at rate $O(1/M)$. Equivalently, this is generalized norm-resolvent convergence in which the copy sector is sent to infinite energy.

Proof. See Appendix A.27.

4.3 A first-order effective operator

Theorem 4.5 (First-order Schur-complement expansion). *Fix $z \in \mathbb{C} \setminus \mathbb{R}$. If $M > \|D - z\|$, then*

$$(D_M - z)^{-1} = \frac{1}{M}\mathbf{1} + R_M^{(2)}(z), \quad \|R_M^{(2)}(z)\| \leq \frac{\|D - z\|}{M(M - \|D - z\|)}.$$

Consequently,

$$K_M(z) = \frac{1}{M}CC^\dagger + \tilde{R}_M(z), \quad \left\| \tilde{R}_M(z) \right\| \leq \frac{\|C\|^2 \|D - z\|}{M(M - \|D - z\|)}.$$

Define the self-adjoint first-order effective operator

$$A_M^{(1)} := A - \frac{1}{M}CC^\dagger.$$

Then

$$\left\| P_{\text{orig}}(H_{\pm, M} - z)^{-1} P_{\text{orig}} - (A_M^{(1)} - z)^{-1} \right\| \leq \frac{\|C\|^2 \|D - z\|}{M(M - \|D - z\|) |\Im z|^2}. \quad (4.3)$$

In particular,

$$P_{\text{orig}}(H_{\pm, M} - z)^{-1} P_{\text{orig}} - (A_M^{(1)} - z)^{-1} = O(1/M^2).$$

Proof. See Appendix A.29.

4.4 Contour spectral projections

For a self-adjoint operator T and a positively oriented simple closed contour $\Gamma \subset \rho(T)$, write

$$P_\Gamma(T) := \frac{1}{2\pi i} \int_\Gamma (z - T)^{-1} dz = -\frac{1}{2\pi i} \int_\Gamma (T - z)^{-1} dz.$$

Thus $P_\Gamma(T)$ is the standard Riesz projection onto the spectral subspace corresponding to the part of $\text{spec}(T)$ contained in the interior of Γ .

Remark 4.6 (Riesz projection convention). For a positively oriented contour, the sign in the definition above is the standard one for the spectral projection onto the spectral subspace enclosed by Γ . We keep the $(T - z)^{-1}$ notation throughout because it matches the resolvent formulas used later.

For a given contour Γ and jam scale M , set

$$\delta_A(\Gamma) := \text{dist}(\Gamma, \text{spec}(A)), \quad \delta_D(\Gamma) := \text{dist}(\Gamma, \text{spec}(D_M)),$$

and

$$\delta_H(\Gamma) := \delta_A(\Gamma) - \frac{\|C\|^2}{\delta_D(\Gamma)}.$$

Theorem 4.7 (General contour-projection comparison). *Let Γ be a positively oriented simple closed contour. Assume*

$$\delta_A(\Gamma) > 0, \quad \delta_D(\Gamma) > 0, \quad \delta_H(\Gamma) > 0.$$

Then:

(i) $\Gamma \subset \rho(H_{\pm, M})$;

(ii)

$$\text{rank} P_\Gamma(H_{\pm, M}) = \text{rank} P_\Gamma(A) + \text{rank} P_\Gamma(D_M);$$

(iii)

$$\left\| (H_{\pm, M} - z)^{-1} - ((A - z)^{-1} \oplus (D_M - z)^{-1}) \right\| \leq \eta_M(\Gamma) \quad (z \in \Gamma), \quad (4.4)$$

where

$$\eta_M(\Gamma) := \frac{\|C\|^2}{\delta_A(\Gamma) \delta_H(\Gamma) \delta_D(\Gamma)} + \frac{2\|C\|}{\delta_H(\Gamma) \delta_D(\Gamma)} + \frac{\|C\|^2}{\delta_H(\Gamma) \delta_D(\Gamma)^2}; \quad (4.5)$$

(iv) consequently,

$$\|P_\Gamma(H_{\pm,M}) - (P_\Gamma(A) \oplus P_\Gamma(D_M))\| \leq \frac{|\Gamma|}{2\pi} \eta_M(\Gamma), \quad (4.6)$$

In particular, for a fixed contour Γ ,

$$P_\Gamma(H_{\pm,M}) - (P_\Gamma(A) \oplus P_\Gamma(D_M)) = O(1/M) \quad \text{as } M \rightarrow \infty.$$

Proof. See Appendix A.30.

4.5 Bounded-energy eigenvalue localization

For $E_{\max} > 0$, define the exact window-separation quantity

$$g_M(E_{\max}) := \text{dist}([-E_{\max}, E_{\max}], \text{spec}(D_M)).$$

Theorem 4.8 (Bounded-energy eigenvalues lie near $\text{spec}(A)$). *Let E be an eigenvalue of $H_{\pm,M}$ with $|E| \leq E_{\max}$. Assume*

$$g_M(E_{\max}) > 0.$$

Then

$$\text{dist}(E, \text{spec}(A)) \leq \frac{\|C\|^2}{g_M(E_{\max})}. \quad (4.7)$$

In particular, if $M > \|D\| + E_{\max}$, then

$$\text{dist}(E, \text{spec}(A)) \leq \frac{\|C\|^2}{M - \|D\| - E_{\max}}. \quad (4.8)$$

Proof. See Appendix A.31.

Corollary 4.9 (Spectral comparison with the relative boundary Hamiltonian). *Under the hypotheses of Theorem 4.8,*

$$\text{dist}(E, \text{spec}(H_{\pm}^{\text{PL}}(\mathcal{K}) + R_{\partial,\pm})) \leq \frac{\|C\|^2}{g_M(E_{\max})}.$$

In particular, if $M > \|D\| + E_{\max}$, then

$$\text{dist}(E, \text{spec}(H_{\pm}^{\text{PL}}(\mathcal{K}) + R_{\partial,\pm})) \leq \frac{\|C\|^2}{M - \|D\| - E_{\max}}.$$

Proof. See Appendix A.32.

Theorem 4.10 (Bounded-energy eigenvalues lie near the first-order effective operator). *Let E be an eigenvalue of $H_{\pm,M}$ with $|E| \leq E_{\max}$. Assume*

$$g_M(E_{\max}) > 0.$$

Then

$$\text{dist}(E, \text{spec}(A_M^{(1)})) \leq \frac{\|C\|^2 (\|D\| + E_{\max})}{M g_M(E_{\max})}. \quad (4.9)$$

In particular, if $M > \|D\| + E_{\max}$, then

$$\text{dist}(E, \text{spec}(A_M^{(1)})) \leq \frac{\|C\|^2 (\|D\| + E_{\max})}{M(M - \|D\| - E_{\max})}. \quad (4.10)$$

Proof. See Appendix A.33.

4.6 Leakage into the jammed sector

Theorem 4.11 (Copy-side leakage bound). *Let (E, ψ) be an eigenpair of $H_{\pm, M}$ with*

$$\psi = (\psi_{\text{orig}}, \psi_{\text{copy}}), \quad |E| \leq E_{\max}.$$

If $g_M(E_{\max}) > 0$, then

$$\|\psi_{\text{copy}}\| \leq \frac{\|C\|}{g_M(E_{\max})} \|\psi_{\text{orig}}\|. \quad (4.11)$$

In particular, if $M > \|D\| + E_{\max}$, then

$$\|\psi_{\text{copy}}\| \leq \frac{\|C\|}{M - \|D\| - E_{\max}} \|\psi_{\text{orig}}\|. \quad (4.12)$$

For normalized ψ this gives

$$\|\psi_{\text{copy}}\| = O(1/M), \quad \|\psi_{\text{copy}}\|^2 = O(1/M^2).$$

Proof. See Appendix A.34.

4.7 Inertia, monotonicity, and no-copy-state bounds

For a finite self-adjoint matrix T , write

$$N_-(T) := \text{rank} \chi_{(-\infty, 0)}(T), \quad N_T(t) := \text{rank} \chi_{(-\infty, t]}(T).$$

Theorem 4.12 (Inertia reduction for large scalar jamming). *Assume $M > \|D\|$, so $D_M = D + M\mathbf{1}$ is positive. Set*

$$S_M(0) := A - CD_M^{-1}C^\dagger.$$

Then

$$N_-(H_{\pm, M}) = N_-(S_M(0)), \quad \dim \ker H_{\pm, M} = \dim \ker S_M(0).$$

Moreover

$$\|S_M(0) - A\| \leq \frac{\|C\|^2}{M - \|D\|}.$$

Consequently, if $0 \notin \text{spec}(A)$, $\delta_A := \text{dist}(0, \text{spec}(A))$, and

$$\frac{\|C\|^2}{M - \|D\|} < \delta_A,$$

then $H_{\pm, M}$ is invertible and

$$N_-(H_{\pm, M}) = N_-(A).$$

Proof. See Appendix A.35.

Corollary 4.13 (Monotonicity in the scalar jam scale). *Let $M_2 > M_1$. If the eigenvalues of $H_{\pm, M}$ are listed in nondecreasing order with multiplicity, then*

$$\lambda_j(H_{\pm, M_1}) \leq \lambda_j(H_{\pm, M_2}) \quad \text{for every } j.$$

Equivalently, for every threshold $t \in \mathbb{R}$,

$$N_{H_{\pm, M_2}}(t) \leq N_{H_{\pm, M_1}}(t).$$

In particular, $M \mapsto N_-(H_{\pm, M})$ is nonincreasing.

Proof. See Appendix A.36.

Corollary 4.14 (No low-energy pure-copy states). *Let $E_{\max} > 0$, and assume $M > \|D\| + E_{\max}$. If*

$$Q_M := \chi_{[-E_{\max}, E_{\max}]}(H_{\pm, M}),$$

then the compression $P_{\text{orig}}Q_M : \text{Ran}Q_M \rightarrow \mathcal{H}_{\text{orig}}$ is injective. Hence

$$\text{rank}Q_M \leq \dim \mathcal{H}_{\text{orig}}.$$

More quantitatively, for every $\psi \in \text{Ran}Q_M$,

$$\|P_{\text{copy}}\psi\| \leq \frac{E_{\max}\|\psi\| + \|C\| \|P_{\text{orig}}\psi\|}{M - \|D\|} \leq \frac{E_{\max} + \|C\|}{M - \|D\|} \|\psi\|.$$

Proof. See Appendix A.37.

Proposition 4.15 (Inherited gap from the original-side block). *Assume*

$$\text{spec}(A) \cap (-g_A, g_A) = \emptyset$$

for some $g_A > 0$. Suppose

$$g_M(g_A) := \text{dist}([-g_A, g_A], \text{spec}(D_M)) > 0,$$

and define

$$\Delta_M := \frac{\|C\|^2}{g_M(g_A)}.$$

If $\Delta_M < g_A$, then

$$\text{spec}(H_{\pm, M}) \cap (-g_A + \Delta_M, g_A - \Delta_M) = \emptyset.$$

Proof. See Appendix A.38.

Corollary 4.16 (Uniform direct-sum large-jam estimates). *Let*

$$H_{n, M} = \begin{pmatrix} A_n & C_n \\ C_n^\dagger & D_n + M\mathbf{1} \end{pmatrix}$$

be a sequence of doubled-and-jammed block Hamiltonians with

$$\bar{C} := \sup_n \|C_n\| < \infty, \quad \bar{D} := \sup_n \|D_n\| < \infty.$$

Set

$$H_M^\oplus := \bigoplus_n H_{n, M}, \quad A^\oplus := \bigoplus_n A_n, \quad C^\oplus := \bigoplus_n C_n,$$

and let $P_{\text{orig}}^\oplus$ be the direct-sum original-side projection. Then the finite- M estimates above hold uniformly in n . In particular, for every $z \in \mathbb{C} \setminus \mathbb{R}$ and $M > \bar{D} + |\Re z|$,

$$\left\| P_{\text{orig}}^\oplus (H_M^\oplus - z)^{-1} P_{\text{orig}}^\oplus - (A^\oplus - z)^{-1} \right\| \leq \frac{\bar{C}^2}{(M - \bar{D} - |\Re z|)|\Im z|^2}.$$

Also

$$(H_M^\oplus - z)^{-1} \longrightarrow (A^\oplus - z)^{-1} \oplus 0$$

in norm, and for every $f \in C_0(\mathbb{R})$,

$$f(H_M^\oplus) \longrightarrow f(A^\oplus) \oplus 0$$

in norm as $M \rightarrow \infty$. The first-order compressed approximation satisfies

$$\left\| P_{\text{orig}}^\oplus (H_M^\oplus - z)^{-1} P_{\text{orig}}^\oplus - \left(A^\oplus - \frac{1}{M} C^\oplus (C^\oplus)^\dagger - z \right)^{-1} \right\| \leq \frac{\bar{C}^2 (\bar{D} + |z|)}{M(M - \bar{D} - |z|) |\Im z|^2}$$

whenever $M > \bar{D} + |z|$. Finally, if E is an eigenvalue of some $H_{n,M}$ with $|E| \leq E_{\max}$ and $M > \bar{D} + E_{\max}$, then

$$\text{dist}(E, \text{spec}(A_n)) \leq \frac{\bar{C}^2}{M - \bar{D} - E_{\max}}, \quad \|\psi_{\text{copy}}\| \leq \frac{\bar{C}}{M - \bar{D} - E_{\max}} \|\psi_{\text{orig}}\|$$

for every corresponding eigenvector ψ .

Proof. See Appendix A.39.

Theorem 4.17 (Internal bulk spectral equivalence of PL and jammed models). *Let $(\mathcal{K}_n)_n$ be finite triangulated oriented surfaces with full boundary subcomplex, and let*

$$d_n := \dim \mathcal{H}(\mathcal{K}_n).$$

For each n , write the doubled-and-jammed Hamiltonian as

$$H_{\pm,n,M_n} = \begin{pmatrix} A_{\pm,n} & C_{\pm,n} \\ C_{\pm,n}^\dagger & D_{\pm,n} + M_n \mathbf{1} \end{pmatrix}, \quad A_{\pm,n} = H_\pm^{\text{PL}}(\mathcal{K}_n) + R_{\partial,\pm,n},$$

where $R_{\partial,\pm,n}$ is the boundary-collar operator from Theorem 3.12. Assume

$$\sup_n \|A_{\pm,n}\| < \infty, \quad \sup_n \|H_\pm^{\text{PL}}(\mathcal{K}_n)\| < \infty, \quad \sup_n \|C_{\pm,n}\| < \infty, \quad \sup_n \|D_{\pm,n}\| < \infty,$$

and $M_n \rightarrow \infty$. Let r_n be the dimension of a subspace supporting $R_{\partial,\pm,n}$, and assume

$$\frac{r_n}{d_n} \longrightarrow 0.$$

Then, for every $f \in C_0(\mathbb{R})$,

$$\frac{1}{d_n} \text{Tr} f(H_{\pm,n,M_n}) - \frac{1}{d_n} \text{Tr} f(H_\pm^{\text{PL}}(\mathcal{K}_n)) \longrightarrow 0.$$

Equivalently, the intrinsic Poincaré–Lefschetz model, the doubled original-side compression, and the finite- M_n jammed model have the same normalized bulk spectral subsequential limits. This statement is finite-dimensional and uses no closed-bulk Chern theorem, no uniform doubled-bulk gap, and no interface-pairing hypothesis.

Proof. See Appendix A.40.

5 A periodic triangular bulk input

One closed-bulk input is checked directly here, rather than imported from Higson–Prodan. Let $z_j = e^{ik_j}$, $k = (k_1, k_2) \in \mathbb{T}^2$, and consider the $\mathbb{Z}^2$ -periodic triangular triangulation with one vertex v , three oriented edges a, b, c , and two oriented faces f, g per fundamental cell. The Bloch fiber is

$$\mathcal{H}(k) = \mathbb{C}\langle v \rangle \oplus \mathbb{C}\langle a, b, c \rangle \oplus \mathbb{C}\langle f, g \rangle.$$

With respect to this ordered basis, the boundary blocks are

$$\partial_1(k) = \begin{bmatrix} z_1 - 1 & z_2 - 1 & z_1 z_2 - 1 \end{bmatrix}, \quad \partial_2(k) = \begin{bmatrix} 1 & -z_2 \\ z_1 & -1 \\ -1 & 1 \end{bmatrix}.$$

A direct multiplication gives $\partial_1(k)\partial_2(k) = 0$, so these blocks form the Bloch boundary operator of a periodic chain complex. The cap-product blocks are

$$P_0(k) = \begin{bmatrix} z_1^{-1} & z_2^{-1} \\ z_1^{-1} & z_2^{-1} \end{bmatrix}, \quad P_1(k) = \begin{bmatrix} 0 & z_1^{-1} & 0 \\ 0 & 0 & 0 \\ -z_2^{-1} & 0 & 0 \end{bmatrix}, \quad P_2(k) = \begin{bmatrix} 1 & 1 \end{bmatrix}.$$

The Hermitian duality block is

$$S_0(k) = \frac{i}{2}(P_2(k)^* + P_0(k)), \quad S_1(k) = \frac{i}{2}(P_1(k)^* - P_1(k)), \quad S_2(k) = -\frac{i}{2}(P_0(k)^* + P_2(k)),$$

and

$$H_{\pm}^{\Delta}(k) = B(k) + B(k)^* \pm S(k),$$

where $B(k)$ has blocks $\partial_1(k)$ and $\partial_2(k)$. The displayed signs give $S(k) = S(k)^*$, hence $H_{\pm}^{\Delta}(k)$ is Hermitian.

Proposition 5.1 (Periodic triangular characteristic polynomial). *Set*

$$F(k) = 28 - 8 \cos k_1 - 6 \cos k_2 - 6 \cos(k_1 + k_2), \quad G(k) = \sin k_2 + \sin(k_1 + k_2).$$

Then

$$\det\left(\lambda \mathbf{1} - H_{\pm}^{\Delta}(k)\right) = \left(\lambda^2 - \frac{F(k)}{4}\right) \left(\lambda^4 - \frac{13}{2}\lambda^2 + \frac{F(k)}{4} \pm \lambda G(k)\right).$$

In particular, $H_{\pm}^{\Delta}(k)$ has one exact pair of eigenvalue branches

$$\lambda = \pm \frac{1}{2} \sqrt{F(k)}.$$

Proof. See Appendix [A.41](#).

Remark 5.2. The factorization is a spectral calculation, not a Chern-number calculation. The quadratic branch can meet branches from the quartic factor at nonzero energies, so this does not split the occupied rank-three Bloch bundle into globally separated line subbundles.

Theorem 5.3 (Periodic triangular bulk gap). *For every $k \in \mathbb{T}^2$, $H_{\pm}^{\Delta}(k)$ is invertible. More precisely,*

$$\det H_{\pm}^{\Delta}(k) = -\frac{1}{16} (-28 + 8 \cos k_1 + 6 \cos k_2 + 6 \cos(k_1 + k_2))^2 \leq -4.$$

Moreover $\|H_{\pm}^{\Delta}(k)\| \leq 8$, hence

$$\text{dist}(0, \text{spec} H_{\pm}^{\Delta}(k)) \geq 2^{-13} \quad (k \in \mathbb{T}^2).$$

The negative Bloch bundle has rank 3.

Proof. See Appendix A.42.

Corollary 5.4 (Small periodic perturbations preserve the triangular gap). *Let $V(k)$ be a continuous self-adjoint 6×6 matrix over $\mathbb{T}^2$. If*

$$\sup_{k \in \mathbb{T}^2} \|V(k)\| < 2^{-13},$$

then $H_{\pm}^{\Delta}(k) + V(k)$ is invertible for every $k \in \mathbb{T}^2$, and its negative Bloch bundle has rank 3. Thus the exact periodic gap is stable under all sufficiently small self-adjoint periodic perturbations, including finite-range periodic perturbations of the triangular lattice model.

Proof. See Appendix A.43.

Proposition 5.5 (Positive duality strength homotopy). *For $m > 0$, set*

$$H_{\pm,m}^{\Delta}(k) = B(k) + B(k)^* \pm mS(k).$$

Then $H_{\pm,m}^{\Delta}(k)$ is invertible for every $k \in \mathbb{T}^2$. In fact, with $c_1 = \cos k_1$, $c_2 = \cos k_2$, and $c_{12} = \cos(k_1 + k_2)$,

$$\det H_{\pm,m}^{\Delta}(k) = -\frac{1}{16} (8(3 - c_1 - c_2 - c_{12}) + 2m^2(2 + c_2 + c_{12}))^2.$$

Consequently all positive values of the duality strength m are connected by a zero-gapped norm-continuous homotopy.

Proof. See Appendix A.44.

Theorem 5.6 (Periodic triangular Chern signs). *For the periodic triangular Bloch Hamiltonians above, the rank-three negative Bloch bundles have*

$$\text{Ch}(E_-(H_+^{\Delta})) = -1, \quad \text{Ch}(E_-(H_-^{\Delta})) = +1.$$

Proof. See Appendix A.45.

Corollary 5.7 (Periodic Chern stability). *For every $m > 0$,*

$$\text{Ch}(E_-(H_{+,m}^{\Delta})) = -1, \quad \text{Ch}(E_-(H_{-,m}^{\Delta})) = +1.$$

For each fixed sign and each fixed $m > 0$, set

$$g_{\pm,m} := \min_{k \in \mathbb{T}^2} \text{dist}(0, \text{spec} H_{\pm,m}^{\Delta}(k)) > 0.$$

If $V(k)$ is a continuous self-adjoint 6×6 matrix over $\mathbb{T}^2$ with

$$\sup_{k \in \mathbb{T}^2} \|V(k)\| < g_{\pm, m},$$

then $H_{\pm, m}^{\Delta}(k) + V(k)$ is gapped at zero and its negative Bloch bundle has the same Chern number as $H_{\pm, m}^{\Delta}$. Moreover, if $V(k)$ is a continuous self-adjoint 6×6 matrix over $\mathbb{T}^2$ with

$$\sup_{k \in \mathbb{T}^2} \|V(k)\| < 2^{-13},$$

then $H_{\pm}^{\Delta}(k) + V(k)$ is gapped at zero and its negative Bloch bundle has the same Chern number as $H_{\pm}^{\Delta}$. In particular, sufficiently small self-adjoint finite-range periodic perturbations of the triangular model do not change the Chern signs.

Proof. See Appendix A.46.

The fixed positive-mass cases $m = 0.5$ for H_+ and $m = 2$ for H_- are included in the reproducible FHS check in Table 4.

Finite-cover pullbacks and finite-index cell enlargements are recorded in Appendix A.8; the following FHS table checks those signs numerically.

As a numerical check on Theorem 5.6, the companion script applies the gauge-invariant lattice formula of Fukui–Hatsugai–Suzuki [7] to the rank-three negative spectral projection. It checks the returned integer, the link denominators, and the plaquette branch margins. The computation records

$$\text{Ch}_{\text{FHS}}(E_-(H_+^{\Delta})) = -1, \quad \text{Ch}_{\text{FHS}}(E_-(H_-^{\Delta})) = +1.$$

Also $H_-^{\Delta}(k) = \overline{H_+^{\Delta}(-k)}$, an antiunitary equivalence that explains the opposite signs in the two FHS computations. The base involution $k \mapsto -k$ on $\mathbb{T}^2$ is orientation-preserving, since its Jacobian has determinant $+1$ in two dimensions; the sign reversal comes from complex conjugation of the Bloch bundle, not from the map $k \mapsto -k$ alone. This symmetry observation is used only to cross-check the two reported FHS computations; the Chern proof is Theorem 5.6.

A second finite-size sanity check, not used as proof, is to diagonalize $H_{+, m}^{\Delta}$ on a large $N \times N$ periodic torus at $m = 1$, insert one flux through the tangential cycle, and track the occupied spectral flow with the same $dk_1 \wedge dk_2$ orientation. The expected crossing direction is the one corresponding to Chern number -1 ; replacing m by -1 or using H_-^{Δ} reverses that direction. This check is a finite-matrix diagnostic of the sign convention, not a substitute for the local crossing proof.

Table 3: Periodic triangular Bloch certificate. The FHS integer is stable across the displayed grids; the positive link and plaquette margins record that the computation is away from gauge singularities and branch-cut crossings.

Hamiltonian	Grid	FHS Chern	min $ E $	min link	max plaquette
H_+	11×11	-1.000000	0.550897	0.905137	0.265005
H_+	21×21	-1.000000	0.540655	0.970926	0.080163
H_+	31×31	-1.000000	0.540954	0.986311	0.037594
H_+	51×51	-1.000000	0.540291	0.994870	0.014056
H_-	11×11	1.000000	0.550897	0.905137	0.265005
H_-	21×21	1.000000	0.540655	0.970926	0.080163
H_-	31×31	1.000000	0.540954	0.986311	0.037594
H_-	51×51	1.000000	0.540291	0.994870	0.014056

Table 4: Periodic triangular extension checks. The same FHS computation is applied to fixed positive-mass models, finite-cover pullbacks, an orientation-reversing cover, and a folded 2×1 supercell. The entries numerically check the Chern signs used in Corollaries 5.7 and A.23 and therefore the straight-periodic Toeplitz results.

Diagnostic	Model	Grid	expected	FHS	min $ E $	min link
positive mass	$H_{+,m}, m = 0.5$	31×31	-1	-1.000000	0.287664	0.959784
positive mass	$H_{-,m}, m = 2$	31×31	1	1.000000	0.806343	0.993537
finite cover	$H_{+,A}, A = \text{diag}(2, 1)$	31×31	-2	-2.000000	0.540954	0.945837
finite cover	$H_{-,A}, A = \text{diag}(2, 1)$	31×31	2	2.000000	0.540954	0.945837
orientation-reversing cover	$H_{+,A}, A = \text{diag}(1, -1)$	31×31	1	1.000000	0.540954	0.986311
cell enlargement	$H_{+}, 2 \times 1$ folded cell	21×21	-1	-1.000000	0.540655	0.965480
cell enlargement	$H_{-}, 2 \times 1$ folded cell	21×21	1	1.000000	0.540655	0.965480

The main straight-periodic Toeplitz evaluation is kept in the body. Finite periodic tori, annular families, positive-copy confinement variants, covers, homotopies, split/corner decompositions, finite defects, and tail-split component maps are collected in Appendix A.5.

The remaining results in this section identify the straight-periodic cases where the interface pairing itself is evaluated in the paper. The Toeplitz theorem gives the edge map, the next corollaries record lattice changes, covers, and homotopies, and the split/corner/tail statements state the additional quotient hypotheses needed for componentwise readings.

Theorem 5.8 (Toeplitz evaluation for a straight periodic interface). *Let $H(k_1, k_2)$ be a continuous finite-range self-adjoint $N \times N$ Bloch Hamiltonian on $\mathbb{T}^2$, gapped at zero, and let*

$$P(k_1, k_2) = \chi_{(-\infty, 0)}(H(k_1, k_2)).$$

Write

$$\text{Ch}(P) = \frac{1}{2\pi i} \int_{\mathbb{T}^2} \text{tr}(P dP \wedge dP) \in \mathbb{Z}$$

with the orientation $dk_1 \wedge dk_2$. Compress the second lattice direction to a half-space and keep the first direction periodic. After Fourier transform in the first direction, the straight-edge extension is the Toeplitz extension

$$0 \longrightarrow C(S^1, \mathcal{K}) \longrightarrow \mathcal{T}_{\text{per}} \xrightarrow{\sigma} C(\mathbb{T}^2) \longrightarrow 0,$$

where σ is the symbol map in the normal variable. If

$$\partial_{\text{T}} : K_0(C(\mathbb{T}^2)) \longrightarrow K_1(C(S^1, \mathcal{K})) \cong K_1(C(S^1))$$

is its connecting map, then

$$\text{wind}(\partial_{\text{T}}([P])) = \text{Ch}(P).$$

The sign is fixed by the above orientation and by taking the compressed coordinate to be the positive normal direction. This is the standard Toeplitz/K-theoretic bulk-edge map in the periodic setting; see, for example, [20, 19, 21, 30].

Proof. See Appendix A.52.

Corollary 5.9 (Primitive-direction straight periodic interfaces). *Let H and P be as in Theorem 5.8. Let $t, n \in \mathbb{Z}^2$ be a lattice basis with t the tangential direction of a straight periodic interface and n the compressed positive normal direction. Set*

$$\varepsilon(t, n) := \det[t \ n] \in \{\pm 1\}.$$

After rewriting the Bloch Hamiltonian in the coordinates

$$k = \kappa t + \eta n, \quad (\kappa, \eta) \in \mathbb{T}^2,$$

the half-space Toeplitz boundary map in the η -direction satisfies

$$\text{wind}(\partial_{\mathbb{T}, t, n}([P])) = \varepsilon(t, n) \text{Ch}(P),$$

where $\text{Ch}(P)$ is computed with the original orientation $dk_1 \wedge dk_2$. In particular, every primitive straight periodic edge direction is covered after a lattice change of variables.

Proof. See Appendix A.53.

Finite-cover Toeplitz formulas are stated in Appendix A.8.

Gapped periodic homotopy invariance is stated in Appendix A.8.

Split componentwise Toeplitz formulas are stated in Appendix A.8.

Finite-corner and K_1 -trivial corner quotient criteria are stated in Appendix A.7.

Corollary 5.10 (Periodic straight-edge winding from Toeplitz pairing). *For the periodic triangular bulk models H_+^Δ and H_-^Δ , suppose the boundary component is a primitive straight periodic interface with tangential/positive-normal basis (t, n) , and use the Toeplitz winding from Theorem 5.8 or Corollary 5.9. Then the doubled-and-jammed interface class has winding*

$$\text{wind} = -\det[t \ n] \quad \text{for } H_+^\Delta, \quad \text{wind} = \det[t \ n] \quad \text{for } H_-^\Delta.$$

The same formula holds for every fixed positive-duality-strength triangular model and for every small finite-range periodic perturbation covered by Corollary 5.7. Finite-cover, finite-defect, finite-corner, split, and tail-split variants are the appendix statements collected in Appendix A.5; their extra sidewise-gap and quotient hypotheses are not part of this body corollary.

Proof. See Appendix A.60.

Finite-defect and fixed-collar Toeplitz robustness variants are stated in Appendix A.6.

Bounded-distance rough collars and eventually periodic finite-defect collars are treated in Appendix A.6.

Tail-split component winding maps and the consolidated list of internally verified straight-periodic classes are stated in Appendix A.8.

6 A coarse interface class construction

We place the doubled-and-jammed refinement family in a Roe-algebra framework that constructs the interface class under the stated uniform bulk hypotheses. We use the quotient extension

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{A} \xrightarrow{\pi} \mathcal{A}/\mathcal{I} \longrightarrow 0, \quad \mathcal{A} := C^*(X), \quad \mathcal{I} := C^*(Y \subset X),$$

and define the interface class from a canonical quotient projection associated to a sidewise realization.

Here $C^*(X)$ denotes the scalar Roe algebra: the norm closure in $\mathcal{B}(\ell^2(X))$ of locally compact finite-propagation operators. We do not use the uniform Roe algebra in this paper. The Ewert–Meyer input is invoked only through the compatibility condition in Definition 6.37, where the relevant coarse Mayer–Vietoris boundary map is paired with the Higson–Prodan Chern pairing.

The conditional structure in this section is explicit. Uniform locality, boundedness, and a uniform closed doubled-bulk gap give the quotient projection and the interface K_1 -class. The closed-bulk Chern value is imported only through Assumption 6.15 or through the periodic Bloch calculation of Section 5; a numerical winding value is then obtained only when the selected coarse Mayer–Vietoris boundary map is paired compatibly with that Chern pairing.

In the present scalar-coefficient setting, the Roe algebra $C^*(X)$ is *unital*: the identity operator on $\ell^2(X)$ has propagation 0 and is locally compact (since every finite-support compression is finite rank), hence belongs to $C^*(X)$. Similarly, every diagonal $\{0, 1\}$ -valued multiplication operator on $\ell^2(X)$ belongs to $C^*(X)$. The quotient $\mathcal{A}/\mathcal{I}$ is therefore also unital, and the continuous functional calculus in both algebras is the standard unital one.

This convention is the scalar, ample-representation version of the Roe algebra. Ewert–Meyer’s Mayer–Vietoris boundary map is used below only through the induced connecting map for the coarsely excisive pair (X^+, X^-) and its compatible pairing with the closed bulk Chern class. Passing between this unital scalar presentation and stabilized or nonunital presentations changes the bookkeeping, not the connecting class, provided the compatibility required in Definition 6.37 is verified.

Remark 6.1 (Notation and conventions). Throughout this section, whenever T is a bounded self-adjoint operator on $\ell^2(X)$, we write

$$\mathrm{spec}_{\mathrm{op}}(T) := \mathrm{spec}_{\mathcal{B}(\ell^2(X))}(T)$$

for its operator spectrum. Because $C^*(X) \subset \mathcal{B}(\ell^2(X))$ is a unital inclusion, the $C^*(X)$ -spectrum and the operator spectrum of any $T \in C^*(X)$ coincide. All spectral-gap statements for concrete operators refer to $\mathrm{spec}_{\mathrm{op}}$.

We define the canonical quotient projection via *continuous* cutoff functions that agree with $\chi_{(-\infty, 0)}$ on the relevant bounded spectral windows. This avoids applying the discontinuous function $\chi_{(-\infty, 0)}$ across the quotient map, since the spectrum of a quotient-algebra element can differ from the operator spectrum of its lift. In the present unital scalar Roe algebra this is ordinary unital functional calculus; in nonunital variants one should use compactly supported cutoffs on the same bounded spectral windows.

6.1 Uniformly regular refinement families

Let $(\mathcal{K}_n)$ be a refinement sequence of a fixed compact oriented surface with boundary. For each n , write

$$X_n := X(D(\mathcal{K}_n))$$

for the set of barycenters of all simplices in the doubled triangulation, equipped with the barycentric graph metric from Definition 2.5.

Definition 6.2 (Uniformly regular refinement family). The refinement sequence $(\mathcal{K}_n)$ is *uniformly regular* if the metric spaces X_n have uniformly bounded geometry: for every $r > 0$ there exists $N(r) < \infty$ such that

$$|B_{X_n}(x, r)| \leq N(r) \quad \text{for all } n \text{ and all } x \in X_n.$$

A sufficient concrete condition is a uniform bound on simplex incidences under refinement.

For each n , let $X_{n, \mathrm{orig}}$ be the subset of simplex barycenters lying on the original side together with the glued boundary simplices, and let $X_{n, \mathrm{copy}}$ be the subset of simplex barycenters strictly on

the reflected side. Let $\Gamma_n \subset X_n$ denote the set of simplex barycenters whose simplices meet the glued boundary subcomplex. If $R > 0$, define the R -collar

$$Y_n^{(R)} := N_R(\Gamma_n) \subset X_n.$$

Definition 6.3 (Coarse disjoint unions and the interface ideal). Let

$$X := \bigsqcup_{n=1}^{\infty} X_n$$

be a coarse disjoint union equipped with a metric that restricts to the given metric on each component and separates distinct components by distances tending to infinity; concretely, for every $r > 0$ there is N_r such that $\text{dist}(X_n, X_m) > r$ whenever $n \neq m$ and $\max\{n, m\} > N_r$. For fixed $R \geq 0$, let

$$Y^{(R)} := \bigsqcup_{n=1}^{\infty} Y_n^{(R)} \subset X.$$

The Roe algebra $C^*(X)$ is the norm closure of bounded finite-propagation operators on $\ell^2(X)$ that are locally compact. The interface ideal $C^*(Y^{(R)} \subset X)$ is the norm closure of bounded finite-propagation operators supported within a bounded neighborhood of $Y^{(R)}$.

Lemma 6.4 (Roe ideal property). *For every $R \geq 0$, the algebra $C^*(Y^{(R)} \subset X)$ is a closed two-sided ideal in $C^*(X)$.*

Proof. See Appendix A.76.

Proposition 6.5 (Independence of fixed collar radius). *Let*

$$\Gamma := \bigsqcup_{n=1}^{\infty} \Gamma_n \subset X, \quad Y^{(R)} := \bigsqcup_{n=1}^{\infty} N_R(\Gamma_n).$$

For any fixed $R, R' \geq 0$,

$$C^*(Y^{(R)} \subset X) = C^*(Y^{(R')} \subset X).$$

Consequently the interface ideal used below is independent of the particular fixed collar radius.

Proof. See Appendix A.77.

Proposition 6.6 (Coarse invariance of the interface ideal). *Let $Y_0, Y_1 \subset X$ be subsets at finite Hausdorff distance: there is $R < \infty$ such that*

$$Y_0 \subset N_R(Y_1), \quad Y_1 \subset N_R(Y_0).$$

Then

$$C^*(Y_0 \subset X) = C^*(Y_1 \subset X).$$

Proof. See Appendix A.78.

Remark 6.7. For background on coarse spaces and Roe algebras, see Roe [35]. Here the only Roe-algebra input is the concrete quotient extension

$$0 \rightarrow C^*(Y \subset X) \rightarrow C^*(X) \rightarrow C^*(X)/C^*(Y \subset X) \rightarrow 0,$$

together with the standard K_1 -class of a unitary in $\mathbf{1} + \mathcal{I}$.

6.2 Uniform locality and the bulk Fermi projection

For each n , write

$$H_{\pm,n} := H_{\pm}(D(\mathcal{K}_n)).$$

Assumption 6.8 (Uniform cap-product locality input). There exist constants $R_S, N_S, c_S > 0$, independent of n , such that for every n and every pair of simplex barycenters $\sigma, \tau \in X_n$:

- (i) if $d_{X_n}(\sigma, \tau) > R_S$, then $S_n(\sigma, \tau) = 0$;
- (ii) each row and each column of the matrix of S_n has at most N_S nonzero entries;
- (iii) $|S_n(\sigma, \tau)| \leq c_S$.

Remark 6.9. Assumption 6.8 is the uniform bounded-geometry locality input used for the Higson–Prodan cap-product operator on the closed doubled family. For the exact Higson–Roe/Higson–Prodan duality operator on a uniformly regular closed family, this is an imported consequence of geometric control: the cap-product operator is a geometrically controlled map in the sense of Higson–Roe [11], hence has uniformly bounded propagation and uniformly bounded matrix coefficients. Equivalently, it may be kept as an explicit hypothesis for abstract variants of the construction. Concretely, the required input is a uniform finite row and column support bound and a uniform entry bound for the cap-product matrix. In the local Higson–Prodan formula this comes from the fact that each term pairs only simplices in a uniformly bounded barycentric neighborhood, with coefficients drawn from a fixed finite set of incidence signs and phase factors; the lemma below records the resulting Schur-test estimate used here.

Lemma 6.10 (Uniform locality and norm bound for S_n). *Assume that the Higson–Prodan cap-product operators S_n satisfy Assumption 6.8. Then*

$$\text{prop}(S_n) \leq R_S, \quad \|S_n\| \leq N_S c_S$$

for all n .

Proof. See Appendix A.79.

Proposition 6.11 (Uniform locality and norm bound). *Let $(\mathcal{K}_n)$ be a uniformly regular refinement family, and assume 6.8. Then there exist constants $R_{\text{loc}}, B_{\text{bulk}} > 0$ such that for all n :*

- (a) $H_{\pm,n}$ has propagation at most R_{loc} on $\ell^2(X_n)$;
- (b) $\|H_{\pm,n}\| \leq B_{\text{bulk}}$.

Proof. See Appendix A.80.

Remark 6.12 (Structure of the gap condition). The operator S_n is degree-reversing, so its $\mathcal{H}_0 \leftrightarrow \mathcal{H}_2$ block is a rectangular matrix of dimensions $|D(\mathcal{K}_n)_2| \times |D(\mathcal{K}_n)_0|$. On a closed surface with Euler characteristic χ , the relation $3|K_2| = 2|K_1|$ gives $|K_0| - |K_2| = \chi - |K_2|/2$, so generically $|K_0| \neq |K_2|$ and S_n has $||K_0| - |K_2||$ zero eigenvalues from the dimension mismatch. Hence the sufficient condition $\text{dist}(0, \text{spec}(S_n)) > 0$ from Proposition 2.9 is *not* available for typical triangulations (it holds only when $|K_0| = |K_2|$, e.g., the tetrahedron).

Nevertheless, $H_{\pm,n}$ can still be gapped because the term $(B_n + B_n^\dagger)$ may be nonzero on $\ker(S_n)$. More precisely:

$$\langle \psi, H_{\pm,n}^2 \psi \rangle = \|(B_n + B_n^\dagger)\psi\|^2 + \|S_n \psi\|^2 \pm \langle \psi, (B_n^\dagger S_n + S_n B_n) \psi \rangle$$

for each ψ . On $\ker(S_n)$, $\|S_n\psi\|^2 = 0$ and the cross term vanishes as a quadratic form: $\langle \psi, (B_n^\dagger S_n + S_n B_n)\psi \rangle = \langle B_n\psi, S_n\psi \rangle + \langle S_n\psi, B_n\psi \rangle = 0$ since $S_n\psi = 0$. So the gap on $\ker(S_n)$ is controlled by $(B_n + B_n^\dagger)^2$. On $\ker(S_n)^\perp$, the positive term S_n^2 also contributes.

The uniform gap therefore depends on a joint spectral property of $B_n + B_n^\dagger$ and S_n together, not on S_n alone. The Higson–Roe controlled-duality machinery supplies the relevant locality and Poincaré-duality input for the closed model, but a refinement-uniform numerical spectral gap is kept here as an explicit hypothesis unless it is supplied by an external Higson–Prodan admissibility or clean-gap theorem for the particular closed refinement family under consideration. The numerics (Figure 14) report the measured gap on the tested refinement levels, but they do not prove a refinement-uniform gap.

We now state the uniform bulk hypothesis used by the coarse interface theorem.

Assumption 6.13 (Uniform bulk hypotheses). The refinement family $(\mathcal{K}_n)$ is uniformly regular and there exist constants $R_{\text{loc}}, B_{\text{bulk}}, g_{\text{bulk}} > 0$ such that for all n :

- (a) $H_{\pm,n}$ has propagation at most R_{loc} on $\ell^2(X_n)$;
- (b) $\|H_{\pm,n}\| \leq B_{\text{bulk}}$;
- (c) $\text{dist}(0, \text{spec}_{\text{op}}(H_{\pm,n})) \geq g_{\text{bulk}}$.

Remark 6.14. Parts (a) and (b) are reduced in Proposition 6.11 to uniform regularity together with the standard cap-product locality input recorded in Assumption 6.8. Part (c) is the only spectral input not proved by the elementary locality argument. As explained in Remark 6.12, this is a joint spectral condition on $B_n + B_n^\dagger$ and S_n that does not reduce to invertibility of S_n alone. For closed doubled families satisfying the Higson–Prodan closed-bulk input below, that imported theorem supplies (c). For a general refinement family, we retain (c) as an explicit assumption so that the interface theorem applies exactly when a uniform gap has been proved by some method.

Assumption 6.15 (Higson–Prodan closed-bulk input). The original refinement family $(\mathcal{K}_n)_n$ is uniformly regular, and for the closed doubled refinement family $L_n = D(\mathcal{K}_n)$ one has identified a specific closed-bulk theorem in the Higson–Prodan/Higson–Roe framework, with all of its stated hypotheses verified for $(L_n)_n$, that supplies the following data:

- (a) the Higson–Roe/Higson–Prodan controlled Poincaré-duality operator S_n with the uniform locality, row-support, and coefficient bounds of Assumption 6.8;
- (b) constants $B_{\text{HP}}, g_{\text{HP}} > 0$, independent of n , such that

$$\|H_{\pm,n}\| \leq B_{\text{HP}}, \quad \text{dist}(0, \text{spec}_{\text{op}}(H_{\pm,n})) \geq g_{\text{HP}};$$

- (c) a compatible Higson–Prodan bulk Chern pairing Ch for which the associated closed-bulk Fermi classes have Chern number ± 1 .

Remark 6.16. Assumption 6.15 is not a new hypothesis about the jammed boundary model. It is the precise imported closed-bulk input needed from the Higson–Prodan/Higson–Roe theory. For any particular refinement family, one must cite the closed theorem being used and verify that its hypotheses apply to the closed doubles $D(\mathcal{K}_n)$. The geometric input is the controlled Poincaré-duality construction of Higson–Roe: bounded-geometry oriented combinatorial manifolds give analytically controlled Hilbert–Poincaré complexes [11, Theorem 3.14 and Proposition 4.1]. Higson and Prodan specialize this controlled construction to the tight-binding Hamiltonians $H_\pm =$

$(B + B^\dagger) \pm S$ and identify the closed-bulk Chern classes in their universal Chern model [8, 9]. The present paper uses that closed theory only as an external input and does not prove the closed doubled gap from uniform regularity alone. Once Assumption 6.15 is verified for a chosen doubled family, the remaining doubled-and-jammed interface conclusions below are proved internally from that input.

Theorem 6.17 (Closed-bulk consequences of the Higson–Prodan input). *If Assumption 6.15 holds, then Assumptions 6.8 and 6.13 hold. Moreover the bulk Fermi projection*

$$P_\pm = \chi_{(-\infty, 0)}(H_\pm^\infty),$$

constructed below in Proposition 6.18, is the Higson–Prodan closed-bulk Fermi projection for the refinement family of doubles, and its imported Chern pairing satisfies

$$\text{Ch}([P_\pm]) = \pm 1.$$

Proof. See Appendix A.81.

Let

$$H_\pm^\infty := \bigoplus_{n=1}^{\infty} H_{\pm, n} \quad \text{on} \quad \ell^2(X) = \bigoplus_{n=1}^{\infty} \ell^2(X_n).$$

Proposition 6.18 (Bulk Roe-algebra class). *Assume 6.13. Then $H_\pm^\infty \in C^*(X)$ is bounded, self-adjoint, and finite-propagation. Let*

$$P_\pm := \chi_{(-\infty, 0)}(H_\pm^\infty)$$

denote the negative operator spectral projection of $H_\pm^\infty$ on $\ell^2(X)$. Then $P_\pm \in C^(X)$ and therefore defines a class*

$$[P_\pm] \in K_0(C^*(X)).$$

Proof. See Appendix A.82.

Proposition 6.19 (Gap-homotopy invariance of the bulk Fermi class). *Let $s \mapsto H_s$, $s \in [0, 1]$, be a norm-continuous path of self-adjoint elements of $C^*(X)$. Assume that there are constants $B, g > 0$ such that*

$$\text{spec}_{\text{op}}(H_s) \subset [-B, -g] \cup [g, B] \quad \text{for every } s \in [0, 1].$$

Set

$$P_s := \chi_{(-\infty, 0)}(H_s).$$

Then $s \mapsto P_s$ is a norm-continuous path of projections in $C^(X)$. In particular,*

$$[P_s] \in K_0(C^*(X))$$

is independent of s .

Proof. See Appendix A.83.

Proposition 6.20 (Quantitative stability of a spectral projection). *Let $H = H^*$ be a bounded self-adjoint operator, let $V = V^*$ be bounded, and let Γ be a positively oriented contour with*

$$\delta_\Gamma := \text{dist}(\Gamma, \text{spec}(H)) > 0.$$

If

$$\|V\| < \delta_\Gamma,$$

then $\Gamma \subset \rho(H + V)$ and

$$\|P_\Gamma(H + V) - P_\Gamma(H)\| \leq \frac{|\Gamma|}{2\pi} \frac{\|V\|}{\delta_\Gamma(\delta_\Gamma - \|V\|)}.$$

Proof. See Appendix A.84.

Small-norm Fermi and positive-side interface stability is a variant statement collected in Appendix A.9.

6.3 A collar-locality theorem

For each n , let $P_{\text{orig},n}$ and $P_{\text{copy},n}$ be the projections associated to the decomposition

$$\ell^2(X_n) = \mathcal{H}_{\text{orig},n} \oplus \mathcal{H}_{\text{copy},n}.$$

Define the split operator

$$H_{\pm,n}^{\text{split}} := P_{\text{orig},n} H_{\pm,n} P_{\text{orig},n} + P_{\text{copy},n} H_{\pm,n} P_{\text{copy},n}.$$

Proposition 6.21 (Finite collar-locality). *Assume that $H_{\pm,n}$ has propagation at most R_{loc} for every n , and set*

$$Y_n := Y_n^{(R_{\text{loc}})} = N_{R_{\text{loc}}}(\Gamma_n).$$

Then

$$\chi_{X_n \setminus Y_n} (H_{\pm,n} - H_{\pm,n}^{\text{split}}) \chi_{X_n \setminus Y_n} = 0. \quad (6.1)$$

Moreover, for every $M > 0$,

$$\chi_{X_n \setminus Y_n} \left(H_{\pm,M,n} - (H_{\pm,n}^{\text{split}} + M P_{\text{copy},n}) \right) \chi_{X_n \setminus Y_n} = 0. \quad (6.2)$$

Proof. See Appendix A.86.

Corollary 6.22 (Support near the coarse interface). *Assume 6.13. Let*

$$H_{\pm}^{\text{split},\infty} := \bigoplus_{n=1}^{\infty} H_{\pm,n}^{\text{split}}, \quad P_{\text{copy}}^{\infty} := \bigoplus_{n=1}^{\infty} P_{\text{copy},n}, \quad J_{\pm,M}^{\infty} := \bigoplus_{n=1}^{\infty} H_{\pm,M,n}.$$

Then

$$H_{\pm}^{\infty} - H_{\pm}^{\text{split},\infty} \in C^*(Y \subset X), \quad J_{\pm,M}^{\infty} - (H_{\pm}^{\text{split},\infty} + M P_{\text{copy}}^{\infty}) \in C^*(Y \subset X),$$

where $Y = Y^{(R_{\text{loc}})}$.

Proof. See Appendix A.87.

For later use define

$$P_{\text{orig}}^{\infty} := \bigoplus_{n=1}^{\infty} P_{\text{orig},n}, \quad H_{\pm,\text{copy}}^{\infty} := \bigoplus_{n=1}^{\infty} P_{\text{copy},n} H_{\pm,n} P_{\text{copy},n},$$

and for each n ,

$$H_{\pm,n}^{\text{copy}} := P_{\text{copy},n} H_{\pm,n} P_{\text{copy},n} \big|_{\mathcal{H}_{\text{copy},n}}.$$

For each fixed sign $\pm$, define the uniform negative copy-side threshold

$$B_{\text{copy},-} := \sup_n \max\{0, -\lambda_{\min}(H_{\pm,n}^{\text{copy}})\}.$$

By Assumption 6.13(b), this quantity is finite and satisfies

$$B_{\text{copy},-} \leq B_{\text{bulk}}.$$

For each $M > 0$, define the auxiliary negative-side comparison operator

$$\tilde{T}_{\pm,M}^{-} := P_{\text{orig}}^{\infty} + H_{\pm,\text{copy}}^{\infty} + M P_{\text{copy}}^{\infty}.$$

6.4 The jammed side is topologically trivial

Proposition 6.23 (Uniform positivity of the auxiliary negative-side model). *Assume 6.13. If*

$$M > B_{\text{copy},-},$$

then the auxiliary comparison operator

$$\tilde{T}_{\pm,M}^{-} = P_{\text{orig}}^{\infty} + H_{\pm,\text{copy}}^{\infty} + M P_{\text{copy}}^{\infty}$$

is bounded, self-adjoint, local, and strictly positive as an operator on $\ell^2(X)$. Moreover,

$$\text{spec}_{\text{op}}(\tilde{T}_{\pm,M}^{-}) \subset \{1\} \cup [M - B_{\text{copy},-}, \infty),$$

so

$$\text{dist}(0, \text{spec}_{\text{op}}(\tilde{T}_{\pm,M}^{-})) \geq \min\{1, M - B_{\text{copy},-}\}.$$

In particular, its negative operator spectral projection vanishes:

$$\chi_{(-\infty,0)}(\tilde{T}_{\pm,M}^{-}) = 0.$$

The stronger condition $M > B_{\text{bulk}}$ is sufficient but not necessary.

Proof. See Appendix A.88.

Remark 6.24. The bare operator $H_{\pm,\text{copy}}^{\infty} + M P_{\text{copy}}^{\infty}$ is positive on the copy sector but has a zero block on the original-side sector when viewed as an operator on all of $\ell^2(X)$. It is therefore *not* gapped at 0 on the full Hilbert space. The added term P_{orig}^{∞} is inserted solely to remove that spurious zero block while preserving the negative-side agreement away from the interface. Any fixed positive scalar multiple $\mu P_{\text{orig}}^{\infty}$ with $\mu > 0$ could be used in place of P_{orig}^{∞} .

6.5 Quotient decomposition away from the interface

For each n , define

$$X_n^+ := X_{n,\text{orig}} \cup Y_n, \quad X_n^- := X_{n,\text{copy}} \cup Y_n, \quad Y_n := Y_n^{(R_{\text{loc}})}.$$

Set

$$X^+ := \bigsqcup_{n=1}^{\infty} X_n^+, \quad X^- := \bigsqcup_{n=1}^{\infty} X_n^-.$$

Then $X = X^+ \cup X^-$ and $X^+ \cap X^- = Y$.

Proposition 6.25 (Coarse excision of the doubled decomposition). *Under Assumption 6.13, the decomposition $X = X^+ \cup X^-$ is coarsely excisive: for every $r > 0$ there exists $s(r) > 0$ such that*

$$N_r(X^+) \cap N_r(X^-) \subset N_{s(r)}(Y).$$

Proof. See Appendix A.89.

Let

$$\mathcal{A} := C^*(X), \quad \mathcal{I} := C^*(Y \subset X), \quad \pi : \mathcal{A} \rightarrow \mathcal{A}/\mathcal{I}$$

be the quotient map. Also set

$$Z^+ := X^+ \setminus Y, \quad Z^- := X^- \setminus Y.$$

Let

$$Q_+ := \chi_{Z^+}, \quad Q_- := \chi_{Z^-}, \quad Q_Y := \chi_Y$$

be the corresponding diagonal multiplication operators on $\ell^2(X)$.

Remark 6.26 (Strict-side projections in $C^*(X)$). Because $C^*(X)$ is unital in the scalar-coefficient setting, the diagonal $\{0, 1\}$ -valued operators Q_+ , Q_- , and Q_Y all belong to $\mathcal{A} = C^*(X)$: each has propagation 0 and is locally compact. They are pairwise orthogonal projections summing to $\mathbf{1}$ and they preserve $\mathcal{I}$ under left and right multiplication.

Proposition 6.27 (Quotient splitting by strict-side compressions). *Assume 6.13. With the notation above:*

(i) *for every algebraic Roe operator T of propagation at most R ,*

$$Q_Y T, T Q_Y, Q_+ T Q_-, Q_- T Q_+ \in \mathcal{I};$$

(ii) *if*

$$e_+ := \pi(Q_+), \quad e_- := \pi(Q_-),$$

then e_+ and e_- are orthogonal central projections in $\mathcal{A}/\mathcal{I}$ with

$$e_+ + e_- = \mathbf{1}_{\mathcal{A}/\mathcal{I}};$$

(iii) *the formulas*

$$E_+(\pi(a)) := \pi(Q_+ a), \quad E_-(\pi(a)) := \pi(Q_- a) \quad (a \in \mathcal{A})$$

equivalently $E_{\pm}(b) = e_{\pm}b$, define well-defined $*$ -homomorphisms

$$E_{\pm} : \mathcal{A}/\mathcal{I} \longrightarrow \mathcal{A}/\mathcal{I}$$

such that

$$E_{\pm}^2 = E_{\pm}, \quad E_+E_- = E_-E_+ = 0, \quad E_+ + E_- = \mathbf{1}_{\mathcal{A}/\mathcal{I}},$$

and

$$E_{\pm}(b_1)b_2 = b_1E_{\pm}(b_2) \quad \text{for all } b_1, b_2 \in \mathcal{A}/\mathcal{I}.$$

Consequently,

$$\mathcal{A}/\mathcal{I} = E_+(\mathcal{A}/\mathcal{I}) \oplus E_-(\mathcal{A}/\mathcal{I})$$

as a direct sum of closed two-sided ideals.

Proof. See Appendix A.90.

Proposition 6.28 (Quotient-side description). *Assume 6.13. Let J, T^+, T^- be bounded self-adjoint local operators on $\ell^2(X)$. Since X is a discrete bounded-geometry scalar Roe space, these operators belong to $\mathcal{A} = C^*(X)$. Define*

$$P^{\pm} := \chi_{(-\infty, 0)}(T^{\pm}),$$

and assume:

$$(a) \ 0 < \gamma_{\pm} := \text{dist}(0, \text{spec}_{\text{op}}(T^{\pm}));$$

(b) there exists $R_{\text{ag}} > 0$ such that

$$\chi_{X^+ \setminus N_{R_{\text{ag}}}(Y)}(J - T^+) \chi_{X^+ \setminus N_{R_{\text{ag}}}(Y)} = 0,$$

$$\chi_{X^- \setminus N_{R_{\text{ag}}}(Y)}(J - T^-) \chi_{X^- \setminus N_{R_{\text{ag}}}(Y)} = 0.$$

Then:

(i)

$$\pi(J) = E_+(\pi(T^+)) + E_-(\pi(T^-));$$

(ii) for every $0 < \gamma < \min\{\gamma_+, \gamma_-\}$, every $B \geq \max\{\|T^+\|, \|T^-\|\}$, and every $h \in C_c(\mathbb{R}, [0, 1])$ satisfying

$$h(t) = 1 \quad \text{for } t \in [-B, -\gamma], \quad h(t) = 0 \quad \text{for } t \in [\gamma, B],$$

one has

$$h(\pi(J)) = E_+(\pi(P^+)) + E_-(\pi(P^-));$$

(iii) the element

$$p_J := E_+(\pi(P^+)) + E_-(\pi(P^-))$$

is a projection in $\mathcal{A}/\mathcal{I}$, independent of the choice of γ and h , and for every such h

$$p_J = h(\pi(J)) = \pi(h(J)).$$

Proof. See Appendix A.91.

Remark 6.29 (Relation to the literal quotient spectral projection). Since $\mathcal{A}/\mathcal{I}$ is unital, the projection p_J from Proposition 6.28 coincides with $\chi_{(-\infty,0)}(\pi(J))$ whenever $0 \notin \text{spec}_{\mathcal{A}/\mathcal{I}}(\pi(J))$. The continuous-cutoff formulation is used because it avoids the need to verify that last spectral condition. The compactly supported version above is also the form compatible with nonunital or stabilized Roe-algebra presentations.

Definition 6.30 (Sidewise class-A interface). Let J , T^+ , and T^- be bounded self-adjoint local operators on $\ell^2(X)$. We say that $(J; T^+, T^-)$ is a *sidewise class-A interface along Y* if:

- (i) 0 lies in a gap of the operator spectra $\text{spec}_{\text{op}}(T^\pm)$;
- (ii) there exists $R > 0$ such that

$$\chi_{X^+ \setminus N_R(Y)}(J - T^+) \chi_{X^+ \setminus N_R(Y)} = 0, \quad \chi_{X^- \setminus N_R(Y)}(J - T^-) \chi_{X^- \setminus N_R(Y)} = 0;$$

- (iii) the operator Fermi projections

$$P^+ := \chi_{(-\infty,0)}(T^+), \quad P^- := \chi_{(-\infty,0)}(T^-)$$

belong to $\mathcal{A} = C^*(X)$.

6.6 Boundary class of a liftable quotient projection

Proposition 6.31 (Boundary class of a liftable quotient projection). *Let $p \in M_n(\mathcal{A}/\mathcal{I})$ be a projection that admits a self-adjoint lift $\tilde{p} \in M_n(\mathcal{A})$. Then*

$$\exp(2\pi i \tilde{p}) - \mathbf{1} \in M_n(\mathcal{I}),$$

so $\exp(2\pi i \tilde{p}) \in \mathbf{1} + M_n(\mathcal{I})$ defines a class in $K_1(\mathcal{I})$. This class depends only on p , not on the chosen self-adjoint lift. We denote it by

$$\partial_Y(p) \in K_1(\mathcal{I}).$$

This fixes our sign convention for the connecting map; some references use the inverse class.

Moreover, if $a \in M_n(\mathcal{A})$ is self-adjoint and $h \in C(\mathbb{R}, [0, 1])$ satisfies

$$h|_{\text{spec}(\pi(a))} = \chi_{(-\infty,0)}|_{\text{spec}(\pi(a))},$$

then $\pi(h(a))$ is a projection in $M_n(\mathcal{A}/\mathcal{I})$ and

$$\partial_Y(\pi(h(a))) = [\exp(2\pi i h(a))] \in K_1(\mathcal{I}).$$

Proof. See Appendix A.92.

Proposition 6.32 (Homotopy invariance of positive-side boundary classes). *Assume 6.13, so that the quotient-side map E_+ is defined. Let $s \mapsto P_s$, $s \in [0, 1]$, be a norm-continuous path of projections in $\mathcal{A} = C^*(X)$. Then*

$$\partial_Y(E_+(\pi(P_s))) \in K_1(\mathcal{I})$$

is independent of s .

Proof. See Appendix A.93.

Definition 6.33 (Interface class of a sidewise realization). If $(J; T^+, T^-)$ is a sidewise class-A interface along Y , let

$$P^+ := \chi_{(-\infty, 0)}(T^+), \quad P^- := \chi_{(-\infty, 0)}(T^-),$$

and define the canonical quotient projection

$$p_J := E_+(\pi(P^+)) + E_-(\pi(P^-)) \in \mathcal{A}/\mathcal{I}.$$

By Proposition 6.28, if

$$0 < \gamma < \min \left\{ \text{dist}(0, \text{spec}_{\text{op}}(T^+)), \text{dist}(0, \text{spec}_{\text{op}}(T^-)) \right\}$$

and, for some $B \geq \max\{\|T^+\|, \|T^-\|\}$, the function $h \in C_c(\mathbb{R}, [0, 1])$ satisfies

$$h(t) = 1 \quad \text{for } t \in [-B, -\gamma], \quad h(t) = 0 \quad \text{for } t \in [\gamma, B],$$

then

$$p_J = h(\pi(J)) = \pi(h(J)).$$

Its *interface class* is

$$\text{Ind}_{\text{int}}(J; T^+, T^-) := \partial_Y(p_J) \in K_1(\mathcal{I}).$$

Equivalently, for every such cutoff h ,

$$\text{Ind}_{\text{int}}(J; T^+, T^-) = [\exp(2\pi i h(J))] \in K_1(\mathcal{I}),$$

and this class is independent of the choice of h .

Sidewise interface-supported changes, finite-component changes, norm-vanishing component defects, sidewise homotopies, and positive copy-confinement variants are collected in Appendix A.9.

Remark 6.34. The interface class is built from the quotient projection p_J and the genuine extension

$$0 \rightarrow C^*(Y \subset X) \rightarrow C^*(X) \rightarrow C^*(X)/C^*(Y \subset X) \rightarrow 0.$$

6.7 Interface class for the jammed family

Theorem 6.35 (Doubled-and-jammed interface class). *Assume 6.13. Let*

$$J_{\pm, M}^\infty = \bigoplus_{n=1}^{\infty} H_{\pm, M, n}$$

be the direct-sum jammed family on the coarse disjoint union X , and let

$$Y = Y^{(R_{\text{loc}})}$$

be the coarse interface from Corollary 6.22. If

$$M > B_{\text{copy}, -},$$

then with

$$T^+ := H_{\pm}^\infty, \quad T^- := \tilde{T}_{\pm, M}^-,$$

the triple $(J_{\pm, M}^\infty; T^+, T^-)$ is a sidewise class-A interface along Y in the sense of Definition 6.30. Its canonical quotient projection is

$$p_{J_{\pm, M}^\infty} = E_+(\pi(P_\pm)), \quad (6.3)$$

where

$$P_\pm := \chi_{(-\infty, 0)}(H_\pm^\infty) \in C^*(X)$$

is the bulk operator Fermi projection from Proposition 6.18. Equivalently, if

$$0 < \gamma < \min\{g_{\text{bulk}}, 1, M - B_{\text{copy}, -}\}$$

and, for some $B \geq \max\{\|T^+\|, \|T^-\|\}$, the function $h \in C_c(\mathbb{R}, [0, 1])$ satisfies

$$h(t) = 1 \text{ for } t \in [-B, -\gamma], \quad h(t) = 0 \text{ for } t \in [\gamma, B],$$

then

$$\pi(h(J_{\pm, M}^\infty)) = E_+(\pi(P_\pm)).$$

Consequently,

$$\text{Ind}_{\text{int}}(J_{\pm, M}^\infty; T^+, T^-) = \partial_Y(E_+(\pi(P_\pm))) \in K_1(C^*(Y \subset X)). \quad (6.4)$$

In particular, the jammed side contributes no negative-energy quotient projection.

Proof. See Appendix A.100.

This theorem uses only the uniform doubled-bulk gap in Assumption 6.13(c), together with the uniform locality and boundedness hypotheses stated there. For Higson–Prodan-admissible closed doubles, these hypotheses are supplied by Theorem 6.17. For other families they remain explicit assumptions. The finite-dimensional decoupling estimates of Section 4 do not use this refinement-limit gap assumption; they require only their stated finite-matrix separation hypotheses.

Large-jam-scale independence is recorded with the other Roe variants in Appendix A.9.

6.8 Identification of the bulk class

The Hamiltonians $H_{\pm, n} = H_\pm(D(\mathcal{K}_n))$ are exactly the Higson–Prodan universal Chern Hamiltonians on the closed oriented surfaces $D(\mathcal{K}_n)$. Under the admissibility hypotheses stated below, the bulk Fermi projection $P_\pm = \chi_{(-\infty, 0)}(H_\pm^\infty)$ is the corresponding Higson–Prodan Fermi projection.

In this subsection Ch denotes the numerical bulk Chern pairing supplied by the Higson–Prodan theorem for admissible closed refinement families. We do not construct a new Chern cocycle on an arbitrary Roe algebra here. Rather, we assume that the coarse-limit class $[P_\pm] \in K_0(C^*(X))$ is evaluated by the Higson–Prodan bulk pairing associated with the closed doubled family, and later require compatibility between this pairing and the external coarse bulk–interface pairing.

Theorem 6.36 (Bulk Chern number of the doubled family, imported from Higson–Prodan). *Assume 6.13. Assume moreover that the refinement family of closed doubles $(D(\mathcal{K}_n))_n$ is an admissible Higson–Prodan refinement family satisfying the hypotheses of the Higson–Prodan bulk theorem, and that the Chern pairing Ch used here is the corresponding Higson–Prodan bulk Chern pairing. Then the K_0 -class $[P_\pm] \in K_0(C^*(X))$ is the Higson–Prodan bulk class for $(D(\mathcal{K}_n))_n$. Consequently, by the Higson–Prodan theorem [8, 9],*

$$\text{Ch}([P_\pm]) = \pm 1.$$

Proof. See Appendix A.102.

The next definition states the extra compatibility needed to evaluate the interface class by a numerical winding. The intermediate Chern- ± 1 and HP-admissible corollaries are collected in Appendix A.9.

Definition 6.37 (HP–Ewert–Meyer admissible boundary family). A boundary refinement family $(\mathcal{K}_n)_n$ is *HP–Ewert–Meyer admissible* if $D(\mathcal{K}_n)$ satisfies the Higson–Prodan closed-bulk input in Assumption 6.15, and if the associated coarse boundary data

$$(X; X^+, X^-, Y)$$

fit the Roe-algebraic bulk–interface framework of Ewert–Meyer [29]. Concretely, the required data are:

- (a) $X = X^+ \cup X^-$ is a coarse partition with interface Y ; in particular, for every $r > 0$ there is $s(r) > 0$ such that

$$N_r(X^+) \cap N_r(X^-) \subset N_{s(r)}(Y).$$

- (b) Away from Y , the positive side agrees with the Higson–Prodan bulk model and the negative side agrees with a trivial gapped comparison model.
- (c) The external theorem identifies the connecting map

$$\partial_Y : K_0(C^*(X)/C^*(Y \subset X)) \longrightarrow K_1(C^*(Y \subset X))$$

with an interface winding pairing compatible with the Higson–Prodan Chern pairing used in Theorem 6.36; a quotient class with Chern number c_+ on the positive side and c_- on the negative side has winding number $c_+ - c_-$.

Admissibility criteria, verification templates, and tail-admissibility variants are collected in Appendix A.9.

The Ewert–Meyer input used here is the coarse Mayer–Vietoris bulk–interface boundary map and its identification with the bulk–edge map [29, Propositions 3.2.1–3.2.2 and Corollary 3.2.3]. Kubota’s controlled framework and the Ludewig–Thiang edge-following theorem give closely related formulations [28, 32]; the theorem below uses the Ewert–Meyer convention fixed in Definition 6.37.

Imported hypotheses for the winding value. The theorem below imports exactly two external inputs: the Higson–Prodan closed-bulk gap and Chern computation in Assumption 6.15, and a compatible Ewert–Meyer bulk–interface pairing for the chosen coarse partition as specified in Definition 6.37. The quotient projection and interface class are constructed inside this paper once the uniform bulk gap is available; the numerical winding value is not asserted without the compatible pairing.

Theorem 6.38 (Winding for HP–Ewert–Meyer admissible boundary families). *Let $(\mathcal{K}_n)_n$ be an HP–Ewert–Meyer admissible boundary refinement family, so the two imported hypotheses in the preceding box hold. Then, for every $M > B_{\text{copy}, -}$,*

$$\text{Ind}_{\text{int}}(J_{\pm, M}^\infty; T^+, T^-) = \partial_Y(E_+(\pi(P_\pm))) \in K_1(C^*(Y \subset X)).$$

The projection $P_\pm$ is the Higson–Prodan closed-bulk Fermi projection and

$$\text{Ch}([P_\pm]) = \pm 1.$$

Moreover the compatible interface winding pairing satisfies

$$\text{Wind}(\text{Ind}_{\text{int}}(J_{\pm,M}^{\infty}; T^+, T^-)) = \pm 1.$$

The displayed winding equality is the compatible Ewert–Meyer pairing evaluation. The same class is independent of the jam scale M for all $M > B_{\text{copy},-}$.

Proof. See Appendix A.108.

Relabeling, orientation reversal, and robustness of the compatible winding are stated in Appendix A.9.

Remark 6.39 (From bulk Chern number to numerical interface index). Theorem 6.38 separates the two ingredients in the strongest form used here. The Roe-algebra construction of the K_1 -class is internal to the paper. The numerical winding pairing is supplied by the chosen coarse bulk–interface theorem through Definition 6.37. This compatibility condition concerns the chosen interface pairing; it is not an additional boundary-gap assumption and does not assert that the negative quotient of the closed doubled bulk projection vanishes. In the corresponding six-term exact sequence

$$K_0(\mathcal{I}) \rightarrow K_0(\mathcal{A}) \rightarrow K_0(\mathcal{A}/\mathcal{I}) \xrightarrow{\partial} K_1(\mathcal{I}) \rightarrow K_1(\mathcal{A}) \rightarrow K_1(\mathcal{A}/\mathcal{I})$$

the connecting map is evaluated by the interface winding pairing. The positive side has Chern number ± 1 , and the jammed negative side is trivial, so the Chern jump is ± 1 .

Remark 6.40 (What the global theorem does and does not say). Theorem 6.35 is a *global* statement about the interface class supported near the entire glued boundary. It does not by itself imply that each boundary component carries its own local index or local boundary map. A componentwise statement requires an additional localized pairing. In the continuum Hall setting, Drouot and Zhu prove a curved-interface formula in which the local edge conductance is an intersection number times the bulk Hall conductance [34, 33]. We view that as the correct model for any future local theorem in the present lattice setting.

7 Numerical checks

The numerical material is collected in Appendix B, with reproducibility details in Appendix B.1. The computations are not proof inputs; they document finite-matrix behavior for the mechanisms proved above.

The checks cited in the body are the relative/compressed boundary comparison, finite- M resolvent and first-order rates, leakage, collar support, flux spectral flow, chirality reversal, copy-side confinement robustness, rough-edge stability, and the periodic triangular FHS checks. The repository URL and snapshot hash are recorded in the reproducibility appendix.

8 Conclusion

We constructed a boundary model for the universal Chern Hamiltonians by replacing a triangulated surface with boundary by its closed double and adding a large positive potential on the reflected side. The relative fundamental class of $(\mathcal{K}, \partial\mathcal{K})$ gave an intrinsic Poincaré–Lefschetz boundary Hamiltonian on the original triangulation. Compressing the doubled Hamiltonian to the original side recovered this relative Hamiltonian up to a boundary-collar operator, and the jammed

Hamiltonian converged to the compressed boundary operator in the large-jam limit by quantitative Schur-complement estimates.

The finite- M estimates controlled resolvents, first-order self-energy corrections, spectral localization, leakage, contour projections, Morse-index stability, monotone jam-scale spectral counts, no-copy low-energy bounds, and inherited gaps under checkable separation conditions. These estimates were finite-dimensional consequences of the block structure and did not use a refinement-limit topological hypothesis. In the sampled families, the computations showed the predicted M^{-1} and M^{-2} scalings, positive contour margins, collar-supported original-copy couplings, boundary-localized low-energy states, and chirality reversal between H_+ and H_- .

In the refinement limit, the Roe-algebra quotient by the interface ideal identified the interface class as the connecting-map image of the positive-side doubled-bulk Fermi projection. For sufficiently large M , the jammed side contributed no negative-energy quotient sector. Thus, for any boundary family satisfying the verification template in Corollary A.44, the resulting interface class has compatible winding number ± 1 .

The periodic triangular-lattice section gives a separate bulk input: its Bloch symbol has an exact determinant gap, a positive-mass homotopy remains gapped, and an isolated-crossing calculation proves the Chern signs, with stability under small self-adjoint periodic perturbations. For annular triangular-lattice families whose doubles are these periodic tori, the boundary/interface construction therefore does not rely on the Higson–Prodan closed-bulk theorem for the gap or Chern input. The same argument is stable under positive duality strength, small finite-range periodic bulk perturbations, gapped finite-range periodic homotopies, finite-cover pullbacks, finite-index cell enlargements, large scalar jamming, and sufficiently positive copy-side confinement; for primitive straight periodic boundary components, including finite split sums, finite or K_1 -trivial finite-corner split sums, bounded-distance straight-edge variants, finite defects, norm-vanishing component defects, and tail-split variants, the Toeplitz edge extension evaluates the interface winding directly and is robust under fixed-collar periodic edge perturbations that preserve the sidewise gaps. For nonperiodic or curved coarse partitions not reducible to these straight Toeplitz or finite-corner quotient models, the compatible coarse interface pairing remains the external topological input. Thus the straight-periodic triangular families listed in Corollary A.32 have gap, Chern value, interface class, and winding proved in this paper, with no Higson–Prodan closed-bulk input and no Ewert–Meyer compatibility assumption.

Outside the closed Higson–Prodan classes used here, the main remaining analytic problem is to prove the uniform Fermi gap for broader closed doubled refinement families. That gap is a property of $H_{\pm} = (B + B^{\dagger}) \pm S$ as a whole, not of the Poincaré-duality block S alone. A second direction is to develop a curved-interface pairing that separates the contributions of individual boundary components.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

A Proofs

A.1 Sign ledger

The table records the orientation conventions used in the periodic triangular crossing and in the Toeplitz edge map. It is a ledger for the proof of Theorem 5.6, not a separate numerical input.

Object	Orientation convention	Sign consequence
Bloch torus	$dk_1 \wedge dk_2$	FHS plaquettes use the same ordered coordinate square.
Crossing space	$dq_1 \wedge dq_2 \wedge d\mu$	The local degree is the sign of $\det Du$.
Self-dual basis	e_1, e_2, e_3, e_4 oriented by $e_1 \wedge e_2 \wedge e_3 \wedge e_4$	The self-dual compression has degree -1 .
Mass reversal	$k \mapsto -k$ on $\mathbb{T}^2$	The base map preserves orientation; complex conjugation reverses c_1 .
Toeplitz edge	Tangential/positive-normal basis (t, n)	Winding is $\det[t \ n] \text{Ch}(P)$.

Orientation and sign ledger. The surface orientation fixes the fundamental and relative fundamental classes. Faces are oriented coherently with that surface orientation, while edge orientations only choose a signed simplex basis. The Brillouin torus in the periodic triangular calculation is oriented by $dk_1 \wedge dk_2$. The local crossing is oriented by $dq_1 \wedge dq_2 \wedge d\mu$, where μ is the duality-strength mass. The Toeplitz calculation uses the same $dk_1 \wedge dk_2$ orientation and compresses the second coordinate in the positive normal direction. With these conventions the analytic triangular signs are $\text{Ch}(E_-(H_+^\triangle)) = -1$ and $\text{Ch}(E_-(H_-^\triangle)) = +1$, the FHS table has the same signs, and the straight Toeplitz winding is the corresponding oriented Chern number.

Lemma A.1 (Local crossing sign convention). *Let $L(q_1, q_2, \mu) = u(q_1, q_2, \mu) \cdot \sigma$ be the Pauli Hamiltonian on $\mathbb{C}^2$, where $u : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is an invertible real linear map. Orient the parameter sphere around the crossing by $dq_1 \wedge dq_2 \wedge d\mu$, and orient the constant-mass tori by $dq_1 \wedge dq_2$. Then the negative line bundle changes by*

$$\text{Ch}(E_-(L_{\mu>0})) - \text{Ch}(E_-(L_{\mu<0})) = \text{sgn} \det(Du).$$

For the direct sum of two identical Pauli blocks, the right side is multiplied by 2.

Proof. See Appendix A.111.

Triangular sign check. In the proof of Theorem 5.6, the compressed harmonic-space model has two identical self-dual Pauli blocks and an orientation-reversing linear map u . Hence the crossing contribution is -2 . The symmetry $H_{-m}^\Delta(k) = \overline{H_m^\Delta(-k)}$ changes the Chern sign, since $k \mapsto -k$ preserves the two-torus orientation and complex conjugation reverses the first Chern class. The two equations force the positive-mass sign -1 and the negative-mass sign $+1$, matching the reported FHS rows in Tables 3 and 4.

A.2 Routine perturbation proofs

The two auxiliary finite-dimensional perturbation statements below are retained for reference and kept out of the body. Their proofs are included with the other appendix proof blocks.

Theorem A.2 (Second-order compressed resolvent expansion). *Let*

$$H_M = \begin{pmatrix} A & C \\ C^* & D + M\mathbf{1} \end{pmatrix}$$

be a finite-dimensional self-adjoint block operator. Fix $z \in \mathbb{C}$ with $\eta = \text{dist}(z, \text{spec}(A)) > 0$, and set

$$R_A(z) = (A - z)^{-1}, \quad c = \|C\|, \quad \beta = \|D - z\|.$$

Assume $M \geq 2\beta$ and $M \geq 4c^2/\eta$. Then the original-side compressed resolvent has the expansion

$$P_{\text{orig}}(H_M - z)^{-1}P_{\text{orig}} = R_A + \frac{1}{M}R_A C C^* R_A \frac{1}{M^2} \left(R_A C C^* R_A C C^* R_A - R_A C (D - z) C^* R_A \right) + E_M(z),$$

where

$$\|E_M(z)\| \leq \frac{4c^2\beta^2}{\eta^2 M^3} \frac{8c^4\beta}{\eta^3 M^3} \frac{16c^6}{\eta^4 M^3}.$$

Proof. See Appendix A.112.

Proposition A.3 (Positive copy-potential finite-dimensional estimates). *Let*

$$H_M^V = \begin{pmatrix} A & C \\ C^* & D + MV \end{pmatrix}$$

be finite-dimensional and self-adjoint, where $V \geq v_0\mathbf{1}$ on the copy sector for some $v_0 > 0$. Fix $z \in \mathbb{C}$ with $\eta = \text{dist}(z, \text{spec}(A)) > 0$, and set $\beta = \|D - z\|$, $c = \|C\|$. If

$$Mv_0 > \beta, \quad \frac{c^2}{Mv_0 - \beta} < \frac{\eta}{2},$$

then $D + MV - z$ is invertible and

$$\|P_{\text{orig}}(H_M^V - z)^{-1}P_{\text{orig}} - (A - z)^{-1}\| \leq \frac{2c^2}{\eta^2(Mv_0 - \beta)}.$$

If $H_M^V \psi = \lambda \psi$, $\psi = \psi_{\text{orig}} \oplus \psi_{\text{copy}}$, and $|\lambda| \leq \Lambda$ with $Mv_0 > \|D\| + \Lambda$, then

$$\|\psi_{\text{copy}}\| \leq \frac{c}{Mv_0 - \|D\| - \Lambda} \|\psi_{\text{orig}}\|.$$

Proof. See Appendix A.113.

A.3 Additional benchmark symbols: hexagonal

A.3.1 Internal hexagonal class: bulk gap

The next symbol is a finite-range two-band representative written in hexagonal reciprocal coordinates. It is used here as an internal benchmark class with a direct gap, Chern, and Toeplitz calculation; it is not a derivation of the full Higson–Prodan universal symbol for arbitrary hexagonal triangulations. Let $k = (k_1, k_2) \in \mathbb{T}^2$ and

$$a_1(k) = \sin k_1, \quad a_2(k) = \sin k_2, \quad a_3^{\text{hex}}(k) = \cos k_1 + \cos k_2 - 1.$$

Definition A.4 (Hexagonal two-band Bloch representative). For $m \in \mathbb{R}$, set

$$H_{\text{hex}}(k; m) = \begin{pmatrix} m a_3^{\text{hex}}(k) & a_1(k) - i a_2(k) \\ a_1(k) + i a_2(k) & -m a_3^{\text{hex}}(k) \end{pmatrix}.$$

Lemma A.5 (Massless hexagonal closure). *The symbol $H_{\text{hex}}(0, 0; 0)$ is gapless at zero.*

Proof. See Appendix A.65.

Lemma A.6 (Hexagonal determinant and bulk gap). For $m \neq 0$,

$$|\det H_{\text{hex}}(k; m)| \geq c_{\text{hex}}(m) := \min \left\{ \frac{1}{4}, (\sqrt{3} - 1)^2 m^2 \right\} \quad (k \in \mathbb{T}^2).$$

Consequently, for $m > 0$,

$$\text{dist}(0, \text{spec} H_{\text{hex}}(k; m)) \geq \gamma_{\text{hex}}(m) := \sqrt{c_{\text{hex}}(m)} \quad (k \in \mathbb{T}^2).$$

Proof. See Appendix A.66.

Lemma A.7 (Hexagonal Chern integer). For $m > 0$, the negative-energy line bundle of $H_{\text{hex}}(k; m)$, with orientation $dk_1 \wedge dk_2$ on $\mathbb{T}^2$, has

$$\text{Ch}(E_-(H_{\text{hex}}(\cdot; m))) = -1.$$

Proof. See Appendix A.67.

Lemma A.8 (Hexagonal mass reversal). For $m < 0$,

$$\text{Ch}(E_-(H_{\text{hex}}(\cdot; m))) = +1.$$

Proof. See Appendix A.68.

A.3.2 Internal hexagonal class: Toeplitz winding

Lemma A.9 (Hexagonal Toeplitz winding). Let $m \neq 0$, set

$$P_{\text{hex}, m}(k) = \chi_{(-\infty, 0)}(H_{\text{hex}}(k; m)),$$

and take a primitive tangential/positive-normal lattice basis (t, n) . For the straight half-space Toeplitz extension in the n -direction,

$$\text{wind}(\partial_{T, t, n}([P_{\text{hex}, m}])) = \det[t \ n] \text{Ch}(P_{\text{hex}, m}).$$

Thus the coordinate-edge winding is -1 for $m > 0$ and $+1$ for $m < 0$.

Proof. See Appendix A.69.

A.3.3 Internal hexagonal bulk-edge correspondence

Theorem A.10 (Hexagonal two-band bulk-edge correspondence). *For every $m \neq 0$, the Bloch family $H_{\text{hex}}(k; m)$ is uniformly gapped at zero with gap at least $\gamma_{\text{hex}}(m)$. Its negative-energy Chern number is*

$$\text{Ch}(P_{\text{hex},m}) = -\text{sgn}(m).$$

For every primitive straight periodic edge with tangential/positive-normal basis (t, n) , the Toeplitz winding is

$$\text{wind}(\partial_{T,t,n}([P_{\text{hex},m}])) = -\det[t \ n] \text{sgn}(m).$$

Proof. See Appendix A.70.

A.4 Additional benchmark symbols: square

A.4.1 Internal square class: bulk gap

Let

$$b_1(k) = \sin k_1, \quad b_2(k) = \sin k_2, \quad b_3^\square(k) = 1 + \cos k_1 + \cos k_2.$$

Definition A.11 (Square two-band Bloch representative). For $m \in \mathbb{R}$, set

$$H_\square(k; m) = \begin{pmatrix} m b_3^\square(k) & b_1(k) - i b_2(k) \\ b_1(k) + i b_2(k) & -m b_3^\square(k) \end{pmatrix}.$$

Lemma A.12 (Massless square closure). *The symbol $H_\square(0, 0; 0)$ is gapless at zero.*

Proof. See Appendix A.71.

Lemma A.13 (Square determinant and bulk gap). *For $m \neq 0$,*

$$|\det H_\square(k; m)| \geq c_\square(m) := \min \left\{ \frac{1}{4}, (\sqrt{3} - 1)^2 m^2 \right\} \quad (k \in \mathbb{T}^2).$$

Consequently, for $m > 0$,

$$\text{dist}(0, \text{spec} H_\square(k; m)) \geq \gamma_\square(m) := \sqrt{c_\square(m)} \quad (k \in \mathbb{T}^2).$$

Proof. See Appendix A.72.

Lemma A.14 (Square Chern integer). *For $m > 0$, the negative-energy line bundle of $H_\square(k; m)$, with orientation $dk_1 \wedge dk_2$, has*

$$\text{Ch}(E_-(H_\square(\cdot; m))) = +1.$$

Proof. See Appendix A.73.

A.4.2 Internal square class: Toeplitz winding

Lemma A.15 (Square Toeplitz winding). *Let $m \neq 0$, set*

$$P_{\square,m}(k) = \chi_{(-\infty, 0)}(H_\square(k; m)),$$

and take a primitive tangential/positive-normal lattice basis (t, n) . For the straight half-space Toeplitz extension in the n -direction,

$$\text{wind}(\partial_{T,t,n}([P_{\square,m}])) = \det[t \ n] \text{Ch}(P_{\square,m}).$$

Thus the coordinate-edge winding is $+1$ for $m > 0$ and -1 for $m < 0$.

Proof. See Appendix A.74.

A.4.3 Internal square bulk-edge correspondence

Theorem A.16 (Square two-band bulk-edge correspondence). *For every $m \neq 0$, the Bloch family $H_{\square}(k; m)$ is uniformly gapped at zero with gap at least $\gamma_{\square}(m)$. Its negative-energy Chern number is*

$$\text{Ch}(P_{\square, m}) = \text{sgn}(m).$$

For every primitive straight periodic edge with tangential/positive-normal basis (t, n) , the Toeplitz winding is

$$\text{wind}(\partial_{T, t, n}([P_{\square, m}])) = \det[t \ n] \text{sgn}(m).$$

Proof. See Appendix A.75.

A.5 Section 5 variants

The periodic triangular section in the body keeps the closed-bulk calculation, the Chern-sign calculation, and the primitive straight Toeplitz edge map. The following subsections collect the variant statements that use the same inputs with extra quotient, tail, finite-defect, finite-corner, or annular hypotheses.

A.6 Robustness variants: finite defects

Corollary A.17 (Robust straight-periodic Toeplitz winding). *In the setting of Corollary 5.10, the Toeplitz winding values are unchanged by the following operations, provided the stated sidewise comparison gaps remain open:*

- (a) replacing the closed periodic triangular bulk by $H_{\pm, m}^{\Delta}$ with fixed $m > 0$;
- (b) adding a self-adjoint finite-range periodic bulk perturbation V to $H_{\pm, m}^{\Delta}$ with $\sup_k \|V(k)\| < g_{\pm, m}$, and in particular adding V to $H_{\pm}^{\Delta}$ with $\sup_k \|V(k)\| < 2^{-13}$;
- (c) adding any bounded finite-propagation self-adjoint perturbation supported in a fixed straight-edge collar, including periodic edge or collar terms that are not norm-small;
- (d) replacing scalar jamming by a bounded componentwise scalar jam above the positivity threshold, or by a sufficiently positive bounded finite-propagation copy-side confining potential;
- (e) changing finitely many initial components, adding norm-vanishing component defects, or passing through a norm-continuous sidewise homotopy with uniformly open comparison gaps.
- (f) replacing the straight interface subset Y by Y' at finite Hausdorff distance from Y , with the same sidewise realization outside a fixed neighborhood of $Y \cup Y'$;
- (g) adding a bounded self-adjoint finite-propagation perturbation to the jammed operator, or to the sidewise data with the comparison gaps still open, supported in a fixed edge collar and in a finite tangential region;
- (h) after deleting finitely many coarse-disjoint-union components, making arbitrary bounded finite-propagation changes on the deleted components when the tail has a split, finite-corner, or K_1 -trivial-corner Toeplitz description.

The same conclusion holds componentwise for every split finite family covered by Theorem A.30 and for every finite-corner split family covered by Theorem A.19. The finite-defect and tail-split cases have the same winding values as the associated straight periodic tail.

Proof. See Appendix A.61.

Theorem A.18 (Bounded-distance rough and eventually periodic collars). *Let Y_0 be a primitive straight periodic interface for a finite-range periodic bulk projection P , and let $Y_1 \subset X$ be another interface subset at finite Hausdorff distance from Y_0 . Then*

$$C^*(Y_0 \subset X) = C^*(Y_1 \subset X) =: \mathcal{I}.$$

Suppose $(J_i; T_i^+, T_i^-)$, $i = 0, 1$, are sidewise class-A interfaces along Y_i , and that

$$J_1 - J_0, \quad T_1^+ - T_0^+, \quad T_1^- - T_0^- \in \mathcal{I}$$

after viewing both realizations on the same Roe space. Then the quotient Fermi data agree:

$$\pi(\chi_{(-\infty, 0)}(T_1^\pm)) = \pi(\chi_{(-\infty, 0)}(T_0^\pm)), \quad p_{J_1} = p_{J_0}.$$

Consequently

$$\text{Ind}_{\text{int}}(J_1; T_1^+, T_1^-) = \text{Ind}_{\text{int}}(J_0; T_0^+, T_0^-) \in K_1(\mathcal{I}),$$

and the rough-edge winding is the straight Toeplitz winding

$$\det[t \ n] \text{ Ch}(P)$$

for the primitive tangential/positive-normal basis (t, n) of the straight tail.

If the difference between the rough collar and the straight periodic collar is supported in a fixed normal collar and in a finite tangential interval, then the displayed differences are finite rank. Thus eventually periodic straight or bounded-distance rough collars with a finite transition region have the same quotient class and the same winding as their periodic tail, provided the sidewise comparison gaps remain open.

Proof. See Appendix A.62.

The fixed-collar robustness checks in Figures 16 and 17 and table 12 are finite-matrix checks of the same ideal-stability mechanism; the winding equality here is the quotient-ideal argument, not a separate finite-volume index.

A.7 Robustness variants: finite corners

Theorem A.19 (Finite-corner split Toeplitz additivity). *Let H and P be as in Theorem 5.8. Suppose an interface ideal $\mathcal{I}_{\text{poly}}$ contains a finite-dimensional corner ideal $\mathcal{F}$ and fits into an exact sequence*

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I}_{\text{poly}} \xrightarrow{q} \bigoplus_{j=1}^r \mathcal{I}_j \longrightarrow 0,$$

where each $\mathcal{I}_j$ is identified with the straight periodic Toeplitz edge ideal $C(S^1, \mathcal{K})$ for a primitive tangential/normal lattice basis (t_j, n_j) . Assume that, after quotienting by $\mathcal{F}$, the corresponding quotient extension is the direct sum of these r Toeplitz extensions. If

$$\partial_{\text{poly}}([P]) \in K_1(\mathcal{I}_{\text{poly}})$$

is the boundary class for the polygonal or cornered interface extension, then

$$q_* \partial_{\text{poly}}([P]) = \bigoplus_{j=1}^r \partial_{\Gamma, t_j, n_j}([P]).$$

Moreover $q_* : K_1(\mathcal{I}_{\text{poly}}) \rightarrow K_1(\bigoplus_j \mathcal{I}_j)$ is injective. Thus the finite corner ideal contributes no K_1 -coordinate, and the j -th component winding is

$$\text{wind}_j(\partial_{\text{poly}}([P])) = \det[t_j \ n_j] \text{Ch}(P).$$

Proof. See Appendix A.57.

Corollary A.20 (K1-trivial corner ideals). *In Theorem A.19, the finite-dimensional hypothesis on $\mathcal{F}$ may be replaced by the weaker condition*

$$K_1(\mathcal{F}) = 0.$$

Under the same quotient-extension hypothesis, q_ is injective on K_1 , the finite-corner class is determined by its straight-edge quotient coordinates, and*

$$\text{wind}_j(\partial_{\text{poly}}([P])) = \det[t_j \ n_j] \text{Ch}(P).$$

This applies, in particular, when $\mathcal{F}$ is finite-dimensional, a c_0 -direct sum of finite-dimensional matrix algebras, or a bounded product of finite-dimensional matrix algebras.

Proof. See Appendix A.58.

Corollary A.21 (Collar-separation criterion for finite-corner quotients). *Let $Y = C \cup Y^{(1)} \cup \dots \cup Y^{(r)}$ be a cornered interface collar in a bounded-geometry Roe space X . Suppose:*

- (i) *the corner-supported ideal $\mathcal{F}_C$, obtained as the closure of operators supported in bounded neighborhoods of C , has $K_1(\mathcal{F}_C) = 0$;*
- (ii) *the edge pieces separate off the corner set: for every propagation bound R there is $S(R)$ such that the sets $N_R(Y^{(i)}) \setminus N_{S(R)}(C)$ and $N_R(Y^{(j)}) \setminus N_{S(R)}(C)$ are disjoint whenever $i \neq j$;*
- (iii) *outside a bounded neighborhood of C , each $Y^{(j)}$ is identified with a primitive straight periodic collar, and restriction to that collar identifies its edge quotient ideal with the Toeplitz ideal $\mathcal{I}_j \cong C(S^1, \mathcal{K})$.*

Then the interface ideal $\mathcal{I}_Y = C^(Y \subset X)$ fits into an exact sequence*

$$0 \longrightarrow \mathcal{F}_C \longrightarrow \mathcal{I}_Y \xrightarrow{q} \bigoplus_{j=1}^r \mathcal{I}_j \longrightarrow 0.$$

If the quotient extension is the corresponding direct sum of primitive straight Toeplitz extensions, the component winding formula of Corollary A.20 applies.

Proof. See Appendix A.59.

Remark A.22. The split hypotheses in Theorems A.30 and A.19 are genuine extra localization inputs. They hold for finite direct sums of independent straight periodic half-space collars and for cornered models where removing a K_1 -trivial corner ideal leaves such a direct sum, for instance under the collar-separation criterion in Corollary A.21. They are not asserted for arbitrary curved or nonperiodic boundary components.

A.8 Robustness variants: tail-split and annular

Corollary A.23 (Finite covers and finite-index cell enlargements). *Let $A \in M_2(\mathbb{Z})$ have $\det A \neq 0$, and define the finite-cover pullback*

$$H_{\pm,A}^{\Delta}(\kappa) := H_{\pm}^{\Delta}(A\kappa), \quad \kappa \in \mathbb{T}^2.$$

Then $H_{\pm,A}^{\Delta}$ is gapped at zero and

$$\text{Ch}(E_{-}(H_{+,A}^{\Delta})) = -\det A, \quad \text{Ch}(E_{-}(H_{-,A}^{\Delta})) = \det A.$$

The same conclusion holds for the positive-mass symbols $H_{\pm,m}^{\Delta}(A\kappa)$, $m > 0$. Small self-adjoint finite-range perturbations with norm below the corresponding pulled-back gap preserve these Chern numbers.

If the same periodic triangular operator is only rewritten with respect to a finite-index sublattice of periods, equivalently with a larger real-space fundamental cell and the folded Bloch Hamiltonian, the zero gap and the Chern signs -1 for H_{+}^{Δ} and $+1$ for H_{-}^{Δ} are unchanged.

Proof. See Appendix A.47.

Corollary A.24 (Finite periodic triangular tori). *Every finite $N_1 \times N_2$ periodic quotient obtained from the symbol above has a zero gap bounded below by 2^{-13} , uniformly in N_1, N_2 . For quotients large enough to be embedded as simplicial tori, this is the corresponding finite periodic triangular torus. Its occupied finite-volume Bloch states are samples of the same rank-three Bloch bundle.*

Proof. See Appendix A.48.

The next interface statement has two layers. The periodic Bloch gap and Chern sign are internal inputs from Theorems 5.3 and 5.6. A nonstraight coarse winding value is asserted only after a compatible coarse interface pairing is supplied. In the straight periodic cases below, the Toeplitz extension supplies that pairing inside the paper.

Corollary A.25 (Periodic annular closed-bulk input and interface class). *Let $(\mathcal{K}_N)_N$ be a sequence of triangular-lattice cylinders whose doubles are the periodic triangular tori of Corollary A.24. Then the uniform closed doubled-bulk gap and closed-bulk Chern input needed for the doubled-and-jammed boundary construction are supplied by Theorems 5.3 and 5.6, not by the Higson–Prodan closed-bulk theorem. Hence, for every sufficiently large jam scale M , the doubled-and-jammed interface class is*

$$\partial_Y(E_{+}(\pi(P_{\pm}))) \in K_1(C^*(Y \subset X)).$$

If the chosen coarse Mayer–Vietoris/interface pairing is compatible with the periodic Bloch Chern pairing, the corresponding winding signs are -1 for H_{+}^{Δ} and $+1$ for H_{-}^{Δ} .

Proof. See Appendix A.49.

The FHS computation in Table 3 independently confirms the Bloch Chern signs in Theorem 5.6 on the displayed grids, and Table 4 checks the finite-cover, folded-cell, and fixed-positive-mass variants used below.

Corollary A.26 (Robust periodic annular input). *The conclusion of Corollary A.25 remains valid if the closed periodic triangular bulk Hamiltonian on the doubled cylinders is replaced by one of the following finite-range periodic bulk models:*

- (i) $H_{\pm,m}^{\Delta}(k) = B(k) + B(k)^* \pm mS(k)$, for any fixed $m > 0$;
- (ii) $H_{\pm,m}^{\Delta}(k) + V(k)$, where $m > 0$ is fixed, V is self-adjoint finite-range periodic, and $\sup_k \|V(k)\| < g_{\pm,m}$ in the notation of Corollary 5.7;
- (iii) $H_{\pm}^{\Delta}(k) + V(k)$, where V is self-adjoint, finite-range periodic, and

$$\sup_{k \in \mathbb{T}^2} \|V(k)\| < 2^{-13}.$$

- (iv) the finite-cover pullbacks and finite-index cell enlargements covered by Corollary A.23.

The coarse interface collar Y is enlarged, if necessary, by the finite propagation radius of the chosen periodic bulk model. For each such model, the closed-bulk gap and the closed-bulk Chern sign are supplied internally by Proposition 5.5 and Corollary 5.7. Therefore, for every jam scale above the corresponding copy-side positivity threshold, the doubled-and-jammed family has the interface class

$$\partial_Y(E_+(\pi(P_{\pm}^{\text{per}}))) \in K_1(C^*(Y \subset X)),$$

where $P_{\pm}^{\text{per}}$ is the negative Fermi projection of the chosen periodic closed-bulk model. This class is independent of the large scalar jam scale and is unchanged if the scalar jam is replaced by any sufficiently positive bounded finite-propagation copy-side confining potential. If the chosen coarse Mayer–Vietoris/interface pairing is compatible with the periodic Bloch Chern pairing, the compatible winding value is the corresponding periodic Bloch Chern number supplied by Theorem 5.6 and Corollary A.23.

Proof. See Appendix A.50.

Corollary A.27 (Periodic boundary families with internal bulk input). *Let $(K_n)_n$ be any sequence of triangular-lattice cylinders whose doubles are finite periodic triangular tori, with arbitrary periods tending to infinity along a coarse disjoint union. The following choices give boundary families for which the closed-bulk locality, gap, and Chern-sign inputs are verified internally in this paper:*

- (a) the unperturbed periodic triangular Hamiltonians $H_{\pm}^{\Delta}$;
- (b) the positive-duality-strength Hamiltonians $H_{\pm,m}^{\Delta}$, for any fixed $m > 0$;
- (c) the perturbed Hamiltonians $H_{\pm,m}^{\Delta} + V$, for any fixed $m > 0$ and any self-adjoint finite-range periodic V satisfying $\sup_k \|V(k)\| < g_{\pm,m}$;
- (d) the explicit-radius perturbations $H_{\pm}^{\Delta} + V$, for any self-adjoint finite-range periodic V satisfying $\sup_k \|V(k)\| < 2^{-13}$;
- (e) finite-cover pullbacks of the triangular symbols, whose Chern numbers are multiplied by the signed covering degree, and finite-index cell enlargements, whose Chern signs are unchanged.

For every family in (a)–(e), every sufficiently large scalar jam scale, every bounded componentwise scalar jam scale bounded below by the corresponding positivity threshold, and every sufficiently positive bounded finite-propagation copy-side confining potential produces the same interface class

$$\partial_Y(E_+(\pi(P_{\pm}^{\text{per}}))) \in K_1(C^*(Y \subset X)),$$

after enlarging the coarse interface collar by the propagation radius of the selected periodic bulk model and copy-side confinement. The class is also unchanged by altering finitely many initial components of the cylinder sequence, or by passing through any norm-continuous sidewise homotopy whose comparison gaps stay uniformly open. If a compatible coarse interface pairing is supplied, its winding value is the corresponding periodic Bloch Chern number from Theorem 5.6 and Corollary A.23.

Proof. See Appendix A.51.

Corollary A.28 (Finite-cover primitive Toeplitz formula). *Let H and P be as in Theorem 5.8, and let $A \in M_2(\mathbb{Z})$ have $\det A \neq 0$. Set*

$$H_A(\kappa) := H(A\kappa), \quad P_A(\kappa) := P(A\kappa).$$

For every primitive tangential/positive-normal basis (t, n) in the cover coordinates,

$$\text{wind}(\partial_{T,t,n}([P_A])) = \det[t \ n] \det A \text{Ch}(P).$$

For the triangular symbols this gives

$$\text{wind}(H_{+,A}^\Delta) = -\det[t \ n] \det A, \quad \text{wind}(H_{-,A}^\Delta) = \det[t \ n] \det A,$$

with the same formula for positive-mass symbols and for the small periodic perturbations covered by Corollary 5.7.

Proof. See Appendix A.54.

Corollary A.29 (Gapped periodic homotopies and Toeplitz winding). *Let $s \mapsto H_s(k)$, $0 \leq s \leq 1$, be a norm-continuous family of finite-range self-adjoint periodic Bloch Hamiltonians with a common finite hopping range. Assume that*

$$\inf_{s,k} \text{dist}(0, \text{spec} H_s(k)) > 0, \quad P_s(k) := \chi_{(-\infty, 0)}(H_s(k)).$$

Then, for every primitive tangential/positive-normal basis (t, n) ,

$$\partial_{T,t,n}([P_s]) \in K_1(C(S^1, \mathcal{K}))$$

is independent of s , and

$$\text{wind}(\partial_{T,t,n}([P_s])) = \det[t \ n] \text{Ch}(P_0) = \det[t \ n] \text{Ch}(P_1).$$

Thus any finite-range periodic model connected to one of the internally verified triangular families by such a zero-gapped homotopy has the same straight-edge Toeplitz winding.

Proof. See Appendix A.55.

Theorem A.30 (Split componentwise Toeplitz winding). *Let H and P be as in Theorem 5.8. Suppose a periodic straight-edge model has finitely many interface components $Y^{(1)}, \dots, Y^{(r)}$, and suppose the edge ideal splits as a C^* -direct sum of the component ideals,*

$$C^*(Y \subset X) \cong \bigoplus_{j=1}^r C^*(Y^{(j)} \subset X),$$

with the j -th summand identified with the Toeplitz edge ideal $C(S^1, \mathcal{K})$ for a primitive tangential/normal lattice basis (t_j, n_j) . Assume also that the quotient extension is the direct sum of these component Toeplitz extensions. Then the global boundary class decomposes as

$$\partial_Y([P]) = \bigoplus_{j=1}^r \partial_{T, t_j, n_j}([P]) \quad \text{in} \quad K_1(C^*(Y \subset X)).$$

Consequently the j -th component winding is

$$\text{wind}_j(\partial_Y([P])) = \det[t_j \ n_j] \text{Ch}(P).$$

If each component is oriented by the induced boundary orientation, the tangential coordinate is reversed exactly when the normal sign is reversed, so each induced-orientation component winding records the same bulk Chern number $\text{Ch}(P)$.

Proof. See Appendix A.56.

Theorem A.31 (Tail split component winding maps). *Let $X = \bigsqcup_n X_n$ be the coarse disjoint union of a straight-periodic boundary family from Corollary A.27. Suppose that, after deleting finitely many initial components, the tail interface ideal has either*

- (i) *a split direct-sum Toeplitz decomposition as in Theorem A.30, or*
- (ii) *a finite-corner or K_1 -trivial-corner quotient decomposition as in Theorem A.19 and Corollary A.20.*

Let (t_j, n_j) , $1 \leq j \leq r$, be the primitive tangential/positive-normal bases of the tail components, and let Ω_j be the component map obtained by deleting the finite initial components, passing to the j -th Toeplitz quotient coordinate, and applying the usual winding homomorphism. Then the maps Ω_j are well defined on the original interface class and

$$\Omega_j(\text{Ind}_{\text{int}}(J; T^+, T^-)) = \det[t_j \ n_j] \text{Ch}(P).$$

For a finite-cover pullback by A , the right-hand side is

$$\det[t_j \ n_j] \det A \text{Ch}(P).$$

The same formulas are unchanged by arbitrary bounded finite-propagation changes on finitely many initial components.

Proof. See Appendix A.63.

Corollary A.32 (Internal straight-periodic classes). *For the following boundary/interface families, with the split, quotient, tail, and sidewise-gap hypotheses stated in the cited results, no Higson–Prodan closed-bulk input and no external Ewert–Meyer pairing assumption are used. Their zero gap, Chern value, interface class, and Toeplitz winding are supplied by the internal periodic calculation and the Toeplitz evaluation above:*

- (a) *primitive straight periodic edges of the unperturbed triangular models $H_{\pm}^{\Delta}$;*
- (b) *primitive straight periodic edges of $H_{\pm, m}^{\Delta}$, for every fixed $m > 0$;*

- (c) *primitive straight periodic edges of $H_{\pm, m}^{\Delta} + V$, for every fixed $m > 0$ and every self-adjoint finite-range periodic perturbation satisfying $\sup_k \|V(k)\| < g_{\pm, m}$; in particular this includes $H_{\pm}^{\Delta} + V$ with $\sup_k \|V(k)\| < 2^{-13}$;*
- (d) *finite-cover pullbacks of the families in (a)–(c), with Chern numbers multiplied by the signed covering degree, and finite-index cell enlargements of the same operators, with the original Chern signs unchanged;*
- (e) *any finite split direct sum of the families in (a)–(d), with component ideals and quotient extension splitting as in Theorem A.30;*
- (f) *any finite-corner or K_1 -trivial-corner split sum of the families in (a)–(d), with quotient extension as in Theorem A.19, Corollary A.20, or Corollary A.21;*
- (g) *finite-defect, bounded-distance straight-edge, and tail-split variants covered by Theorem A.18, Corollary A.17, and Theorem A.31;*
- (h) *the same families after replacing scalar jamming by any bounded componentwise scalar jam uniformly above the positivity threshold, or by any sufficiently positive bounded finite-propagation copy-side confinement;*
- (i) *the same families after adding norm-vanishing component defects, with the sidewise comparison gaps still open.*

For the triangular families before applying a finite cover, the winding values are -1 for H_+ and $+1$ for H_- , up to the orientation factor $\det[t_j \ n_j]$ on each primitive component. For a finite-cover pullback, these values are further multiplied by the signed covering degree. Fixed-collar finite-propagation edge perturbations, norm-vanishing component defects, and norm-continuous sidewise homotopies preserve these values whenever the sidewise comparison gaps remain open; this last gap condition is a stability condition for the perturbation path, not an imported bulk–interface theorem.

Proof. See Appendix A.64.

A.9 Roe interface variants

The main body keeps the conditional Roe interface theorem and the conditional winding theorem. This subsection records the stability, admissibility, relabeling, tail, and confinement variants that share those hypotheses.

Corollary A.33 (Small-norm stability of Fermi and interface classes). *Let $H = H^* \in C^*(X)$ satisfy*

$$\text{spec}_{\text{op}}(H) \subset [-B, -g] \cup [g, B]$$

for some $B, g > 0$. Let $V = V^ \in C^*(X)$ with*

$$\|V\| < g.$$

Then

$$H + V$$

is gapped at zero, with

$$\text{spec}_{\text{op}}(H + V) \cap (-(g - \|V\|), g - \|V\|) = \emptyset,$$

and

$$[\chi_{(-\infty,0)}(H+V)] = [\chi_{(-\infty,0)}(H)] \quad \text{in } K_0(C^*(X)).$$

If, in addition, $H = H_{\pm}^{\infty}$ under Assumption 6.13 and $V = V^* \in C^*(X)$ satisfies

$$\|V\| < g_{\text{bulk}},$$

then, with

$$P_V := \chi_{(-\infty,0)}(H_{\pm}^{\infty} + V).$$

one has

$$\partial_Y(E_+(\pi(P_V))) = \partial_Y(E_+(\pi(P_{\pm}))) \quad \text{in } K_1(C^*(Y \subset X)).$$

Proof. See Appendix A.85.

Proposition A.34 (Interface-supported sidewise changes). *Let $(J; T^+, T^-)$ be a sidewise class-A interface along Y . Let*

$$J' = J + V, \quad T'^+ = T^+ + V^+, \quad T'^- = T^- + V^-,$$

where $V, V^+, V^- \in \mathcal{I}$ are self-adjoint. Assume that $(J'; T'^+, T'^-)$ is again a sidewise class-A interface along Y . Then

$$p_{J'} = p_J, \quad \text{Ind}_{\text{int}}(J'; T'^+, T'^-) = \text{Ind}_{\text{int}}(J; T^+, T^-).$$

The special case $V^+ = V^- = 0$ gives invariance under perturbing only the interface Hamiltonian J by a self-adjoint element of $\mathcal{I}$.

Proof. See Appendix A.94.

Proposition A.35 (Stability under finite-component changes). *Let $X = \bigsqcup_n X_n$ be the coarse disjoint union, and assume each Y_n is nonempty. Let $V = V^*$ be a bounded finite-propagation operator supported on only finitely many components: there is $N < \infty$ such that*

$$V\chi_{X_n} = \chi_{X_n}V = 0 \quad \text{for all } n > N.$$

Then $V \in C^*(Y \subset X)$. Consequently, changing any of the sidewise data J, T^+, T^- on finitely many initial components does not change the interface class, provided the sidewise gap hypotheses remain valid after the change.

Proof. See Appendix A.95.

Proposition A.36 (Norm-vanishing component defects). *Let $X = \bigsqcup_n X_n$ be the coarse disjoint union and assume each Y_n is nonempty. Let*

$$V = \bigoplus_{n=1}^{\infty} V_n$$

be a bounded self-adjoint block-diagonal operator with $V_n \in \mathcal{B}(\ell^2(X_n))$ and $\|V_n\| \rightarrow 0$. Then

$$V \in C^*(Y \subset X).$$

Consequently, adding such norm-vanishing component defects to J, T^+ , or T^- does not change the sidewise interface class, provided the resulting comparison operators remain gapped and the sidewise class-A hypotheses remain valid.

Proof. See Appendix A.96.

Proposition A.37 (Homotopy invariance of the sidewise interface class). *Let*

$$s \mapsto (J_s; T_s^+, T_s^-), \quad s \in [0, 1],$$

be a family of sidewise class-A interfaces along the same interface Y . Assume that $s \mapsto T_s^\pm$ are norm-continuous and that there is a uniform comparison gap

$$\text{dist}(0, \text{spec}_{\text{op}}(T_s^\pm)) \geq \gamma > 0$$

for all $s \in [0, 1]$. Then

$$\text{Ind}_{\text{int}}(J_s; T_s^+, T_s^-)$$

is independent of s .

Proof. See Appendix A.97.

Corollary A.38 (Topological independence of the confining copy potential). *Assume 6.13. Let $s \mapsto V_s$, $s \in [0, 1]$, be a norm-continuous family of bounded self-adjoint finite-propagation operators on $\ell^2(X)$ such that*

$$V_s = P_{\text{copy}}^\infty V_s P_{\text{copy}}^\infty$$

and, on the copy sector,

$$V_s \geq M_0 \mathbf{1} \quad \text{for a constant } M_0 > B_{\text{copy},-}.$$

Define

$$J_s := H_\pm^\infty + V_s, \quad T_s^+ := H_\pm^\infty, \quad T_s^- := P_{\text{orig}}^\infty + H_{\pm, \text{copy}}^\infty + V_s.$$

Then the sidewise interface class of $(J_s; T_s^+, T_s^-)$ is independent of s .

Proof. See Appendix A.98.

Theorem A.39 (Interface theorem for general confining copy potentials). *Assume 6.13. Let $V = V^*$ be a bounded finite-propagation operator on $\ell^2(X)$ such that*

$$V = P_{\text{copy}}^\infty V P_{\text{copy}}^\infty$$

and, on the copy sector,

$$V \geq M_0 \mathbf{1} \quad \text{for some } M_0 > B_{\text{copy},-}.$$

Define

$$J_V := H_\pm^\infty + V, \quad T_V^+ := H_\pm^\infty, \quad T_V^- := P_{\text{orig}}^\infty + H_{\pm, \text{copy}}^\infty + V.$$

Then $(J_V; T_V^+, T_V^-)$ is a sidewise class-A interface along Y . Its canonical quotient projection is

$$p_{J_V} = E_+(\pi(P_\pm)),$$

the negative-side comparison has the quantitative positivity margin

$$\text{dist}(0, \text{spec}_{\text{op}}(T_V^-)) \geq \min\{1, M_0 - B_{\text{copy},-}\},$$

and hence

$$\text{Ind}_{\text{int}}(J_V; T_V^+, T_V^-) = \partial_Y(E_+(\pi(P_\pm))) \in K_1(C^*(Y \subset X)).$$

In particular, the interface class is unchanged by replacing scalar jamming by any sufficiently positive bounded finite-propagation copy-side confining potential.

Proof. See Appendix A.99.

Corollary A.40 (Independence of the interface class for large jam scales). *Assume 6.13. If $M_1, M_2 > B_{\text{copy},-}$, then*

$$\text{Ind}_{\text{int}}(J_{\pm, M_1}^\infty; T_1^+, T_1^-) = \text{Ind}_{\text{int}}(J_{\pm, M_2}^\infty; T_2^+, T_2^-) = \partial_Y(E_+(\pi(P_\pm))),$$

where

$$T_j^+ = H_\pm^\infty, \quad T_j^- = \tilde{T}_{\pm, M_j}^-.$$

Proof. See Appendix A.101.

Combining Theorems 6.35 and 6.36 with Corollary A.40, we obtain the following explicit statement.

Corollary A.41 (Interface class from a Chern- ± 1 bulk projection). *Assume 6.13, the additional Higson–Prodan hypotheses from Theorem 6.36, and $M > B_{\text{copy},-}$. Then*

$$\text{Ind}_{\text{int}}(J_{\pm, M}^\infty; T^+, T^-) = \partial_Y(E_+(\pi(P_\pm))) \in K_1(C^*(Y \subset X)),$$

where $P_\pm$ is the Higson–Prodan bulk Fermi projection with $\text{Ch}([P_\pm]) = \pm 1$. The interface class depends only on the bulk projection and the quotient decomposition, not on the jam scale M .

Proof. See Appendix A.103.

Corollary A.42 (Higson–Prodan input implies the doubled-and-jammed interface class). *Assume that the closed doubled refinement family satisfies the Higson–Prodan closed-bulk input in Assumption 6.15. Then Assumptions 6.8 and 6.13 hold. Consequently, for every $M > B_{\text{copy},-}$,*

$$\text{Ind}_{\text{int}}(J_{\pm, M}^\infty; T^+, T^-) = \partial_Y(E_+(\pi(P_\pm))) \in K_1(C^*(Y \subset X)),$$

where $P_\pm$ is the Higson–Prodan bulk Fermi projection and

$$\text{Ch}([P_\pm]) = \pm 1.$$

The construction of this K_1 -class is internal to the present paper; its numerical winding-number interpretation is obtained by applying the external coarse bulk–interface correspondence under its compatibility hypotheses.

Proof. See Appendix A.104.

Proposition A.43 (Admissibility criterion). *Let $(\mathcal{K}_n)_n$ be a uniformly regular full-boundary refinement family. Suppose that the closed doubled family $D(\mathcal{K}_n)$ satisfies the Higson–Prodan closed-bulk input in Assumption 6.15, and that the Ewert–Meyer Mayer–Vietoris boundary map is paired compatibly with the Higson–Prodan Chern pairing as in Definition 6.37. Then $(\mathcal{K}_n)_n$ is HP–Ewert–Meyer admissible. No boundary spectral gap is required.*

Proof. See Appendix A.105.

Corollary A.44 (Verification template for boundary refinement families). *Let $(\mathcal{K}_n)_n$ be a refinement sequence of a fixed compact oriented PL surface with full boundary subcomplex. Suppose the following checks have been completed after the standard rescaling used in the closed Higson–Prodan refinement limit:*

- (i) the closed doubles $D(\mathcal{K}_n)$ satisfy a cited Higson–Prodan/Higson–Roe closed-bulk theorem supplying Assumption 6.15;
- (ii) the original/copy decomposition gives the coarse partition required in Definition 6.37;
- (iii) the chosen Ewert–Meyer Mayer–Vietoris boundary map is paired compatibly with the Higson–Prodan Chern pairing used in Theorem 6.36.

Then $(\mathcal{K}_n)_n$ is HP–Ewert–Meyer admissible. The following common refinement schemes are therefore not asserted to be admissible by name; rather, they are natural schemes to which this template may be applied once the three checks above have been verified:

- (a) iterated barycentric refinements;
- (b) bounded-valence stellar or edge refinements;
- (c) quasi-uniform shape-regular triangulations used in finite-element refinements.

For any verified subclass, and every $M > B_{\text{copy},-}$, the doubled-and-jammed interface class has compatible winding number ± 1 .

Proof. See Appendix A.106.

Corollary A.45 (Only the tail of a refinement family matters). *Suppose that, after deleting finitely many initial refinements, the tail $(\mathcal{K}_n)_{n \geq N_0}$ is HP–Ewert–Meyer admissible. Extend the tail by any finite set of initial components with bounded local sidewise data for which the comparison operators are gapped at zero. Then the doubled-and-jammed interface class and its compatible winding value are the same as those of the tail. In particular, admissibility only has to be checked eventually along a refinement sequence.*

Proof. See Appendix A.107.

Corollary A.46 (Relabeling and orientation reversal). *The HP–Ewert–Meyer admissibility condition, the interface class, and the compatible winding value are invariant under orientation-preserving boundary-compatible simplicial isomorphisms of refinement families. If the surface orientation is reversed, then the roles of H_+ and H_- are interchanged and the compatible winding sign is reversed.*

Proof. See Appendix A.109.

Corollary A.47 (Robustness of the interface class and winding). *Under the hypotheses of Theorem 6.38, the interface class and its compatible winding value are unchanged by the following operations, whenever the stated sidewise comparison gaps remain open:*

- (a) replacing the scalar jam scale M by any other scalar $M' > B_{\text{copy},-}$;
- (b) replacing scalar jamming by a bounded finite-propagation copy-side confinement V satisfying

$$V = P_{\text{copy}}^\infty V P_{\text{copy}}^\infty, \quad V \geq M_0 \mathbf{1}, \quad M_0 > B_{\text{copy},-};$$

this includes bounded componentwise scalar jam scales $V = \bigoplus_n M_n P_{\text{copy},n}$ with $B_{\text{copy},-} < \inf_n M_n \leq \sup_n M_n < \infty$;

- (c) changing the sidewise data by self-adjoint interface-supported operators;
- (d) changing the boundary/copy coupling by uniformly bounded finite-propagation perturbations supported in any fixed boundary collar;
- (e) replacing the chosen fixed collar model for Y by another fixed-radius collar;
- (f) changing the operators on finitely many initial refinement components;
- (g) adding norm-vanishing component defects;
- (h) passing through a norm-continuous sidewise homotopy with a uniform comparison gap.

Proof. See Appendix [A.110](#).

A.10 Symbolic verification appendix

This subsection describes a minimal symbolic check for the local crossing calculation in Theorem [5.6](#). It is a verification aid only; the proof is the analytic harmonic-compression and crossing-orientation argument above. The script records the Bloch matrix, the harmonic projector, the compressed local crossing, and the expected sign output.

```
import sympy as sp
sqrt = sp.sqrt
q1, q2, mu = sp.symbols('q1 q2 mu', real=True)
i = sp.I

# Ordered basis: v, a, b, c, f, g.
z1 = sp.exp(i*q1)
z2 = sp.exp(i*q2)
d1 = sp.Matrix([[z1-1, z2-1, z1*z2-1]])
d2 = sp.Matrix([[1, -z2], [z1, -1], [-1, 1]])
B = sp.zeros(6)
B[0,1:4] = d1
B[1:4,4:6] = d2

P0 = sp.Matrix([[z1**-1*z2**-1], [z2**-1]])
P1 = sp.Matrix([[0, z1**-1, 0], [0, 0, 0], [-z2**-1, 0, 0]])
P2 = sp.Matrix([[1, 1]])
S = sp.zeros(6)
S[0,4:6] = (i/2) * P2
S[4:6,0] = (i/2) * P0
S[1:4,1:4] = (i/2) * (P1.conjugate().T - P1)
S[4:6,0] = (-i/2) * (P0.conjugate().T + P2.T)
H = B + B.conjugate().T + mu*S

h0 = sp.Matrix([1,0,0,0,0,0])
h1 = sp.Matrix([0,1/sqrt(2),-1/sqrt(2),0,0,0])
h2 = sp.Matrix([0,1/sqrt(6),1/sqrt(6),2/sqrt(6),0,0])
h3 = sp.Matrix([0,0,0,0,1/sqrt(2),1/sqrt(2)])
```

```

U = sp.Matrix.hstack(h0,h1,h2,h3)
Pi_h = U*U.T

L = sp.zeros(4)
for x in (q1, q2, mu):
    L += x * (U.T * H.diff(x).subs({q1:0,q2:0,mu:0}) * U)
print(sp.simplify(L))

a = (1 + 1/sqrt(6))/2
b = sqrt(3)/2 + 1/(2*sqrt(2))
c = 1 + 1/sqrt(6)
J = sp.Matrix([[a, -a, 0], [b, b, 0], [0, 0, -c]])
print(sp.factor(J.det())) # negative
print('expected Chern signs: H_plus -> -1, H_minus -> +1')

```

The proof of each result in the main text is collected here in the order in which the results appear.

A.11 Proof of Result 2.9

Proof of Proposition 2.9. Expand $H_{\pm}^2$ and use the assumed anticommutation to eliminate the cross term. Since $(B + B^\dagger)^2 \succeq 0$ and $S^2 \succeq \mu_S^2 \mathbf{1}$, we obtain $H_{\pm}^2 \succeq \mu_S^2 \mathbf{1}$. $\square$

A.12 Proof of Result 2.10

Proof of Proposition 2.10. Since

$$H_{\pm}^2 = Q^2 + S^2 \pm (QS + SQ),$$

we have $Q^2 \succeq 0$ and $S^2 \succeq \mu_S^2 \mathbf{1}$. Also,

$$\pm(QS + SQ) \succeq -\|QS + SQ\| \mathbf{1} = -\varepsilon_{ac} \mathbf{1}.$$

Hence $H_{\pm}^2 \succeq (\mu_S^2 - \varepsilon_{ac}) \mathbf{1}$. The spectral-gap conclusion follows immediately. $\square$

A.13 Proof of Result 3.2

Proof of Proposition 3.2. We verify that $D(\mathcal{K})$ is a well-defined abstract simplicial complex. A simplex is determined by its vertex set, so we must check that no two distinct simplices (of any dimension) in the doubled complex share the same vertex set.

Faces. Let $F = \{v_0, v_1, v_2\}$ be a face of $\mathcal{K}$. By the fullness condition in Definition 2.2, F has at least one interior vertex v_0 . Every reflected face contains at least one copy-side vertex $\bar{w} \in \bar{\mathcal{K}}_0^{\text{int}}$, and no original face contains any copy-side vertex. Hence no original face coincides with any reflected face. Since faces within each copy are distinct, all 2-simplices of $D(\mathcal{K})$ have distinct vertex sets.

Edges. Interior edges of $\mathcal{K}$ have at least one interior endpoint (by fullness: both endpoints on $\partial\mathcal{K}$ would force the edge to be a boundary edge). Hence every reflected interior edge acquires at least one copy-side vertex and has a different vertex set from the original. Boundary edges are identified (not duplicated), so they appear once. Thus all 1-simplices of $D(\mathcal{K})$ have distinct vertex sets.

Edge valences. Every interior edge of $\mathcal{K}$ is incident to exactly two faces in $\mathcal{K}$. Its reflected copy (with a different vertex set) is incident to exactly two reflected faces. So each such pair contributes two edges, each incident to two faces. Every boundary edge is incident to one original face and one reflected face, hence to exactly two faces. Thus every edge of $D(\mathcal{K})$ is incident to exactly two faces.

At an interior vertex of either copy, the link is already a cycle. At a boundary vertex v , the link in $\mathcal{K}$ is a path and the link in $\bar{\mathcal{K}}$ is a second path with the opposite boundary orientation. After the boundary identification these two paths glue end-to-end to form a cycle. Thus every vertex link in $D(\mathcal{K})$ is a cycle, so $D(\mathcal{K})$ is a closed triangulated surface.

Finally, orient the faces in the original copy by the given surface orientation and orient the faces in the reflected copy by the opposite orientation. Along every glued boundary edge, the two induced edge orientations are opposite, so the face orientations are compatible across the gluing. Hence $D(\mathcal{K})$ is oriented. $\square$

A.14 Proof of Result 3.3

Proof of Theorem 3.3. Faces are duplicated and never identified (each reflected face has a distinct vertex set by the fullness condition), so $|D(\mathcal{K})_2| = 2|\mathcal{K}_2|$. Every non-boundary edge has at least one interior endpoint (by fullness), so its reflected copy has a distinct vertex set; non-boundary edges are therefore duplicated. Boundary edges are identified, not duplicated. Hence $|D(\mathcal{K})_1| = 2|\mathcal{K}_1^{\text{int}}| + |\partial\mathcal{K}_1| = 2(|\mathcal{K}_1| - |\partial\mathcal{K}_1|) + |\partial\mathcal{K}_1| = 2|\mathcal{K}_1| - |\partial\mathcal{K}_1|$. The same reasoning applies to vertices: interior vertices are duplicated, boundary vertices are identified, giving $|D(\mathcal{K})_0| = 2|\mathcal{K}_0| - |\partial\mathcal{K}_0|$. Since $\partial\mathcal{K}$ is a disjoint union of cycles, each connected component has as many vertices as edges, so $|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|$. $\square$

A.15 Proof of Result 3.4

Proof of Corollary 3.4. Use

$$\dim \mathcal{H}(\mathcal{K}) = |\mathcal{K}_0| + |\mathcal{K}_1| + |\mathcal{K}_2|, \quad \dim \mathcal{H}(D(\mathcal{K})) = |D(\mathcal{K})_0| + |D(\mathcal{K})_1| + |D(\mathcal{K})_2|,$$

insert Theorem 3.3, and simplify with $|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|$. $\square$

A.16 Proof of Result 3.5

Proof of Proposition 3.5. Insert the identities from Theorem 3.3 into $\chi = |\mathcal{K}_0| - |\mathcal{K}_1| + |\mathcal{K}_2|$ and use $|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|$. $\square$

A.17 Proof of Result 3.6

Proof of Corollary 3.6. For an orientable surface with boundary, $\chi(\mathcal{K}) = 2 - 2g - b$ [15]. By Proposition 3.5,

$$\chi(D(\mathcal{K})) = 4 - 4g - 2b.$$

Since a closed orientable surface of genus G has Euler characteristic $2 - 2G$, one gets $G = 2g + b - 1$. $\square$

A.18 Proof of Result 3.10

Proof of Lemma 3.10. For a p -cochain α , identified with a p -chain by the simplex basis, the standard cap-product boundary formula [15, Section 3.3] is

$$\partial(c \cap \alpha) = (-1)^p((\partial c) \cap \alpha - c \cap \delta \alpha).$$

Taking $c = [\mathcal{K}, \partial\mathcal{K}]$, whose boundary is zero in the relative chain complex, gives (3.5). Applying j_{1-p} to that identity and subtracting from $B_{\mathcal{K}, 2-p} j_{2-p} P_p^{\text{rel}}$ gives (3.6).

The operator $B_{\mathcal{K}, r} j_r - j_{r-1} \partial_r^{\text{rel}}$ is zero on every relative simplex whose codimension-one faces are all represented by non-boundary simplices. When it is nonzero, its output is a boundary simplex, and its input simplex meets $\partial\mathcal{K}$. Since the cap-product maps P_p^{rel} have a fixed finite propagation radius, a nonzero coefficient in (3.6) can occur only with source and target inside a uniformly bounded collar of $\partial\mathcal{K}$. The operator $S_{\mathcal{K}}^{\text{PL}}$ is obtained from $\hat{P}$ and $\hat{P}^\dagger$ by fixed degree phases and Hermitian symmetrization; taking adjoints preserves the same support condition. Hence the displayed duality defect is collar-supported. $\square$

A.19 Proof of Result 3.11

Proof of Lemma 3.11. With the coefficient convention in Section 2.6, a matrix coefficient of either cap-product operator is a signed count of oriented faces for which the source simplex appears as the prescribed suffix and the target simplex appears as the complementary prefix. If both simplices are outside the R_- -collar, every face contributing to the coefficient is an original-side face disjoint from the glued boundary. On such faces the closed fundamental cycle of $D(\mathcal{K})$ restricts exactly to the relative fundamental chain $[\mathcal{K}, \partial\mathcal{K}]$, and the quotient map q does not remove the complementary prefix. Hence the two signed coefficient sums are identical. $\square$

A.20 Proof of Result 3.12

Proof of Theorem 3.12. Self-adjointness is built into the Hermitian symmetrization. Locality follows from the chain-level cap product: a matrix coefficient of P^{rel} can be nonzero only when the two simplices are incident inside a common face or star. The representative map j removes boundary representatives but does not enlarge support, so $S_{\mathcal{K}}^{\text{PL}}$, and hence $H_{\pm}^{\text{PL}}(\mathcal{K})$, has finite propagation.

The relative duality defect is collar-supported by Lemma 3.10.

The $B + B^\dagger$ part of $H_{\pm}^\partial(\mathcal{K})$ agrees exactly with $B_{\mathcal{K}} + B_{\mathcal{K}}^\dagger$. Compressing the doubled boundary operator to the original side keeps the original faces, edges, and vertices and removes only reflected simplices not contained in $\partial\mathcal{K}$. Thus the difference between $H_{\pm}^\partial(\mathcal{K})$ and $H_{\pm}^{\text{PL}}(\mathcal{K})$ is entirely in the cap-product term.

Lemma 3.11 gives coefficient-by-coefficient equality between the doubled cap-product operator, compressed to the original side, and the lifted relative cap-product operator away from a fixed boundary collar. Thus any discrepancy can occur only when the local cap-product calculation involves a simplex meeting $\partial\mathcal{K}$, or when the relative quotient has discarded a boundary representative. Since the cap-product rule has a fixed finite propagation radius, all such discrepancies have both source and target inside a fixed boundary collar. Defining

$$R_{\partial, \pm} := H_{\pm}^\partial(\mathcal{K}) - H_{\pm}^{\text{PL}}(\mathcal{K})$$

gives (3.7). The difference of two self-adjoint operators is self-adjoint, and the collar-support statement follows from the preceding support analysis. The uniform statement follows because the cap-product propagation radius is fixed over a uniformly regular refinement family. $\square$

A.21 Proof of Result 3.14

Proof of Corollary 3.14. The difference on the n -th component is $R_{\partial, \pm, n}$, supported in the fixed boundary collar from Theorem 3.12. The direct sum has uniformly bounded propagation and

uniformly bounded matrix coefficients by uniform regularity and the fixed local cap-product convention. Hence it belongs to the algebraic interface ideal, and therefore to $C^*(Y \subset X)$. The quotient equality follows immediately. The same theorem gives the fixed collar support of the relative duality defect, so its direct sum is in the same ideal. Finally, quotient maps commute with continuous functional calculus for bounded self-adjoint elements. $\square$

A.22 Proof of Result 3.15

Proof of Lemma 3.15. Since $R = P_W R P_W$, $\text{rank} R \leq r$. Weyl's rank inequality for spectral distribution functions [17, Chapter III] gives

$$N_A(t) - \text{rank} R \leq N_{A+R}(t) \leq N_A(t) + \text{rank} R,$$

which proves the first estimate. The interval estimate follows by subtracting the distribution functions at a and b . The final statement is Theorem 3.12. $\square$

A.23 Proof of Result 3.17

Proof of Corollary 3.17. Apply Lemma 3.15 with $W = \ell^2(N_{\text{RPL}}(\partial\mathcal{K}_n))$, divide by d_n , and take the supremum in t . The final assertion is the usual boundary-versus-area scaling for quasi-uniform fixed-surface two-dimensional refinements; outside such geometric refinement regimes it should be checked as the explicit condition $r_n/d_n \rightarrow 0$. The statement about continuous test functions follows because the operators have uniformly bounded norm under uniform regularity, so the empirical spectral measures are supported in a common compact interval and the distribution-function estimate forces the same weak subsequential limits. $\square$

A.24 Proof of Result 4.1

Proof of Lemma 4.1. Since D is self-adjoint and $\Im z \neq 0$, the operator $D - z$ is invertible. Standard block Gaussian elimination gives the Schur-complement formula. $\square$

A.25 Proof of Result 4.2

Proof of Theorem 4.2. By Lemma 4.1,

$$P_{\text{orig}}(H_{\pm,M} - z)^{-1}P_{\text{orig}} = (A - z - K_M(z))^{-1}, \quad K_M(z) = C(D_M - z)^{-1}C^\dagger.$$

Hence

$$\|K_M(z)\| \leq \frac{\|C\|^2}{d_M(z)}.$$

Now set $R_0 = (A - z)^{-1}$ and $R_M = (A - z - K_M(z))^{-1}$. The resolvent identity gives

$$R_M - R_0 = R_M K_M(z) R_0.$$

Moreover R_M is the $(1,1)$ block of $(H_{\pm,M} - z)^{-1}$, so

$$\|R_M\| \leq \|(H_{\pm,M} - z)^{-1}\| \leq \frac{1}{|\Im z|}, \quad \|R_0\| \leq \frac{1}{|\Im z|}.$$

Combining these bounds yields (4.1). Since

$$\text{spec}(D_M) \subset [M - \|D\|, M + \|D\|],$$

we also have $d_M(z) \geq M - \|D\| - |\Re z|$ whenever $M > \|D\| + |\Re z|$, which gives (4.2). $\square$

A.26 Proof of Result 4.3

Proof of Corollary 4.3. This is the first case in Appendix A.28. $\square$

A.27 Proof of Result 4.4

Proof of Corollary 4.4. This is the second case in Appendix A.28. $\square$

A.28 Routine Schur-complement corollaries

Proof. For Corollary 4.3, apply Theorem 4.2 with $A = H_{\pm}^{\partial}(\mathcal{K})$.

For Corollary 4.4, set $R_D(z) = (D_M - z)^{-1}$ and $S_M(z) = A - z - CR_D(z)C^{\dagger}$. The block inverse formula gives

$$(H_{\pm, M} - z)^{-1} = \begin{pmatrix} S_M(z)^{-1} & -S_M(z)^{-1}CR_D(z) \\ -R_D(z)C^{\dagger}S_M(z)^{-1} & R_D(z) + R_D(z)C^{\dagger}S_M(z)^{-1}CR_D(z) \end{pmatrix}.$$

The upper-left block is controlled by Theorem 4.2. The off-diagonal blocks have norm at most $\|C\|/(|\Im z| d_M(z))$, while the lower-right block is bounded by

$$\frac{1}{d_M(z)} + \frac{\|C\|^2}{d_M(z)^2 |\Im z|}.$$

Adding these four bounds gives Corollary 4.4. $\square$

A.29 Proof of Result 4.5

Proof of Theorem 4.5. Write

$$D_M - z = M \left(\mathbf{1} + \frac{D - z}{M} \right).$$

If $M > \|D - z\|$, the Neumann series gives

$$\left(\mathbf{1} + \frac{D - z}{M} \right)^{-1} = \mathbf{1} - \frac{D - z}{M} + \left(\frac{D - z}{M} \right)^2 - \dots,$$

which yields the stated expansion and remainder estimate. Multiplying on both sides by C and $C^{\dagger}$ gives the formula for $K_M(z)$.

Set

$$R_M = P_{\text{orig}}(H_{\pm, M} - z)^{-1}P_{\text{orig}} = (A - z - K_M(z))^{-1}, \quad R_M^{(1)} = (A_M^{(1)} - z)^{-1}.$$

Then

$$R_M - R_M^{(1)} = R_M \tilde{R}_M(z) R_M^{(1)}.$$

Since both $H_{\pm, M}$ and $A_M^{(1)}$ are self-adjoint,

$$\|R_M\| \leq \frac{1}{|\Im z|}, \quad \|R_M^{(1)}\| \leq \frac{1}{|\Im z|}.$$

Combining these bounds with the estimate on $\tilde{R}_M(z)$ gives (4.3). $\square$

A.30 Proof of Result 4.7

Proof of Theorem 4.7. For $t \in [0, 1]$ define the homotopy

$$H_{t,M} := \begin{pmatrix} A & tC \\ tC^\dagger & D_M \end{pmatrix}.$$

Fix $z \in \Gamma$. Since $\delta_D(\Gamma) > 0$, the operator $D_M - z$ is invertible. The Schur complement of $H_{t,M} - z$ with respect to the lower-right block is

$$S_{t,M}(z) = A - z - t^2 C(D_M - z)^{-1} C^\dagger.$$

Because $z \in \Gamma$, we have

$$\|(A - z)^{-1}\| \leq \frac{1}{\delta_A(\Gamma)}, \quad \|(D_M - z)^{-1}\| \leq \frac{1}{\delta_D(\Gamma)}.$$

Hence

$$\|(A - z)^{-1} t^2 C(D_M - z)^{-1} C^\dagger\| \leq \frac{\|C\|^2}{\delta_A(\Gamma) \delta_D(\Gamma)} < 1$$

by the hypothesis $\delta_H(\Gamma) > 0$. Therefore

$$S_{t,M}(z) = (A - z) \left[\mathbf{1} - (A - z)^{-1} t^2 C(D_M - z)^{-1} C^\dagger \right]$$

is invertible, because the bracketed term is invertible by the Neumann series. Since $D_M - z$ is invertible as well, the block matrix $H_{t,M} - z$ is invertible. This proves $\Gamma \subset \rho(H_{t,M})$ for all $t \in [0, 1]$, and hence $\Gamma \subset \rho(H_{\pm,M})$.

The map $t \mapsto P_\Gamma(H_{t,M})$ is norm-continuous, so its rank is constant on the connected interval $[0, 1]$. At $t = 0$,

$$H_{0,M} = A \oplus D_M,$$

and therefore

$$P_\Gamma(H_{0,M}) = P_\Gamma(A) \oplus P_\Gamma(D_M).$$

This proves the rank identity.

For the norm estimate, abbreviate

$$K_M(z) := C(D_M - z)^{-1} C^\dagger, \quad S_M(z) := A - z - K_M(z).$$

The Schur-complement inverse formula gives Set $R_D(z) = (D_M - z)^{-1}$. The block inverse formula gives

$$(H_{\pm,M} - z)^{-1} = \begin{pmatrix} S_M(z)^{-1} & -S_M(z)^{-1} C R_D(z) \\ -R_D(z) C^\dagger S_M(z)^{-1} & R_D(z) + R_D(z) C^\dagger S_M(z)^{-1} C R_D(z) \end{pmatrix}.$$

Because $\delta_H(\Gamma) > 0$,

$$\|S_M(z)^{-1}\| \leq \frac{1}{\delta_H(\Gamma)}, \quad \|(A - z)^{-1}\| \leq \frac{1}{\delta_A(\Gamma)}, \quad \|(D_M - z)^{-1}\| \leq \frac{1}{\delta_D(\Gamma)}.$$

Also,

$$S_M(z)^{-1} - (A - z)^{-1} = S_M(z)^{-1} K_M(z) (A - z)^{-1},$$

so

$$\|S_M(z)^{-1} - (A - z)^{-1}\| \leq \frac{\|C\|^2}{\delta_A(\Gamma) \delta_H(\Gamma) \delta_D(\Gamma)}.$$

The off-diagonal blocks satisfy

$$\|S_M(z)^{-1} C (D_M - z)^{-1}\| \leq \frac{\|C\|}{\delta_H(\Gamma) \delta_D(\Gamma)},$$

and the extra lower-right block is bounded by

$$\|(D_M - z)^{-1} C^\dagger S_M(z)^{-1} C (D_M - z)^{-1}\| \leq \frac{\|C\|^2}{\delta_H(\Gamma) \delta_D(\Gamma)^2}.$$

Subtracting the block-diagonal resolvent

$$\begin{pmatrix} (A - z)^{-1} & 0 \\ 0 & (D_M - z)^{-1} \end{pmatrix}$$

gives the pointwise bound (4.4). Integrating around Γ then gives (4.6). $\square$

A.31 Proof of Result 4.8

Proof of Theorem 4.8. Let $\psi = (\psi_{\text{orig}}, \psi_{\text{copy}})$ be a normalized eigenvector with eigenvalue E . If $\psi_{\text{orig}} = 0$, then $(D_M)\psi_{\text{copy}} = E\psi_{\text{copy}}$, which is impossible because $|E| \leq E_{\text{max}}$ and $g_M(E_{\text{max}}) > 0$. Thus $\psi_{\text{orig}} \neq 0$.

Solving the second block equation gives

$$\psi_{\text{copy}} = -(D_M - E)^{-1} C^\dagger \psi_{\text{orig}}.$$

Substituting into the original-block equation yields

$$(A - \Sigma_M(E))\psi_{\text{orig}} = E\psi_{\text{orig}}, \quad \Sigma_M(E) = C(D_M - E)^{-1} C^\dagger.$$

Hence $E \in \text{spec}(A - \Sigma_M(E))$. Since

$$\|\Sigma_M(E)\| \leq \|C\|^2 \|(D_M - E)^{-1}\| \leq \frac{\|C\|^2}{g_M(E_{\text{max}})},$$

Weyl's inequality implies (4.7). If $M > \|D\| + E_{\text{max}}$, then

$$g_M(E_{\text{max}}) \geq M - \|D\| - E_{\text{max}},$$

which gives (4.8). $\square$

A.32 Proof of Result 4.9

Proof of Corollary 4.9. By Theorem 3.12, the original-side block is $A = H_{\pm}^{\text{PL}}(\mathcal{K}) + R_{\partial, \pm}$. The result is therefore Theorem 4.8 with this identification of A . $\square$

A.33 Proof of Result 4.10

Proof of Theorem 4.10. Let $\psi = (\psi_{\text{orig}}, \psi_{\text{copy}})$ be a normalized eigenvector of $H_{\pm, M}$ with eigenvalue E , where $|E| \leq E_{\max}$. As in the proof of Theorem 4.8,

$$(A - \Sigma_M(E))\psi_{\text{orig}} = E\psi_{\text{orig}}, \quad \Sigma_M(E) = C(D_M - E)^{-1}C^\dagger.$$

Now

$$(D_M - E)^{-1} - \frac{1}{M}\mathbf{1} = -\frac{1}{M}(D - E)(D_M - E)^{-1},$$

so

$$\left\| (D_M - E)^{-1} - \frac{1}{M}\mathbf{1} \right\| \leq \frac{\|D - E\|}{M} \|(D_M - E)^{-1}\| \leq \frac{\|D\| + E_{\max}}{M g_M(E_{\max})}.$$

Hence

$$\Sigma_M(E) = \frac{1}{M}CC^\dagger + \tilde{\Sigma}_M(E), \quad \|\tilde{\Sigma}_M(E)\| \leq \frac{\|C\|^2(\|D\| + E_{\max})}{M g_M(E_{\max})}.$$

Therefore

$$A - \Sigma_M(E) = A_M^{(1)} - \tilde{\Sigma}_M(E).$$

Since $E \in \text{spec}(A_M^{(1)} - \tilde{\Sigma}_M(E))$, Weyl's inequality yields (4.9). The crude bound (4.10) follows from $g_M(E_{\max}) \geq M - \|D\| - E_{\max}$ when $M > \|D\| + E_{\max}$. $\square$

A.34 Proof of Result 4.11

Proof of Theorem 4.11. From the second block equation,

$$(D_M - E)\psi_{\text{copy}} = -C^\dagger\psi_{\text{orig}}.$$

Because $g_M(E_{\max}) > 0$, the operator $D_M - E$ is invertible for every $|E| \leq E_{\max}$, so

$$\|\psi_{\text{copy}}\| \leq \|(D_M - E)^{-1}\| \|C\| \|\psi_{\text{orig}}\| \leq \frac{\|C\|}{g_M(E_{\max})} \|\psi_{\text{orig}}\|.$$

If $M > \|D\| + E_{\max}$, then $g_M(E_{\max}) \geq M - \|D\| - E_{\max}$, which gives (4.12). $\square$

A.35 Proof of Result 4.12

Proof of Theorem 4.12. Since $D_M > 0$, block Gaussian elimination gives the congruence

$$\begin{pmatrix} \mathbf{1} & -CD_M^{-1} \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} A & C \\ C^\dagger & D_M \end{pmatrix} \begin{pmatrix} \mathbf{1} & 0 \\ -D_M^{-1}C^\dagger & \mathbf{1} \end{pmatrix} = \begin{pmatrix} S_M(0) & 0 \\ 0 & D_M \end{pmatrix}.$$

Sylvester's law of inertia gives the first two identities because the lower-right block is positive. The norm estimate follows from $\|D_M^{-1}\| \leq (M - \|D\|)^{-1}$. If this norm is strictly smaller than δ_A , then the straight-line path $A - tCD_M^{-1}C^\dagger$, $0 \leq t \leq 1$, remains invertible by Weyl's inequality, so its negative spectral count is constant. Combining this with the first identity proves the final claim. $\square$

A.36 Proof of Result 4.13

Proof of Corollary 4.13. One has

$$H_{\pm, M_2} - H_{\pm, M_1} = (M_2 - M_1)P_{\text{copy}} \succeq 0.$$

The min-max principle therefore gives the monotonicity of each ordered eigenvalue. The spectral count statement is the same assertion written for distribution functions. $\square$

A.37 Proof of Result 4.14

Proof of Corollary 4.14. Write $\psi = \psi_{\text{orig}} \oplus \psi_{\text{copy}}$. Since $\psi \in \text{Ran} Q_M$, the spectral theorem gives $\|H_{\pm, M}\psi\| \leq E_{\max}\|\psi\|$. Taking the copy block of $H_{\pm, M}\psi$ yields

$$D_M\psi_{\text{copy}} + C^\dagger\psi_{\text{orig}} = P_{\text{copy}}H_{\pm, M}\psi.$$

Because $D_M \geq (M - \|D\|)\mathbf{1}$,

$$(M - \|D\|)\|\psi_{\text{copy}}\| \leq E_{\max}\|\psi\| + \|C\| \|\psi_{\text{orig}}\|,$$

which proves the quantitative estimate. If $\psi_{\text{orig}} = 0$, the same block equation gives

$$(M - \|D\|)\|\psi_{\text{copy}}\| \leq E_{\max}\|\psi_{\text{copy}}\|.$$

The strict inequality $M - \|D\| > E_{\max}$ forces $\psi_{\text{copy}} = 0$, so $\psi = 0$. Thus P_{orig} is injective on $\text{Ran} Q_M$, and the rank bound follows. $\square$

A.38 Proof of Result 4.15

Proof of Proposition 4.15. Suppose, toward a contradiction, that $E \in \text{spec}(H_{\pm, M})$ and $|E| < g_A - \Delta_M$. Then $|E| \leq g_A$, so Theorem 4.8 with $E_{\max} = g_A$ gives

$$\text{dist}(E, \text{spec}(A)) \leq \Delta_M.$$

Since $\text{spec}(A) \cap (-g_A, g_A) = \emptyset$ and $|E| < g_A$, we also have

$$\text{dist}(E, \text{spec}(A)) \geq g_A - |E| > \Delta_M,$$

a contradiction. $\square$

A.39 Proof of Result 4.16

Proof of Corollary 4.16. Apply Theorems 4.2, 4.5, 4.8, and 4.11 componentwise and take suprema over n , using the bounds $\|C_n\| \leq \bar{C}$ and $\|D_n\| \leq \bar{D}$. The full resolvent convergence follows similarly from Corollary 4.4. Since the resolvent functions $(t - z)^{-1}$ generate $C_0(\mathbb{R})$ as a C^* -algebra, the C_0 -functional calculus convergence follows from the norm convergence of the resolvents. $\square$

A.40 Proof of Result 4.17

Proof of Theorem 4.17. Set

$$\bar{C} = \sup_n \|C_{\pm, n}\|, \quad \bar{D} = \sup_n \|D_{\pm, n}\|.$$

Since $M_n \rightarrow \infty$, the full block-resolvent estimate in Corollary 4.4, applied with the uniform bounds $\bar{C}, \bar{D}$, gives

$$\|(H_{\pm, n, M_n} - z)^{-1} - ((A_{\pm, n} - z)^{-1} \oplus 0)\| \rightarrow 0$$

for every $z \in \mathbb{C} \setminus \mathbb{R}$. The algebra generated by the resolvent functions is dense in $C_0(\mathbb{R})$, so

$$\|f(H_{\pm, n, M_n}) - (f(A_{\pm, n}) \oplus 0)\| \rightarrow 0 \quad (f \in C_0(\mathbb{R})).$$

The doubled Hilbert space has dimension at most $2d_n$. Therefore

$$\left| \frac{1}{d_n} \operatorname{Tr} f(H_{\pm,n,M_n}) - \frac{1}{d_n} \operatorname{Tr} f(A_{\pm,n}) \right| \leq 2 \|f(H_{\pm,n,M_n}) - (f(A_{\pm,n}) \oplus 0)\| \longrightarrow 0.$$

It remains to compare $A_{\pm,n}$ with $H_{\pm}^{\text{PL}}(\mathcal{K}_n)$. By Theorem 3.12, their difference is $R_{\partial,\pm,n} = P_{W_n} R_{\partial,\pm,n} P_{W_n}$ for a collar subspace W_n of dimension r_n . Lemma 3.15 gives

$$\sup_{t \in \mathbb{R}} \left| \frac{1}{d_n} N_{A_{\pm,n}}(t) - \frac{1}{d_n} N_{H_{\pm}^{\text{PL}}(\mathcal{K}_n)}(t) \right| \leq \frac{r_n}{d_n} \longrightarrow 0.$$

The two sequences of operators have uniformly bounded norm by hypothesis, so their normalized spectral measures are supported in a common compact interval. Uniform convergence of the distribution functions up to an $o(1)$ error implies that the integrals of every continuous function on this compact interval have the same limit; applying this to the restriction of $f \in C_0(\mathbb{R})$ proves

$$\frac{1}{d_n} \operatorname{Tr} f(A_{\pm,n}) - \frac{1}{d_n} \operatorname{Tr} f(H_{\pm}^{\text{PL}}(\mathcal{K}_n)) \longrightarrow 0.$$

Combining the two limits gives the claimed normalized-trace equivalence. $\square$

A.41 Proof of Result 5.1

Proof of Proposition 5.1. Substituting the displayed matrices for $B(k)$ and $S(k)$ and expanding $\det(\lambda \mathbf{1} - H_{\pm}^{\Delta}(k))$ gives the stated factorization. Equivalently, with $z_j = e^{ik_j}$, clearing denominators first factors the characteristic polynomial into

$$4\lambda^2 - F(k)$$

times

$$4\lambda^4 - 26\lambda^2 + F(k) \pm 4\lambda G(k),$$

and division by the leading coefficient gives the monic form above. $\square$

A.42 Proof of Result 5.3

Proof of Theorem 5.3. Proposition 5.1 gives

$$\det H_{\pm}^{\Delta}(k) = -\frac{1}{16} (-28 + 8 \cos k_1 + 6 \cos k_2 + 6 \cos(k_1 + k_2))^2.$$

Since each cosine is at most 1,

$$-28 + 8 \cos k_1 + 6 \cos k_2 + 6 \cos(k_1 + k_2) \leq -8,$$

so the determinant is at most -4 . The absolute row sums of $H_{\pm}^{\Delta}(k)$ are bounded by 8, so $\|H_{\pm}^{\Delta}(k)\| \leq 8$. If $\lambda_1, \dots, \lambda_6$ are the eigenvalues, then $|\lambda_j| \leq 8$ and

$$4 \leq |\det H_{\pm}^{\Delta}(k)| \leq \operatorname{dist}(0, \operatorname{spec} H_{\pm}^{\Delta}(k)) 8^5,$$

which gives the stated gap. At $k = 0$, the characteristic polynomial is

$$(\lambda^2 - 2)(2\lambda^4 - 13\lambda^2 + 4)/2,$$

so exactly three eigenvalues are negative. The gap prevents the rank from changing over $\mathbb{T}^2$.

The same matrices satisfy $B(k)S(k) + S(k)B(k)^* = 0$. The stronger anticommutator $(B(k) + B(k)^*)S(k) + S(k)(B(k) + B(k)^*)$ is not used here. $\square$

A.43 Proof of Result 5.4

Proof of Corollary 5.4. Theorem 5.3 gives

$$\text{dist}(0, \text{spec} H_{\pm}^{\Delta}(k)) \geq 2^{-13} \quad (k \in \mathbb{T}^2).$$

Weyl's inequality gives

$$\text{dist}(0, \text{spec}(H_{\pm}^{\Delta}(k) + V(k))) \geq 2^{-13} - \|V(k)\| > 0.$$

The negative spectral projection therefore varies continuously in k , and its rank is constant. At $V = 0$ this rank is 3, so it remains 3 along the straight-line homotopy $H_{\pm}^{\Delta}(k) + sV(k)$, $0 \leq s \leq 1$. $\square$

A.44 Proof of Result 5.5

Proof of Proposition 5.5. The determinant identity is an exact computation from the displayed Bloch matrices. Since $c_1 + c_2 + c_{12} \leq 3$ and $2 + c_2 + c_{12} \geq 0$, the bracket is nonnegative. If the first term vanishes, then $c_1 = c_2 = c_{12} = 1$, and the second term equals $8m^2 > 0$. Thus the bracket is positive for every $m > 0$, so the determinant never vanishes. The path $(m, k) \mapsto H_{\pm, m}^{\Delta}(k)$ is norm-continuous on each compact interval $m \in [m_0, m_1] \subset (0, \infty)$. $\square$

A.45 Proof of Result 5.6

Proof of Theorem 5.6. The proof uses the orientation conventions fixed in Section 2.5. The Brillouin torus is oriented by $dk_1 \wedge dk_2$, the crossing parameter space is oriented by $dq_1 \wedge dq_2 \wedge d\mu$, and Chern numbers are computed as

$$\text{Ch}(P) = \frac{1}{2\pi i} \int_{\mathbb{T}^2} \text{tr}(P dP \wedge dP).$$

The parameter μ is the coefficient of S in

$$H_{\mu}^{\Delta}(k) = B(k) + B(k)^* + \mu S(k).$$

Proposition 5.5 shows that $H_{\mu}^{\Delta}(k)$ is gapped at zero for $\mu \neq 0$, so the Chern number can change only at the crossing $(k, \mu) = (0, 0)$.

Step 1: harmonic compression. At $k = 0$, the boundary matrices are

$$\partial_1(0) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}, \quad \partial_2(0) = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Thus $\text{rank } \partial_1(0) = 0$ and $\text{rank } \partial_2(0) = 1$. The kernel of the Hodge–Dirac operator $D_0 = B(0) + B(0)^*$ is the direct sum of the harmonic parts in degrees 0, 1, 2: one vertex vector, the two-dimensional orthogonal complement of $\text{im } \partial_2(0) \subset \mathbb{C}\langle a, b, c \rangle$, and the kernel of $\partial_2(0)$ in degree 2. With the ordered basis v, a, b, c, f, g , choose the orthonormal harmonic basis

$$h_0 = v, \quad h_1 = \frac{a - b}{\sqrt{2}}, \quad h_2 = \frac{a + b + 2c}{\sqrt{6}}, \quad h_3 = \frac{f + g}{\sqrt{2}}.$$

Let $U : \mathbb{C}^4 \rightarrow \mathcal{H}(0)$ be the isometry $Ue_j = h_{j-1}$, and let

$$\Pi_h = UU^*$$

be the harmonic projector. The complement $(1 - \Pi_h)\mathcal{H}(0)$ is separated from zero for D_0 , because D_0 is self-adjoint and has no kernel on that finite-dimensional complement.

Put $q_1 = k_1$, $q_2 = k_2$. Since $z_j = e^{iq_j} = 1 + iq_j + O(|q|^2)$, the first-order harmonic compression is

$$L(q_1, q_2, \mu) = \Pi_h(q_1 \partial_{q_1} H_0^\Delta(0) + q_2 \partial_{q_2} H_0^\Delta(0) + \mu \partial_\mu H_\mu^\Delta(0)|_{\mu=0}) \Pi_h|_{\mathcal{H}(0)}.$$

Equivalently,

$$\langle h_r, L(q_1, q_2, \mu) h_s \rangle = i(q_1 X_1 + q_2 X_2 + \mu X_3)_{r+1, s+1},$$

where the real skew-symmetric matrices are

$$X_1 = \begin{bmatrix} 0 & 1/\sqrt{2} & \sqrt{3}/2 & 0 \\ -1/\sqrt{2} & 0 & 0 & -1/2 \\ -\sqrt{3}/2 & 0 & 0 & 1/(2\sqrt{3}) \\ 0 & 1/2 & -1/(2\sqrt{3}) & 0 \end{bmatrix},$$

$$X_2 = \begin{bmatrix} 0 & -1/\sqrt{2} & \sqrt{3}/2 & 0 \\ 1/\sqrt{2} & 0 & 0 & -1/2 \\ -\sqrt{3}/2 & 0 & 0 & -1/(2\sqrt{3}) \\ 0 & 1/2 & 1/(2\sqrt{3}) & 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 0 & 0 & 0 & -\sqrt{2} \\ 0 & 0 & -1/\sqrt{3} & 0 \\ 0 & 1/\sqrt{3} & 0 & 0 \\ \sqrt{2} & 0 & 0 & 0 \end{bmatrix}.$$

These entries are obtained by differentiating the displayed Bloch matrices in Section 5 and applying the matrix elements $\langle h_r, (\partial H) h_s \rangle$. The factor i is forced by Hermiticity: the derivatives of the unitary phases at 1 are imaginary, while X_j are real skew matrices.

It remains to justify that this compression has the same local Chern jump as the full crossing. Write the full local operator, relative to $\Pi_h \mathcal{H} \oplus (1 - \Pi_h)\mathcal{H}$, as

$$\begin{pmatrix} A(x) & K(x) \\ K(x)^* & G(x) \end{pmatrix}, \quad x = (q_1, q_2, \mu).$$

Here $A(x) = L(x) + O(|x|^2)$, $K(x) = O(|x|)$, and $G(0)$ is invertible. Shrinking to $|x| \leq r$, $G(x)$ remains invertible and

$$A(x) - K(x)G(x)^{-1}K(x)^* = L(x) + O(|x|^2).$$

On the sphere $|x| = r$, the linear model below is invertible with a lower bound cr for some $c > 0$. Choose r so that the quadratic error has norm at most $cr/2$. Then the straight-line homotopy from the Schur complement to $L(x)$ is invertible on the sphere, and the complementary block remains invertible throughout. Therefore the full crossing and the displayed harmonic model have the same linking Chern number.

Step 2: self-dual and anti-self-dual splitting. Identify a real skew 4×4 matrix with a two-form by the ordered basis

$$e_{12}, e_{13}, e_{14}, e_{23}, e_{24}, e_{34}$$

for $\Lambda^2 \mathbb{R}^4$, where $e_{rs} = e_r \wedge e_s$ and $e_1 \wedge e_2 \wedge e_3 \wedge e_4$ is the positive orientation. The Hodge involution is

$$*e_{12} = e_{34}, \quad *e_{13} = -e_{24}, \quad *e_{14} = e_{23},$$

together with the same equations read backwards. Hence the self-dual basis is

$$\omega_1^+ = \frac{e_{12} + e_{34}}{\sqrt{2}}, \quad \omega_2^+ = \frac{e_{13} - e_{24}}{\sqrt{2}}, \quad \omega_3^+ = \frac{e_{14} + e_{23}}{\sqrt{2}},$$

and the anti-self-dual basis is

$$\omega_1^- = \frac{e_{12} - e_{34}}{\sqrt{2}}, \quad \omega_2^- = \frac{e_{13} + e_{24}}{\sqrt{2}}, \quad \omega_3^- = \frac{e_{14} - e_{23}}{\sqrt{2}}.$$

For

$$X(q_1, q_2, \mu) = q_1 X_1 + q_2 X_2 + \mu X_3,$$

write $X = X_+(u) + X_-(\beta)$ in these bases. Direct projection onto the ω_j^+ coordinates gives

$$u(q_1, q_2, \mu) = \left(\frac{1 + 1/\sqrt{6}}{2} (q_1 - q_2), \left(\frac{\sqrt{3}}{2} + \frac{1}{2\sqrt{2}} \right) (q_1 + q_2), -(1 + 1/\sqrt{6})\mu \right).$$

The anti-self-dual vector is a scalar multiple of the same coordinate vector,

$$\beta(q_1, q_2, \mu) = \rho u(q_1, q_2, \mu), \quad \rho = \frac{\sqrt{6} - 1}{\sqrt{6} + 1} < 1.$$

For a skew form with self-dual and anti-self-dual coordinates (α, β) , the Hermitian matrix $iX(\alpha, \beta)$ has eigenvalues

$$\pm(|\alpha| + |\beta|), \quad \pm(|\alpha| - |\beta|).$$

Along the homotopy $(\alpha, \beta) = (u, t\rho u)$, $0 \leq t \leq 1$, the smallest absolute eigenvalue is $(1 - t\rho)|u|$. This is positive on every linking sphere because $\rho < 1$ and $u \neq 0$ there. Thus the anti-self-dual part can be removed without changing the local Chern jump.

The self-dual coordinate map is linear with Jacobian

$$\frac{\partial u}{\partial(q_1, q_2, \mu)} = \begin{bmatrix} a & -a & 0 \\ b & b & 0 \\ 0 & 0 & -c \end{bmatrix}, \quad a = \frac{1 + 1/\sqrt{6}}{2}, \quad b = \frac{\sqrt{3}}{2} + \frac{1}{2\sqrt{2}}, \quad c = 1 + \frac{1}{\sqrt{6}}.$$

Therefore

$$\det \frac{\partial u}{\partial(q_1, q_2, \mu)} = -2abc < 0.$$

With the crossing orientation $dq_1 \wedge dq_2 \wedge d\mu$, the self-dual local model is orientation-reversing.

Step 3: Chern jump and FHS sign convention. Under the standard spinor identification of self-dual two-forms, $iX_+(u)$ is the direct sum of two identical Pauli Hamiltonians $u \cdot \sigma$. For the identity crossing $u = (q_1, q_2, \mu)$, the negative line over the linking sphere has Chern number +1 with the boundary orientation induced by $dq_1 \wedge dq_2 \wedge d\mu$. Pullback by a linear map multiplies this number by the degree of that map. Since $\det Du < 0$, one Pauli block contributes -1 , and two identical blocks contribute

$$\text{Ch}(E_-(H_{\mu>0}^\Delta)) - \text{Ch}(E_-(H_{\mu<0}^\Delta)) = -2.$$

The FHS convention used in Tables 3 and 4 orders each plaquette by the positively oriented coordinate square in the k_1, k_2 directions. Its lattice integer therefore approximates the same $(2\pi i)^{-1} \int \text{tr}(P dP \wedge dP)$ used here; no sign conversion is inserted between the proof and the tables.

Finally,

$$H_{-\mu}^\Delta(k) = \overline{H_\mu^\Delta(-k)}.$$

The base involution $k \mapsto -k$ has Jacobian determinant $+1$ on the two-torus, so it preserves the orientation $dk_1 \wedge dk_2$. Complex conjugation of a complex vector bundle reverses the first Chern class. Hence

$$\text{Ch}(E_-(H_{-\mu}^\Delta)) = -\text{Ch}(E_-(H_\mu^\Delta)).$$

Combining this symmetry with the jump -2 gives

$$\text{Ch}(E_-(H_\mu^\Delta)) = -1 \quad (\mu > 0), \quad \text{Ch}(E_-(H_\mu^\Delta)) = +1 \quad (\mu < 0).$$

Taking $\mu = 1$ gives the stated signs for H_+^Δ and H_-^Δ . $\square$

A.46 Proof of Result 5.7

Proof of Corollary 5.7. The positive-mass statement follows from Proposition 5.5 and Theorem 5.6: changing $m > 0$ gives a norm-continuous gapped homotopy for $H_{+,m}^\Delta$, and $H_{-,m}^\Delta = H_{-m}^\Delta$ is handled by the same argument on the negative-mass side. Thus the Chern number is unchanged. The number $g_{\pm,m}$ is positive by compactness of $\mathbb{T}^2$ and the positive-mass gap. If $\|V\|_\infty < g_{\pm,m}$, Weyl's inequality keeps the straight-line homotopy $H_{\pm,m}^\Delta + sV$ gapped at zero, so the Chern number is unchanged along it.

For the uniform perturbation statement at $m = 1$, the straight-line homotopy

$$H_{\pm,s}^\Delta(k) = H_\pm^\Delta(k) + sV(k), \quad 0 \leq s \leq 1,$$

is gapped by Theorem 5.3 and Weyl's inequality:

$$\text{dist}(0, \text{spec} H_{\pm,s}^\Delta(k)) \geq 2^{-13} - s \sup_{q \in \mathbb{T}^2} \|V(q)\| > 0.$$

The Chern number of the negative spectral projection is constant along this gapped homotopy. $\square$

A.47 Proof of Result A.23

Proof of Corollary A.23. The map $p_A : \mathbb{T}^2 \rightarrow \mathbb{T}^2$, $p_A(\kappa) = A\kappa$, is a finite covering of degree $\det A$ with sign [15]. The pulled-back Hamiltonian has the same fiber spectra as the original one, so the gap is unchanged. Its negative Bloch bundle is $p_A^*E_-(H_\pm^\Delta)$, and naturality of the Chern form gives

$$\text{Ch}(p_A^*E) = \deg(p_A) \text{Ch}(E) = \det A \text{Ch}(E).$$

The displayed signs follow from Theorem 5.6; the positive-mass and small-perturbation statements follow in the same way from Corollary 5.7.

For a finite-index sublattice $\Lambda' \subset \mathbb{Z}^2$, the folded Bloch fiber over the smaller Brillouin torus is the direct sum of the original fibers over the finitely many momenta lying above it. The Chern form of this direct sum is the sum of the pulled-back Chern forms over the sheets, and integration over the smaller Brillouin torus gives exactly the integral over the original Brillouin torus. The gap is the minimum of the same original fiber gaps. Thus changing the period cell does not change the Chern number of the underlying periodic operator. $\square$

A.48 Proof of Result A.24

Proof of Corollary A.24. The finite periodic torus diagonalizes by the discrete Bloch transform. Its fiber momenta are points of $\mathbb{T}^2$, so the gap bound follows from Theorem 5.3. The Chern data are thermodynamic Bloch-bundle data proved in Theorem 5.6 and numerically checked in Table 3; they are not finite-volume invariants of the discrete sample alone. $\square$

A.49 Proof of Result A.25

Proof of Corollary A.25. The exact gap, finite-range symbol, and analytic Chern calculation provide the locality, norm, closed-bulk gap, and closed-bulk Chern inputs needed in Assumptions 6.8 and 6.13, and they supply the periodic closed-bulk Chern input used here in place of the Higson–Prodan-specific input from Assumption 6.15. With the gap and sidewise inputs fixed, the side-wise and quotient-interface conclusions are exactly Theorem 6.35. The last sentence is the same interface-pairing evaluation used in Theorem 6.38, with the periodic Chern numbers supplied by Theorem 5.6. $\square$

A.50 Proof of Result A.26

Proof of Corollary A.26. For $H_{\pm,m}^{\Delta}$, the gap follows from Proposition 5.5 and the Chern signs from Corollary 5.7. The fixed- m perturbation case and the $m = 1$ explicit-radius perturbation case are also exactly Corollary 5.7. Finite-cover and finite-index cell choices are Corollary A.23. Finite-range periodicity gives the same uniform locality and bounded-geometry hypotheses as in the unperturbed periodic annular family, after replacing the fixed collar by the enlarged collar in the statement. The proofs of Theorem 6.35, Corollary A.40, and Theorem A.39 then apply verbatim with $H_{\pm}^{\infty}$ replaced by the chosen finite-range periodic bulk family: the only inputs used there are uniform locality, boundedness, the closed-bulk gap, sidewise agreement away from the collar, and positivity of the copy-side comparison. The final winding statement is the same compatible interface-pairing evaluation used in Corollary A.25. $\square$

A.51 Proof of Result A.27

Proof of Corollary A.27. The unperturbed case is Corollary A.25. The positive-mass, small finite-range perturbation, finite-cover, and cell-enlargement cases are Corollary A.26. Independence of the scalar jam scale is Corollary A.40; bounded componentwise scalar jams and nonconstant positive copy-side confinements are covered by Theorem A.39 and Corollary A.38. Changing finitely many initial components changes the data by a finite-component perturbation, so Proposition A.35 applies. Uniformly gapped sidewise homotopies are covered by Proposition A.37. The last sentence is the compatible interface-pairing evaluation already isolated in Corollaries A.25 and A.26. $\square$

A.52 Proof of Result 5.8

Proof of Theorem 5.8. The proof is the standard Toeplitz calculation, included here to make the periodic family independent of the general Ewert–Meyer compatibility hypothesis. The extension above is the external tensor product of $C(S^1)$, in the tangential variable k_1 , with the one-variable Toeplitz extension in the normal variable k_2 . Hence its boundary map is a group homomorphism

$$K^0(\mathbb{T}^2) \longrightarrow K^1(S^1).$$

The group $K^0(\mathbb{T}^2)$ is generated by the trivial line bundle and one Bott class [18, 19]. A constant projection lifts to a constant projection in $\mathcal{T}_{\text{per}}$, so the boundary map vanishes on the rank summand.

It remains to compute the Bott generator. Use the clutching description of a Chern-one line bundle: split the k_2 -circle into upper and lower arcs, trivialize the bundle over the two cylinders $S_{k_1}^1 \times I_{\pm}$, and take transition function $z(k_1) = e^{ik_1}$ on one overlap component and 1 on the other. The first Chern number is the winding difference of these two transition functions, hence 1 with

orientation $dk_1 \wedge dk_2$. In the Toeplitz extension, lifting the projection to the two Toeplitz charts and comparing the lifts on the overlap gives exactly this boundary unitary z . Therefore

$$\partial_T([\text{Bott}]) = [z] \in K_1(C(S^1)), \quad \text{wind}(z) = 1.$$

By additivity and the standard $K^0(\mathbb{T}^2)$ decomposition, any projection class decomposes as

$$[P] = \text{rank}(P)[1] + \text{Ch}(P)[\text{Bott}] \quad \text{in } K^0(\mathbb{T}^2),$$

and applying ∂_T gives

$$\partial_T([P]) = \text{Ch}(P)[z].$$

Pairing with the standard winding homomorphism on $K_1(C(S^1))$ gives the claimed equality. $\square$

A.53 Proof of Result 5.9

Proof of Corollary 5.9. The map $(\kappa, \eta) \mapsto \kappa t + \eta n$ is a torus automorphism. Pulling H back by this automorphism preserves finite range, self-adjointness, and the zero gap, so Theorem 5.8 applies in the (κ, η) -coordinates. The first Chern number of the pulled-back projection is multiplied by the degree of the torus automorphism, namely $\det[t \ n]$. Therefore the coordinate Toeplitz winding equals $\varepsilon(t, n) \text{Ch}(P)$. $\square$

A.54 Proof of Result A.28

Proof of Corollary A.28. The map $p_A : \mathbb{T}^2 \rightarrow \mathbb{T}^2$, $p_A(\kappa) = A\kappa$, has signed degree $\det A$ [15], and $P_A = p_A^* P$. Hence

$$\text{Ch}(P_A) = \det A \text{Ch}(P).$$

Applying Corollary 5.9 to P_A in the primitive cover coordinates gives the displayed formula. The triangular values are Theorem 5.6 and Corollaries 5.7 and A.23. $\square$

A.55 Proof of Result A.29

Proof of Corollary A.29. The uniform gap gives a norm-continuous Riesz projection path $s \mapsto P_s$ in $C(\mathbb{T}^2, M_N)$. Hence $[P_s] \in K_0(C(\mathbb{T}^2))$ is constant. Naturality of the Toeplitz connecting map gives a constant class $\partial_{T,t,n}([P_s])$, and Corollary 5.9 evaluates its winding. The common finite hopping range is the locality condition needed to realize the whole homotopy by the same straight periodic edge extension. $\square$

A.56 Proof of Result A.30

Proof of Theorem A.30. For a direct sum of extensions, the six-term exact sequence is natural and the connecting map is the direct sum of the connecting maps of the summands. Hence the boundary map for the split edge ideal sends the quotient projection class to

$$\bigoplus_{j=1}^r \partial_{T,t_j,n_j}([P]).$$

The j -th coordinate projection

$$K_1 \left(\bigoplus_{m=1}^r C^*(Y^{(m)} \subset X) \right) \longrightarrow K_1(C^*(Y^{(j)} \subset X)) \cong K_1(C(S^1))$$

therefore recovers exactly the j -th Toeplitz class. Applying Corollary 5.9 gives the displayed component winding. Reversing the normal direction changes the sign of the Toeplitz extension for a fixed tangential coordinate; the induced boundary orientation reverses the tangential coordinate at the same time, which multiplies the winding homomorphism by the opposite sign and gives the final statement. $\square$

A.57 Proof of Result A.19

Proof of Theorem A.19. Naturality of the six-term exact sequence for the quotient map of extensions [19] gives

$$q_* \partial_{\text{poly}}([P]) = \partial_{\oplus T}([P]),$$

where $\partial_{\oplus T}$ is the connecting map for the quotient extension obtained after deleting the finite corner ideal. By hypothesis this quotient extension is the direct sum of the r primitive straight Toeplitz extensions, so

$$\partial_{\oplus T}([P]) = \bigoplus_{j=1}^r \partial_{T, t_j, n_j}([P]).$$

Since $\mathcal{F}$ is finite-dimensional, $K_1(\mathcal{F}) = 0$. Exactness of the six-term sequence for

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}_{\text{poly}} \xrightarrow{q} \bigoplus_j \mathcal{I}_j \rightarrow 0$$

therefore makes q_* injective on $K_1(\mathcal{I}_{\text{poly}})$. Applying the j -th coordinate projection and then the standard winding homomorphism gives Corollary 5.9, hence the displayed formula. $\square$

A.58 Proof of Result A.20

Proof of Corollary A.20. The preceding proof used finite-dimensionality only to obtain $K_1(\mathcal{F}) = 0$. With that condition assumed directly, exactness of

$$0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}_{\text{poly}} \xrightarrow{q} \bigoplus_j \mathcal{I}_j \rightarrow 0$$

again makes

$$q_* : K_1(\mathcal{I}_{\text{poly}}) \rightarrow K_1\left(\bigoplus_j \mathcal{I}_j\right)$$

injective, and naturality identifies the quotient boundary class with the direct sum of the straight Toeplitz classes. Finite-dimensional algebras and c_0 -sums of them have vanishing K_1 . For a bounded product of finite-dimensional algebras, every unitary has a bounded self-adjoint logarithm with norm at most π in each finite-dimensional factor, so the unitary group is path connected after stabilization and K_1 again vanishes. $\square$

A.59 Proof of Result A.21

Proof of Corollary A.21. For an algebraic finite-propagation operator T supported near Y , condition (ii) implies that all matrix coefficients coupling two different edge pieces are supported inside a bounded neighborhood of C . Modulo the corner ideal, T is therefore the direct sum of its edge restrictions. Condition (iii) identifies these restrictions with elements of the Toeplitz edge ideals and gives the quotient map q . Extension by zero from the separated edge collars gives surjectivity

of q , again modulo corner-supported terms. If all edge restrictions vanish, then the support of T is contained in a bounded neighborhood of C , so the kernel is exactly $\mathcal{F}_C$ after taking norm closures. The final statement is Corollary A.20. $\square$

A.60 Proof of Result 5.10

Proof of Corollary 5.10. Corollary A.27 identifies the doubled-and-jammed straight-interface class with the boundary image of the periodic bulk Fermi projection. For a straight periodic component, this boundary image is exactly the Toeplitz boundary map in Theorem 5.8; for a primitive direction it is the lattice-change version in Corollary 5.9. The Chern signs of the unperturbed triangular model, the positive-duality-strength models, and the allowed small periodic perturbations are supplied by Theorem 5.6 and Corollary 5.7; finite-cover and finite-index cell variants are Corollary A.23. Substituting those signs into Theorem 5.8 gives the displayed winding values.

The invariance under scalar jams, componentwise scalar jams, positive copy-side confinement, finite initial-component changes, and uniformly gapped sidewise homotopies is exactly the robustness statement in Corollary A.27; norm-vanishing component defects are Proposition A.36. The split componentwise statement is Theorem A.30; the finite-corner version is Theorem A.19. $\square$

A.61 Proof of Result A.17

Proof of Corollary A.17. Parts (a) and (b) preserve the closed-bulk Chern signs by Corollary 5.7. A fixed-collar perturbation in part (c) belongs to the straight-edge interface ideal, so Proposition A.34 and Proposition A.34 leave the interface class unchanged whenever the sidewise comparison gaps remain open. Part (d) is Theorem A.39 together with Corollary A.38, and part (e) is Propositions A.35, A.36, and A.37. For part (f), Proposition 6.6 identifies the two fixed-collar ideals. For part (g), a finite tangential defect inside a fixed edge collar is finite rank, and every bounded finite-propagation perturbation supported in a fixed collar belongs to the algebraic interface ideal; the invariance follows from Propositions A.34 and A.34. Part (h) is Proposition A.35 on the deleted components, followed by Theorem A.30, Theorem A.19, or Corollary A.20 on the tail. The Toeplitz winding homomorphism, and each component coordinate winding in the split or finite-corner split case, depends only on the resulting K_1 -class, so the winding values remain those of Corollary 5.10. $\square$

A.62 Proof of Result A.18

Proof of Theorem A.18. The equality of ideals is Proposition 6.6. Since the sidewise differences belong to this common ideal, the quotient images of J_i and $T_i^\pm$ are the same. The proof of Proposition A.34 then gives the displayed equality of quotient Fermi projections and of p_{J_i} ; applying the connecting map gives equality of interface classes. The class for the straight tail is evaluated by Corollary 5.9. A perturbation supported in a finite tangential interval and a fixed normal collar acts on only finitely many lattice degrees of freedom, hence is finite rank and in particular belongs to $\mathcal{I}$. An eventually periodic collar differs from its periodic tail exactly by such a transition defect. $\square$

A.63 Proof of Result A.31

Proof of Theorem A.31. Deleting or changing finitely many initial components changes the sidewise data by an operator in the interface ideal by Proposition A.35; hence the full interface class and the tail class have the same image under the quotient that discards those components. On the tail, the split case is Theorem A.30. The finite-corner case is Theorem A.19, and the K_1 -trivial

version is Corollary A.20. In all cases, applying the j -th coordinate projection and the winding homomorphism gives $\det[t_j \ n_j] \text{Ch}(P)$. The finite-cover formula is Corollary A.28. $\square$

A.64 Proof of Result A.32

Proof of Corollary A.32. The zero gap and Chern signs for (a)–(c) are Theorem 5.3, Theorem 5.6, and Corollary 5.7; finite covers and cell enlargements are Corollary A.23. The straight-edge winding is Theorem 5.8 in coordinate directions and Corollary 5.9 in arbitrary primitive directions. Finite split sums are Theorem A.30; finite-corner split sums are Theorem A.19 and Corollary A.20; the collar-separation criterion is Corollary A.21. Bounded-distance and eventually periodic collars are Theorem A.18; finite-defect and tail-split variants are Corollary A.17 and Theorem A.31. The scalar-jam, componentwise-jam, and positive copy-confinement invariance follows from Corollary 5.10 and Corollary A.17. Norm-vanishing component defects are Proposition A.36. The final sentence is the fixed-collar and homotopy part of Corollary A.17. $\square$

A.65 Proof of Result A.5

Proof of Lemma A.5. At $k = (0, 0)$, $a_1 = a_2 = 0$ and $a_3^{\text{hex}} = 1$. Thus

$$H_{\text{hex}}(0, 0; 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

so $\det H_{\text{hex}}(0, 0; 0) = 0$ and zero belongs to the spectrum. $\square$

A.66 Proof of Result A.6

Proof of Lemma A.6. The trace of $H_{\text{hex}}(k; m)$ is zero, and a direct 2×2 determinant calculation gives

$$\det H_{\text{hex}}(k; m) = -\sin^2 k_1 - \sin^2 k_2 - m^2(\cos k_1 + \cos k_2 - 1)^2.$$

Set $s = \sin^2 k_1 + \sin^2 k_2$. If $s \geq 1/4$, then $|\det H_{\text{hex}}(k; m)| \geq 1/4$. If $s < 1/4$, then $|\sin k_j| < 1/2$, hence $|\cos k_j| > \sqrt{3}/2$, for $j = 1, 2$. Checking the four sign choices for $(\cos k_1, \cos k_2)$ gives

$$|\cos k_1 + \cos k_2 - 1| \geq \sqrt{3} - 1.$$

Therefore

$$|\det H_{\text{hex}}(k; m)| \geq \min \left\{ \frac{1}{4}, (\sqrt{3} - 1)^2 m^2 \right\}.$$

Since the eigenvalues of a trace-zero 2×2 Hermitian matrix are $\pm \sqrt{|\det H_{\text{hex}}(k; m)|}$, the stated spectral gap follows. $\square$

A.67 Proof of Result A.7

Proof of Lemma A.7. For $m > 0$, the path replacing m by $(1 - s)m + s$, $0 \leq s \leq 1$, stays in the positive-mass region and is gapped by Lemma A.6. Hence the Chern number equals the value at $m = 1$. Write

$$d_{\text{hex}}(k) = (\sin k_1, \sin k_2, \cos k_1 + \cos k_2 - 1), \quad \hat{d}_{\text{hex}} = d_{\text{hex}}/|d_{\text{hex}}|.$$

For a two-band Hamiltonian $d \cdot \sigma$, the negative-energy Chern number is

$$\text{Ch}(E_-) = -\frac{1}{4\pi} \int_{\mathbb{T}^2} \hat{d} \cdot (\partial_{k_1} \hat{d} \times \partial_{k_2} \hat{d}) dk_1 dk_2 = -\deg(\hat{d}),$$

with the same convention used in the Bloch-bundle formula; see [1, Chapters 4–5] and the Bellissard–van Elst–Schulz-Baldes formulation of the Chern character [2].

The north pole $(0, 0, 1)$ is a regular value. Its preimages satisfy $\sin k_1 = \sin k_2 = 0$ and $\cos k_1 + \cos k_2 - 1 > 0$. The only such point is $(0, 0)$. The local degree there is

$$\operatorname{sgn} \det \begin{pmatrix} \cos k_1 & 0 \\ 0 & \cos k_2 \end{pmatrix}_{(0,0)} = +1.$$

Thus $\deg(\widehat{d}_{\text{hex}}) = 1$, and the negative-energy Chern number is -1 . $\square$

A.68 Proof of Result A.8

Proof of Lemma A.8. Let $U = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. Since a_1 and a_2 are odd under $k \mapsto -k$, while a_3^{hex} is even,

$$H_{\text{hex}}(k; -m) = U \overline{H_{\text{hex}}(-k; m)} U^*.$$

The base involution $k \mapsto -k$ preserves the orientation $dk_1 \wedge dk_2$. Unitary conjugation preserves the Bloch line bundle, while complex conjugation changes the sign of its first Chern class. Therefore the negative-energy Chern number at $-m$ is the negative of the value at $m > 0$. By Lemma A.7, it is $+1$. $\square$

A.69 Proof of Result A.9

Proof of Lemma A.9. The entries of $H_{\text{hex}}(k; m)$ are Laurent polynomials in e^{ik_1} and e^{ik_2} , so the symbol is finite range. By Lemma A.6, it is gapped at zero for $m \neq 0$. The straight half-space Toeplitz extension therefore applies. The coordinate statement is Theorem 5.8, and the primitive-basis statement is Corollary 5.9. Substituting Lemmas A.7 and A.8 gives the displayed signs. $\square$

A.70 Proof of Result A.10

Proof of Theorem A.10. The gap estimate is Lemma A.6. The Chern value is Lemma A.7 for $m > 0$ and Lemma A.8 for $m < 0$, which is the formula $-\operatorname{sgn}(m)$. The Toeplitz formula is Lemma A.9. $\square$

A.71 Proof of Result A.12

Proof of Lemma A.12. At $k = (0, 0)$, $b_1 = b_2 = 0$ and $b_3^{\square} = 3$. Hence

$$H_{\square}(0, 0; 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

so $\det H_{\square}(0, 0; 0) = 0$. $\square$

A.72 Proof of Result A.13

Proof of Lemma A.13. Again the matrix is trace zero, and

$$\det H_{\square}(k; m) = -\sin^2 k_1 - \sin^2 k_2 - m^2(1 + \cos k_1 + \cos k_2)^2.$$

If $s = \sin^2 k_1 + \sin^2 k_2 \geq 1/4$, then $|\det H_{\square}(k; m)| \geq 1/4$. If $s < 1/4$, then $|\cos k_j| > \sqrt{3}/2$, $j = 1, 2$, and the four possible sign choices give

$$|1 + \cos k_1 + \cos k_2| \geq \sqrt{3} - 1.$$

This gives the lower bound by $c_\square(m)$. The eigenvalues are $\pm\sqrt{|\det H_\square(k; m)|}$, so the distance from zero to the spectrum is at least $\sqrt{c_\square(m)}$. $\square$

A.73 Proof of Result A.14

Proof of Lemma A.14. For $m > 0$, the positive-mass path to $m = 1$ stays gapped by Lemma A.13. At $m = 1$, set

$$d_\square(k) = (\sin k_1, \sin k_2, 1 + \cos k_1 + \cos k_2), \quad \widehat{d}_\square = d_\square/|d_\square|.$$

As in the two-band Chern formula cited in the proof of Lemma A.7, $\text{Ch}(E_-) = -\deg(\widehat{d}_\square)$. The north-pole preimages are

$$(0, 0), \quad (\pi, 0), \quad (0, \pi),$$

since $1 + \cos k_1 + \cos k_2 > 0$ at exactly those three points among the four points with $\sin k_1 = \sin k_2 = 0$. The local signs are respectively

$$\text{sgn}(\cos k_1 \cos k_2) = +1, -1, -1.$$

Thus $\deg(\widehat{d}_\square) = -1$, and the negative-energy Chern number is $+1$. $\square$

A.74 Proof of Result A.15

Proof of Lemma A.15. The entries of $H_\square(k; m)$ are Laurent polynomials, and Lemma A.13 gives a zero gap for $m \neq 0$. The straight-edge Toeplitz index formula in Theorem 5.8 and the primitive-coordinate version Corollary 5.9 give

$$\text{wind}(\partial_{T,t,n}([P_{\square,m}])) = \det[t \ n] \text{Ch}(P_{\square,m}).$$

For $m > 0$, Lemma A.14 gives $+1$. For $m < 0$, the same antiunitary identity used in Lemma A.8, with $b_3^\square(-k) = b_3^\square(k)$, changes the Chern sign. This gives -1 . $\square$

A.75 Proof of Result A.16

Proof of Theorem A.16. Lemma A.13 gives the zero gap. Lemma A.14 gives the Chern number $+1$ for $m > 0$, and the antiunitary mass-reversal argument in the proof of Lemma A.15 gives -1 for $m < 0$. The Toeplitz winding formula is Lemma A.15. $\square$

A.76 Proof of Result 6.4

Proof of Lemma 6.4. At the algebraic level, operators supported within a bounded neighborhood of $Y^{(R)}$ are stable under addition and adjoint. They are also stable under multiplication on either side by bounded finite-propagation operators in $C^*(X)$, because finite propagation enlarges the support neighborhood by at most a bounded amount. Taking norm closure gives a closed two-sided ideal. $\square$

A.77 Proof of Result 6.5

Proof of Proposition 6.5. Assume $R' \geq R$. Then

$$Y^{(R)} \subset Y^{(R')} \subset N_{R'}(Y^{(R)}).$$

Thus an operator supported within a bounded neighborhood of $Y^{(R')}$ is also supported within a bounded neighborhood of $Y^{(R)}$, and the converse inclusion is immediate from $Y^{(R)} \subset Y^{(R')}$. The corresponding algebraic ideals therefore coincide, and taking norm closures gives the equality. The case $R > R'$ is symmetric. $\square$

A.78 Proof of Result 6.6

Proof of Proposition 6.6. Let T be an algebraic Roe operator whose matrix support is contained in $N_s(Y_0) \times N_s(Y_0)$ for some $s < \infty$. Since $Y_0 \subset N_R(Y_1)$, the same matrix support is contained in $N_{s+R}(Y_1) \times N_{s+R}(Y_1)$. Hence the algebraic interface ideal for Y_0 is contained in the algebraic interface ideal for Y_1 . The reverse inclusion is identical, using $Y_1 \subset N_R(Y_0)$. Taking norm closures gives the equality. $\square$

A.79 Proof of Result 6.10

Proof of Lemma 6.10. The propagation bound is exactly Assumption 6.8(i). For the norm bound, every row sum and every column sum of absolute values is at most $N_S c_S$ by Assumption 6.8(ii)–(iii). The Schur test therefore gives

$$\|S_n\| \leq N_S c_S.$$

$\square$

A.80 Proof of Result 6.11

Proof of Proposition 6.11. By the Higson–Prodan construction, $B + B^\dagger$ maps a simplex barycenter to barycenters of simplices that share an incidence relation. In the barycentric graph metric, every such pair is at distance 1. Hence $B + B^\dagger$ has propagation exactly 1 on every $\ell^2(X_n)$. The operator S is handled by Lemma 6.10. Since $H_{\pm,n} = (B + B^\dagger) \pm S$, the propagation satisfies

$$\text{prop}(H_{\pm,n}) \leq \max\{1, R_S\},$$

which is bounded uniformly in n . This proves (a).

For (b), $\|B + B^\dagger\|$ is bounded by twice the maximum simplex degree in the barycentric graph. Under uniform regularity (Definition 6.2), the ball sizes $|B_{X_n}(x, 1)|$ are uniformly bounded, so the maximum degree is uniformly bounded. Similarly, $\|S_n\|$ is bounded uniformly in n by Lemma 6.10. Hence

$$\|H_{\pm,n}\| \leq \|B + B^\dagger\| + \|S_n\| \leq B_{\text{bulk}}$$

for a constant B_{bulk} independent of n . $\square$

A.81 Proof of Result 6.17

Proof of Theorem 6.17. Assumption 6.15 includes uniform regularity of $(\mathcal{K}_n)_n$, and its part (a) gives Assumption 6.8. Its part (b) gives the uniform norm bound and the uniform spectral gap in Assumption 6.13; the propagation bound follows from Proposition 6.11. The identification of $P_\pm$ with the Higson–Prodan Fermi projection and the Chern computation are precisely part (c) of the imported closed-bulk input. $\square$

A.82 Proof of Result 6.18

Proof of Proposition 6.18. Boundedness and self-adjointness are immediate from the direct-sum construction and Assumption 6.13(b). The propagation bound in Assumption 6.13(a) passes to the direct sum, so $H_\pm^\infty$ has propagation at most R_{loc} on $\ell^2(X)$.

It remains to verify local compactness. Let $F \subset X$ be finite and write χ_F for multiplication by its characteristic function. Because F is finite and $H_\pm^\infty$ has propagation at most R_{loc} , every

vector in the range of $\chi_F H_\pm^\infty$ is supported inside the finite set $N_{R_{\text{loc}}}(F)$. Hence $\chi_F H_\pm^\infty$ has finite-dimensional range and is therefore finite rank. The same argument applied to the adjoint shows that $H_\pm^\infty \chi_F$ is finite rank as well. Thus $H_\pm^\infty$ is locally compact and belongs to the algebraic Roe algebra; therefore $H_\pm^\infty \in C^*(X)$ after taking norm closure.

By Assumption 6.13(b),

$$\text{spec}_{\text{op}}(H_\pm^\infty) \subset [-B_{\text{bulk}}, B_{\text{bulk}}],$$

and by Assumption 6.13(c),

$$\text{spec}_{\text{op}}(H_\pm^\infty) \subset (-\infty, -g_{\text{bulk}}] \cup [g_{\text{bulk}}, \infty).$$

Choose a compactly supported continuous function $f \in C_c(\mathbb{R}) \subset C_0(\mathbb{R})$ such that

$$f(t) = 1 \quad \text{for } t \in [-B_{\text{bulk}}, -g_{\text{bulk}}/2], \quad f(t) = 0 \quad \text{for } t \in [g_{\text{bulk}}/2, B_{\text{bulk}}],$$

and, for example, $f(t) = 0$ for $|t| \geq B_{\text{bulk}} + 1$. Then f agrees with $\chi_{(-\infty, 0)}$ on $\text{spec}_{\text{op}}(H_\pm^\infty)$ (which avoids the interval $(-g_{\text{bulk}}, g_{\text{bulk}})$), so the operator functional calculus on $\ell^2(X)$ gives

$$f(H_\pm^\infty) = \chi_{(-\infty, 0)}(H_\pm^\infty) = P_\pm.$$

Since $H_\pm^\infty \in C^*(X)$ and $f \in C_0(\mathbb{R})$, continuous functional calculus in $C^*(X)$ gives $f(H_\pm^\infty) \in C^*(X)$. Hence $P_\pm \in C^*(X)$, and so $P_\pm$ defines a class

$$[P_\pm] \in K_0(C^*(X)).$$

□

A.83 Proof of Result 6.19

Proof of Proposition 6.19. Choose one function $f \in C_c(\mathbb{R})$ such that

$$f(t) = 1 \quad \text{for } t \in [-B, -g/2], \quad f(t) = 0 \quad \text{for } t \in [g/2, B].$$

Then $P_s = f(H_s)$ for every s . Continuous functional calculus is norm-continuous for a fixed continuous function on a norm-continuous path of self-adjoint elements, so $s \mapsto P_s$ is a norm-continuous path of projections in $C^*(X)$. A norm-continuous path of projections represents a constant K_0 -class. □

A.84 Proof of Result 6.20

Proof of Proposition 6.20. For $z \in \Gamma$,

$$\|(H - z)^{-1}\| \leq \delta_\Gamma^{-1}.$$

Since

$$\|V(H - z)^{-1}\| \leq \frac{\|V\|}{\delta_\Gamma} < 1,$$

the operator $H + V - z$ is invertible and

$$\|(H + V - z)^{-1}\| \leq \frac{1}{\delta_\Gamma - \|V\|}.$$

The resolvent identity gives

$$(H + V - z)^{-1} - (H - z)^{-1} = -(H + V - z)^{-1}V(H - z)^{-1}.$$

Integrating this identity around Γ gives the stated projection estimate. □

A.85 Proof of Result A.33

Proof of Corollary A.33. For $t \in [0, 1]$, set $H_t := H + tV$. Weyl's spectral-stability estimate gives

$$\text{spec}_{\text{op}}(H_t) \cap (-(g - t\|V\|), g - t\|V\|) = \emptyset.$$

Since $g - \|V\| > 0$, the path H_t is uniformly gapped at zero. Proposition 6.19 therefore applies, with spectral bound $B + \|V\|$ and gap $g - \|V\|$, and gives equality of the endpoint Fermi-projection classes. In the interface case, the path

$$s \mapsto \chi_{(-\infty, 0)}(H_{\pm}^{\infty} + sV)$$

is a norm-continuous path of projections joining $P_{\pm}$ to P_V . Applying Proposition 6.32 gives the stated equality. $\square$

A.86 Proof of Result 6.21

Proof of Proposition 6.21. The difference $H_{\pm, n} - H_{\pm, n}^{\text{split}}$ is exactly the sum of the two off-diagonal blocks

$$P_{\text{orig}, n} H_{\pm, n} P_{\text{copy}, n} + P_{\text{copy}, n} H_{\pm, n} P_{\text{orig}, n}.$$

Thus it suffices to prove that every nonzero matrix element between the original side and the strict copy side is supported inside Y_n .

A vertex of the barycentric graph is the barycenter of a simplex of $D(\mathcal{K}_n)$, and an edge joins two vertices precisely when the corresponding simplices are incident by inclusion. Let

$$U_{n, \text{orig}} := X_{n, \text{orig}} \setminus \Gamma_n, \quad U_{n, \text{copy}} := X_{n, \text{copy}} \setminus \Gamma_n.$$

We claim that no graph edge joins a vertex of $U_{n, \text{orig}}$ directly to a vertex of $U_{n, \text{copy}}$. An edge in the barycentric graph corresponds to a pair of simplices $\sigma \subset \tau$ or $\tau \subset \sigma$. If one of these simplices lies strictly on the original side and the other lies strictly on the copy side, then they cannot be nested: a simplex containing vertices from both sides necessarily meets the glued boundary, and a boundary simplex itself already belongs to Γ_n . Hence every adjacency from the original side to the copy side must pass through a simplex meeting the glued boundary.

As a consequence, every graph path from $X_{n, \text{orig}}$ to $X_{n, \text{copy}}$ must pass through Γ_n . Now let

$$x \in X_{n, \text{orig}}, \quad y \in X_{n, \text{copy}}, \quad \langle x, H_{\pm, n} y \rangle \neq 0.$$

Since $H_{\pm, n}$ has propagation at most R_{loc} , we have

$$d_{X_n}(x, y) \leq R_{\text{loc}}.$$

Choose a geodesic path from x to y in the barycentric graph. By the previous paragraph, this path meets Γ_n at some vertex z . Because the path has total length at most R_{loc} , we have

$$d_{X_n}(x, z) \leq R_{\text{loc}}, \quad d_{X_n}(y, z) \leq R_{\text{loc}}.$$

Hence both x and y belong to $N_{R_{\text{loc}}}(\Gamma_n) = Y_n$. Therefore every original-copy matrix coefficient of $H_{\pm, n}$ vanishes after compressing to $X_n \setminus Y_n$, which is exactly (6.1).

For (6.2), observe that the jam term is already diagonal with respect to the decomposition $\mathcal{H}_{\text{orig}, n} \oplus \mathcal{H}_{\text{copy}, n}$. Consequently,

$$H_{\pm, M, n} - (H_{\pm, n}^{\text{split}} + MP_{\text{copy}, n}) = H_{\pm, n} - H_{\pm, n}^{\text{split}},$$

and (6.2) follows from (6.1). $\square$

A.87 Proof of Result 6.22

Proof of Corollary 6.22. By Proposition 6.21, each summand of either difference is supported inside $Y_n \times Y_n$. In both differences the jam term is absent or cancels exactly, so their propagations are uniformly bounded by R_{loc} and their norms are uniformly bounded by the uniform bound on $(H_{\pm,n})_n$. Hence the direct sums are bounded operators on $\ell^2(X)$ with uniformly finite propagation and support contained in a bounded neighborhood of Y .

Because X has bounded geometry and we work with scalar coefficients, such operators are locally compact: if $F \subset X$ is finite, then finite propagation implies that both $T\chi_F$ and $\chi_F T$ have range contained in ℓ^2 of a finite neighborhood of F , hence are finite rank. Therefore the two direct sums belong to the algebraic interface ideal, and thus to its norm closure $C^*(Y \subset X)$. $\square$

A.88 Proof of Result 6.23

Proof of Proposition 6.23. The operator $\tilde{T}_{\pm,M}^-$ is a direct sum of bounded local self-adjoint operators, so it is bounded, self-adjoint, and local. On the original-side sector it equals the identity, hence has spectral value 1.

On the copy-side sector it equals $H_{\pm,\text{copy}}^\infty + MP_{\text{copy}}^\infty$. For each n every spectral value λ of $H_{\pm,n}^{\text{copy}} + M\mathbf{1}$ satisfies

$$\lambda \geq M + \lambda_{\min}(H_{\pm,n}^{\text{copy}}).$$

By the definition of $B_{\text{copy},-}$, the right-hand side is strictly positive when $M > B_{\text{copy},-}$. Thus the whole operator spectrum of $\tilde{T}_{\pm,M}^-$ is contained in $\{1\} \cup [M - B_{\text{copy},-}, \infty)$, which gives the stated spectral-distance bound. In particular, the spectrum is contained in $(0, \infty)$, so the negative operator spectral projection vanishes. $\square$

A.89 Proof of Result 6.25

Proof of Proposition 6.25. Fix $r > 0$. By the definition of coarse disjoint union, there exists N_r such that

$$\text{dist}(X_n, X_m) > 2r \quad \text{whenever } n \neq m \text{ and } \max\{n, m\} > N_r.$$

Let

$$x \in N_r(X^+) \cap N_r(X^-).$$

Choose $x_+ \in X^+$ and $x_- \in X^-$ with

$$d(x, x_+) \leq r, \quad d(x, x_-) \leq r.$$

Then $d(x_+, x_-) \leq 2r$. Suppose first that either x_+ or x_- lies in a component X_n with $n > N_r$. The separation condition forces both points to lie in that same component. If either point lies in Y_n , then $x \in N_r(Y)$. Otherwise

$$x_+ \in X_{n,\text{orig}} \setminus Y_n, \quad x_- \in X_{n,\text{copy}} \setminus Y_n.$$

By the same separation argument used in Proposition 6.21, every geodesic in the barycentric graph from x_+ to x_- meets $\Gamma_n \subset Y_n$. Therefore $d(x_+, Y_n) \leq 2r$, and consequently

$$d(x, Y) \leq d(x, x_+) + d(x_+, Y_n) \leq 3r.$$

Thus every such x lies in $N_{3r}(Y)$.

It remains to treat the case in which neither x_+ nor x_- lies in a component of index larger than N_r . Let

$$F_r := \bigsqcup_{n=1}^{N_r} X_n.$$

This is a finite set. Because each $\mathcal{K}_n$ has nonempty boundary, every Y_n is nonempty, so

$$c_r := \max_{x \in N_r(F_r)} d(x, Y)$$

is finite. Therefore

$$N_r(X^+) \cap N_r(X^-) \subset N_{s(r)}(Y), \quad s(r) := \max\{3r, c_r\},$$

which is exactly coarse excision. □

A.90 Proof of Result 6.27

Proof of Proposition 6.27. For $Q_Y T$ and $T Q_Y$, finite propagation implies

$$Q_Y T = \chi_Y T \chi_{N_R(Y)}, \quad T Q_Y = \chi_{N_R(Y)} T \chi_Y,$$

so both are supported in $N_R(Y) \times N_R(Y)$ and hence belong to $\mathcal{I}$.

Now consider $Q_+ T Q_-$. If

$$\langle x, Q_+ T Q_- y \rangle \neq 0,$$

then $x \in Z^+$, $y \in Z^-$, and $d(x, y) \leq R$. Therefore

$$x \in N_R(X^-), \quad y \in N_R(X^+).$$

Since also $x \in X^+$ and $y \in X^-$, Proposition 6.25 gives

$$x, y \in N_{s(R)}(Y).$$

Hence $Q_+ T Q_-$ is supported in $N_{s(R)}(Y) \times N_{s(R)}(Y)$, so $Q_+ T Q_- \in \mathcal{I}$. The same argument gives $Q_- T Q_+ \in \mathcal{I}$.

Because $\mathcal{I}$ is closed and algebraic Roe operators are dense in $\mathcal{A}$, the same containments hold for every $T \in \mathcal{A}$.

We also record that multiplication by Q_+ , Q_- , or Q_Y preserves $\mathcal{I}$: algebraic Roe operators supported near Y remain supported near Y after such left or right multiplication, and the claim follows by norm closure.

We now prove part (ii). Since $Q_Y \in \mathcal{I}$ by part (i), with $T = \mathbf{1}$, we have

$$e_+ + e_- = \pi(Q_+ + Q_-) = \pi(\mathbf{1} - Q_Y) = \mathbf{1}_{\mathcal{A}/\mathcal{I}}.$$

Orthogonality is immediate from $Q_+ Q_- = 0$. The centrality identity proved below shows that $e_\pm$ are central projections.

We next prove part (iii). If $\pi(a) = \pi(a')$, then $a - a' \in \mathcal{I}$, hence

$$Q_\pm(a - a') \in \mathcal{I}.$$

Therefore $E_\pm$ are well-defined.

Idempotence and orthogonality are immediate:

$$E_{\pm}^2(\pi(a)) = \pi(Q_{\pm}Q_{\pm}a) = E_{\pm}(\pi(a)), \quad E_+E_-(\pi(a)) = \pi(Q_+Q_-a) = 0,$$

and similarly for E_-E_+ . Also,

$$\pi(a) = \pi((Q_+ + Q_- + Q_Y)a) = E_+(\pi(a)) + E_-(\pi(a)),$$

because $Q_Ya \in \mathcal{I}$. Thus $E_+ + E_- = \mathbf{1}_{\mathcal{A}/\mathcal{I}}$.

To prove multiplicativity for E_+ , let $a, b \in \mathcal{A}$. Then

$$E_+(\pi(a))E_+(\pi(b)) = \pi(Q_+a)\pi(Q_+b) = \pi(Q_+aQ_+b).$$

Now

$$Q_+ab - Q_+aQ_+b = Q_+a(Q_- + Q_Y)b.$$

The term Q_+aQ_-b lies in $\mathcal{I}$ because $Q_+aQ_- \in \mathcal{I}$ and $\mathcal{I}$ is an ideal. The term Q_+aQ_Yb also lies in $\mathcal{I}$ because $aQ_Y \in \mathcal{I}$. Hence

$$\pi(Q_+aQ_+b) = \pi(Q_+ab) = E_+(\pi(ab)).$$

So E_+ is multiplicative. The same proof applies to E_- . The $*$ -property is immediate:

$$E_+(\pi(a))^* = \pi((Q_+a)^*) = \pi(a^*Q_+) = E_+(\pi(a^*)),$$

because

$$a^*Q_+ - Q_+a^* = (Q_+a - aQ_+)^* \in \mathcal{I},$$

as shown next.

To prove the centrality identity, let $a, b \in \mathcal{A}$. Then

$$E_+(\pi(a))\pi(b) - \pi(a)E_+(\pi(b)) = \pi((Q_+a - aQ_+)b).$$

Now

$$Q_+a - aQ_+ = Q_+a(Q_- + Q_Y) - (Q_- + Q_Y)aQ_+.$$

Each term on the right belongs to $\mathcal{I}$: the Q_+aQ_- and Q_-aQ_+ terms are in $\mathcal{I}$ by part (i), and the terms involving Q_Y are in $\mathcal{I}$ because aQ_Y and Q_Ya are in $\mathcal{I}$ and $\mathcal{I}$ is stable under multiplication by Q_+ . Hence $Q_+a - aQ_+ \in \mathcal{I}$, so

$$E_+(\pi(a))\pi(b) = \pi(a)E_+(\pi(b)).$$

Thus E_+ is central in the sense stated. The same holds for E_- .

Finally, every $b \in \mathcal{A}/\mathcal{I}$ decomposes as

$$b = E_+(b) + E_-(b),$$

and orthogonality $E_+E_- = E_-E_+ = 0$ implies uniqueness of this decomposition. Since $E_{\pm}(\mathcal{A}/\mathcal{I})$ are closed two-sided ideals, this proves the direct-sum decomposition. $\square$

A.91 Proof of Result 6.28

Proof of Proposition 6.28. The operators J, T^+, T^- are locally compact: for every finite $F \subset X$, the operators $\chi_F T$ and $T \chi_F$ have finite-dimensional range, with T equal to J, T^+ , or T^- . Hence

$$J, T^+, T^- \in \mathcal{A} = C^*(X),$$

so $\pi(J)$ and $\pi(T^\pm)$ are well-defined.

Since 0 lies in an operator spectral gap of $T^\pm$, choose compactly supported continuous functions $h_\pm$ that agree with $\chi_{(-\infty, 0)}$ on $\text{spec}_{\text{op}}(T^\pm)$. Continuous functional calculus in $\mathcal{A}$ gives

$$P^\pm = h_\pm(T^\pm) \in \mathcal{A}.$$

Thus $\pi(P^\pm)$ is well-defined.

Set

$$D := J - Q_+ T^+ Q_+ - Q_- T^- Q_-.$$

We prove $D \in \mathcal{I}$.

Expand J using $Q_+ + Q_- + Q_Y = \mathbf{1}$ on $\ell^2(X)$. Every term in this expansion containing Q_Y lies in $\mathcal{I}$ by Proposition 6.27(i). The same is true for the cross terms $Q_+ J Q_-$ and $Q_- J Q_+$.

It remains to treat the diagonal difference terms

$$Q_+(J - T^+)Q_+, \quad Q_-(J - T^-)Q_-.$$

Let

$$S_+ := \text{prop}(J - T^+).$$

Suppose

$$\langle x, Q_+(J - T^+)Q_+ y \rangle \neq 0.$$

Then $x, y \in Z^+$ and $d(x, y) \leq S_+$. If $x \notin N_{R_{\text{ag}} + S_+}(Y)$, then

$$d(x, Y) > R_{\text{ag}} + S_+,$$

so

$$d(y, Y) \geq d(x, Y) - d(x, y) > R_{\text{ag}}.$$

Hence both x and y lie in $X^+ \setminus N_{R_{\text{ag}}}(Y)$, and the assumed sidewise agreement forces the matrix coefficient to vanish, a contradiction. Thus every nonzero matrix coefficient of $Q_+(J - T^+)Q_+$ has

$$x \in N_{R_{\text{ag}} + S_+}(Y).$$

By symmetry the same holds for y . Therefore

$$Q_+(J - T^+)Q_+ \in \mathcal{I}.$$

The same argument gives

$$Q_-(J - T^-)Q_- \in \mathcal{I}.$$

Hence $D \in \mathcal{I}$.

Applying π gives

$$\pi(J) = \pi(Q_+ T^+ Q_+) + \pi(Q_- T^- Q_-).$$

By Proposition 6.27,

$$\pi(Q_+T^+Q_+) = E_+(\pi(T^+)), \quad \pi(Q_-T^-Q_-) = E_-(\pi(T^-)),$$

which proves (i).

Now fix $0 < \gamma < \min\{\gamma_+, \gamma_-\}$, choose $B \geq \max\{\|T^+\|, \|T^-\|\}$, and let h be as in part (ii). Because

$$\mathcal{A}/\mathcal{I} = E_+(\mathcal{A}/\mathcal{I}) \oplus E_-(\mathcal{A}/\mathcal{I}),$$

continuous functional calculus is componentwise on this direct sum. Therefore part (i) implies

$$h(\pi(J)) = E_+(h(\pi(T^+))) + E_-(h(\pi(T^-))).$$

Since π is a $*$ -homomorphism, it commutes with continuous functional calculus:

$$h(\pi(T^\pm)) = \pi(h(T^\pm)).$$

On the operator spectrum of $T^\pm$, the function h agrees with $\chi_{(-\infty, 0]}$ because $\text{spec}_{\text{op}}(T^\pm) \subset [-B, B]$ and $\text{spec}_{\text{op}}(T^\pm) \cap (-\gamma, \gamma) = \emptyset$. Hence the operator functional calculus on $\ell^2(X)$ gives

$$h(T^\pm) = \chi_{(-\infty, 0]}(T^\pm) = P^\pm.$$

Consequently,

$$h(\pi(J)) = E_+(\pi(P^+)) + E_-(\pi(P^-)),$$

which proves (ii).

For part (iii), each $\pi(P^\pm)$ is a projection, $E_\pm$ are $*$ -homomorphisms, and their ranges are orthogonal ideals. Therefore

$$p_J^2 = E_+(\pi(P^+))^2 + E_-(\pi(P^-))^2 = p_J, \quad p_J^* = p_J.$$

So p_J is a projection. The identity $p_J = h(\pi(J)) = \pi(h(J))$ follows from part (ii) and the continuous functional calculus. Since the right-hand side is the same for every admissible cutoff h , the projection p_J is independent of the choice of γ and h . $\square$

A.92 Proof of Result 6.31

Proof of Proposition 6.31. Because $\tilde{p}$ is a self-adjoint lift of the projection p ,

$$\pi(\exp(2\pi i \tilde{p})) = \exp(2\pi i p) = \mathbf{1}.$$

Hence

$$\exp(2\pi i \tilde{p}) - \mathbf{1} \in M_n(\mathcal{I}),$$

so $\exp(2\pi i \tilde{p})$ is a unitary in $\mathbf{1} + M_n(\mathcal{I})$ and therefore defines a class in $K_1(\mathcal{I})$.

If $\tilde{p}_0$ and $\tilde{p}_1$ are two self-adjoint lifts of the same projection p , then

$$\tilde{p}_t := (1-t)\tilde{p}_0 + t\tilde{p}_1 \quad (0 \leq t \leq 1)$$

is a norm-continuous path of self-adjoint lifts of p . Therefore

$$t \longmapsto \exp(2\pi i \tilde{p}_t)$$

is a homotopy of unitaries in $\mathbf{1} + M_n(\mathcal{I})$, so the K_1 -class is independent of the chosen lift.

For the last statement, because $\mathcal{A}$ is unital, $h(a) \in M_n(\mathcal{A})$ for every continuous h . Functional calculus in $\mathcal{A}/\mathcal{I}$ gives

$$\pi(h(a)) = h(\pi(a)).$$

Since h agrees with $\chi_{(-\infty, 0)}$ on $\text{spec}(\pi(a))$, the element $h(\pi(a))$ is a projection. Applying the first part with lift $h(a)$ proves

$$\partial_Y(\pi(h(a))) = [\exp(2\pi i h(a))].$$

□

A.93 Proof of Result 6.32

Proof of Proposition 6.32. The map

$$s \longmapsto E_+(\pi(P_s))$$

is a norm-continuous path of projections in $\mathcal{A}/\mathcal{I}$, because π and E_+ are bounded $*$ -homomorphisms. Hence the $K_0(\mathcal{A}/\mathcal{I})$ -class of $E_+(\pi(P_s))$ is constant. Applying the connecting map

$$\partial_Y : K_0(\mathcal{A}/\mathcal{I}) \longrightarrow K_1(\mathcal{I})$$

gives the claim. □

A.94 Proof of Result A.34

Proof of Proposition A.34. Since $V, V^\pm \in \mathcal{I}$, the quotient images satisfy

$$\pi(J') = \pi(J), \quad \pi(T'^\pm) = \pi(T^\pm).$$

Let

$$P^\pm := \chi_{(-\infty, 0)}(T^\pm), \quad P'^\pm := \chi_{(-\infty, 0)}(T'^\pm).$$

Choose continuous compactly supported cutoffs $h^\pm$ and $h'^\pm$ agreeing with $\chi_{(-\infty, 0)}$ on the operator spectra of $T^\pm$ and $T'^\pm$, respectively. Since

$$\text{spec}(\pi(T^\pm)) = \text{spec}(\pi(T'^\pm))$$

and this quotient spectrum is contained in both operator spectra, both cutoffs give the same functional calculus element in the quotient:

$$\pi(P'^\pm) = h'^\pm(\pi(T'^\pm)) = h'^\pm(\pi(T^\pm)) = h^\pm(\pi(T^\pm)) = \pi(P^\pm).$$

Therefore

$$p_{J'} = E_+(\pi(P'^+)) + E_-(\pi(P'^-)) = E_+(\pi(P^+)) + E_-(\pi(P^-)) = p_J.$$

Applying ∂_Y gives equality of the interface classes. □

A.95 Proof of Result A.35

Proof of Proposition A.35. Let

$$F := \bigsqcup_{n=1}^N X_n.$$

This is finite. Since each Y_n is nonempty,

$$R_F := \max_{x \in F} \text{dist}(x, Y)$$

is finite, so $F \subset N_{R_F}(Y)$. An operator supported on $F \times F$ is therefore supported inside $N_{R_F}(Y) \times N_{R_F}(Y)$. Since V is bounded and finite-propagation, it belongs to the algebraic interface ideal, hence to $C^*(Y \subset X)$. Applying this observation to each finite-component change in the sidewise data, the final statement follows from Proposition A.34. $\square$

A.96 Proof of Result A.36

Proof of Proposition A.36. Let $V^{(N)} := \bigoplus_{n \leq N} V_n$. This operator is supported on the finite set $\bigsqcup_{n \leq N} X_n$. Since each Y_n is nonempty, the proof of Proposition A.35 shows that $V^{(N)} \in C^*(Y \subset X)$. Also

$$\|V - V^{(N)}\| = \sup_{n > N} \|V_n\| \longrightarrow 0.$$

The interface ideal is norm closed, so $V \in C^*(Y \subset X)$. The final statement is Proposition A.34. $\square$

A.97 Proof of Result A.37

Proof of Proposition A.37. Let

$$B := \sup_{s \in [0,1]} \max\{\|T_s^+\|, \|T_s^-\|\} < \infty.$$

Choose one compactly supported continuous cutoff h that equals 1 on $[-B, -\gamma/2]$ and 0 on $[\gamma/2, B]$. Then h agrees with $\chi_{(-\infty, 0)}$ on the operator spectra of all $T_s^\pm$, and

$$P_s^\pm := h(T_s^\pm)$$

are norm-continuous paths of projections in $\mathcal{A}$. Hence

$$p_s := E_+(\pi(P_s^+)) + E_-(\pi(P_s^-))$$

is a norm-continuous path of projections in $\mathcal{A}/\mathcal{I}$, so its K_0 -class is constant. Applying the boundary map

$$\partial_Y : K_0(\mathcal{A}/\mathcal{I}) \rightarrow K_1(\mathcal{I})$$

shows that the interface class is independent of s . $\square$

A.98 Proof of Result A.38

Proof of Corollary A.38. The positive-side comparison is fixed and has the uniform bulk gap from Assumption 6.13. On the copy sector,

$$H_{\pm, \text{copy}}^\infty + V_s \geq (M_0 - B_{\text{copy}, -})\mathbf{1},$$

while on the original sector T_s^- equals $\mathbf{1}$. Hence the negative-side comparison has a uniform positive gap at zero. The collar-locality argument is unchanged: away from the glued boundary, J_s agrees with T_s^+ on the positive side and with T_s^- on the negative side. Thus $(J_s; T_s^+, T_s^-)$ is a norm-continuous family of sidewise class-A interfaces with a uniform comparison gap, and Proposition A.37 applies. $\square$

A.99 Proof of Result A.39

Proof of Theorem A.39. The operators are bounded, self-adjoint, finite-propagation Roe operators. The positive comparison $T^+ = H_\pm^\infty$ has the bulk gap and Fermi projection $P_\pm$ by Proposition 6.18. On the original sector, T_V^- is the identity. On the copy sector,

$$H_{\pm, \text{copy}}^\infty + V \geq (M_0 - B_{\text{copy}, -})\mathbf{1} > 0,$$

so

$$\text{dist}(0, \text{spec}_{\text{op}}(T_V^-)) \geq \min\{1, M_0 - B_{\text{copy}, -}\}.$$

Thus T_V^- is strictly positive and its negative Fermi projection is zero.

It remains to check sidewise agreement. On the positive side outside Y we are strictly in the original sector, so V vanishes and $J_V = T^+$. On the negative side outside Y , the original sector projection vanishes, and Proposition 6.21 kills the original-copy cross block. Thus $J_V = T_V^-$ there. Therefore Proposition 6.28 applies, and since the negative-side Fermi projection is zero it gives

$$p_{J_V} = E_+(\pi(P_\pm)).$$

The formula for the interface class is Definition 6.33. □

A.100 Proof of Result 6.35

Proof of Theorem 6.35. The operators $J_{\pm, M}^\infty$, $T^+ = H_\pm^\infty$, and $T^- = \tilde{T}_{\pm, M}^-$ are bounded self-adjoint local operators on the scalar bounded-geometry space X , hence belong to $\mathcal{A} = C^*(X)$.

By Proposition 6.18,

$$T^+ = H_\pm^\infty$$

is bounded, self-adjoint, local, has an operator gap at 0, and has operator Fermi projection

$$P^+ = P_\pm \in C^*(X).$$

By Proposition 6.23,

$$T^- = \tilde{T}_{\pm, M}^-$$

is bounded, self-adjoint, local, and has

$$\text{dist}(0, \text{spec}_{\text{op}}(T^-)) \geq \min\{1, M - B_{\text{copy}, -}\}$$

when $M > B_{\text{copy}, -}$. Therefore

$$P^- := \chi_{(-\infty, 0)}(T^-) = 0.$$

It remains to verify the sidewise agreement conditions. On the positive side outside Y we are strictly on the original side, so the jam term vanishes and

$$\chi_{X^+ \setminus Y}(J_{\pm, M}^\infty - T^+)\chi_{X^+ \setminus Y} = 0.$$

On the negative side outside Y we are strictly on the copy side. There P_{orig}^∞ vanishes, and Proposition 6.21 shows that all original-copy matrix coefficients vanish there. Hence

$$\chi_{X^- \setminus Y}(J_{\pm, M}^\infty - T^-)\chi_{X^- \setminus Y} = 0.$$

Thus $(J_{\pm, M}^\infty; T^+, T^-)$ is a sidewise class-A interface along Y .

Now Proposition 6.28 applies. Since $P^- = 0$, it yields

$$p_{J_{\pm,M}^\infty} = E_+(\pi(P^+)) + E_-(\pi(P^-)) = E_+(\pi(P_\pm)),$$

which is (6.3). The equivalent cutoff statement follows from the same proposition.

Finally, Definition 6.33 gives

$$\text{Ind}_{\text{int}}(J_{\pm,M}^\infty; T^+, T^-) = \partial_Y(p_{J_{\pm,M}^\infty}) = \partial_Y(E_+(\pi(P_\pm))),$$

which is (6.4). The statement about the jammed side is exactly the vanishing of P^- and hence of the negative-side quotient contribution. $\square$

A.101 Proof of Result A.40

Proof of Corollary A.40. The right-hand side depends only on the bulk projection $P_\pm$ and the positive quotient endomorphism E_+ , not on M . Apply Theorem 6.35. $\square$

A.102 Proof of Result 6.36

Proof of Theorem 6.36. Each $H_{\pm,n} = (B_n + B_n^\dagger) \pm S_n$ is the closed-case Higson–Prodan Hamiltonian on $D(\mathcal{K}_n)$, so the direct-sum family $H_\pm^\infty$ is the Higson–Prodan family for the refinement sequence of closed doubles. The bulk projection $P_\pm$ constructed in Proposition 6.18 is therefore the Higson–Prodan Fermi projection, and the Chern-number computation is the imported content of the Higson–Prodan construction and its Higson–Roe analytic-surgery input [8, 9, 10, 11, 12]. $\square$

A.103 Proof of Result A.41

Proof of Corollary A.41. The identity is Theorem 6.35 under the additional Chern-number input recorded in Theorem 6.36. The independence of the large scalar jam scale is Corollary A.40. This gives exactly the displayed formula and the stated identification of the bulk projection. $\square$

A.104 Proof of Result A.42

Proof of Corollary A.42. Theorem 6.17 gives Assumptions 6.8 and 6.13. The interface-class formula is then Theorem 6.35, and the Chern-number statement is the imported Higson–Prodan bulk theorem as recorded in Theorem 6.36. The final sentence is upgraded to the winding statement in Theorem 6.38 for HP–Ewert–Meyer admissible families. $\square$

A.105 Proof of Result A.43

Proof of Proposition A.43. The closed-bulk input gives the positive-side model and its Chern pairing. Proposition 6.25 proves that the original/copy decomposition gives a coarse partition with interface Y . Corollary 6.22 gives sidewise agreement away from Y : the positive side agrees with the closed Higson–Prodan model and the copy side agrees with the jammed comparison model. Proposition 6.23 shows that this comparison model is trivial on the negative side once $M > B_{\text{copy},-}$. These are exactly the geometric and sidewise hypotheses in Definition 6.37; the last pairing condition is the stated Ewert–Meyer compatibility. None of these steps assumes a separate boundary gap. $\square$

A.106 Proof of Result A.44

Proof of Corollary A.44. The double of a full-boundary triangulated oriented surface is a closed oriented triangulated surface by Proposition 3.2. The listed refinement schemes have uniformly bounded incidence under the stated hypotheses, and doubling preserves that bounded geometry, but bounded incidence alone is not the topological input. The first check supplies the controlled cap-product operator, the uniform closed doubled-bulk gap, and the Chern pairing $\text{Ch}([P_{\pm}]) = \pm 1$. Proposition 6.25 supplies the second check for the original/copy decomposition, and the third check is exactly the external pairing compatibility in Definition 6.37. Proposition A.43 then gives HP–Ewert–Meyer admissibility, and Theorem 6.38 gives the last assertion. $\square$

A.107 Proof of Result A.45

Proof of Corollary A.45. Changing or deleting finitely many components gives an operator supported on a finite subset of the coarse disjoint union. Since each interface component is nonempty, such an operator belongs to the interface ideal by Proposition A.35. The sidewise interface class is therefore unchanged by Proposition A.34, and the compatible winding pairing has the same value on the resulting K_1 -class. $\square$

A.108 Proof of Result 6.38

Proof of Theorem 6.38. HP–Ewert–Meyer admissibility includes the Higson–Prodan closed-bulk input, so Theorem 6.17 gives Assumptions 6.8 and 6.13. It also identifies $P_{\pm}$ as the Higson–Prodan Fermi projection with $\text{Ch}([P_{\pm}]) = \pm 1$. Theorem 6.35 gives the sidewise interface realization and the formula

$$\text{Ind}_{\text{int}}(J_{\pm, M}^{\infty}; T^+, T^-) = \partial_Y(E_+(\pi(P_{\pm}))).$$

The negative comparison operator has no negative-energy Fermi projection by Proposition 6.23, so the negative side contributes Chern number 0. Therefore the Ewert–Meyer pairing compatibility in Definition 6.37 applies with $c_+ = \pm 1$ and $c_- = 0$, giving winding number ± 1 . Independence of M is Corollary A.40. $\square$

A.109 Proof of Result A.46

Proof of Corollary A.46. An orientation-preserving boundary-compatible simplicial isomorphism induces a unitary identification of the simplex Hilbert spaces. It conjugates the boundary operator, cap-product operator, original/copy projections, collar ideals, and quotient maps, so the corresponding K -classes and pairings are unchanged. Reversing orientation changes the sign of the fundamental and relative fundamental classes. Hence the Poincaré-duality operator S changes sign, up to the same unitary relabeling, and $H_+ = (B + B^{\dagger}) + S$ is identified with the old H_- . The Higson–Prodan Chern number and the compatible interface winding therefore change sign. $\square$

A.110 Proof of Result A.47

Proof of Corollary A.47. The scalar jam-scale statement is Corollary A.40. Positive copy-side confinements are covered by Theorem A.39 and Corollary A.38. Interface-supported changes are Proposition A.34, finite-component changes are Proposition A.35, norm-vanishing component defects are Proposition A.36, and norm-continuous sidewise homotopies are Proposition A.37. Each operation preserves the class $\partial_Y(E_+(\pi(P_{\pm})))$. The compatible winding pairing in Theorem 6.38 is a homomorphism on this K_1 -class, so it preserves the value ± 1 as well. $\square$

A.111 Proof of Result A.1

Proof of Lemma A.1. First take $u(q, \mu) = (q_1, q_2, \mu)$. The negative eigenspace of $q_1\sigma_1 + q_2\sigma_2 + \mu\sigma_3$ over a small sphere is the tautological Hopf line with first Chern number +1 for the boundary orientation induced by $dq_1 \wedge dq_2 \wedge d\mu$. Restricting the sphere to two constant-mass caps and gluing along the equator identifies this Chern number with the difference between the positive-mass and negative-mass Chern numbers, computed with the $dq_1 \wedge dq_2$ orientation on each cap. Thus the crossing contribution is +1 for the identity map.

For a general invertible real linear u , the family factors through the identity crossing by the map $u/|u|$ on the linking sphere. This map has degree $\text{sgn det}(Du)$. Pullback of the Hopf line multiplies the Chern number by the degree, giving the displayed formula. For a direct sum of two identical Pauli blocks, the negative rank-two bundle is the direct sum of two pulled-back Hopf lines, so first Chern classes add and the jump is multiplied by 2. $\square$

A.112 Proof of Result A.2

Proof of Theorem A.2. Put $E = D - z$. Since $M \geq 2\beta$, the copy block $D + M - z = M\mathbf{1} + E$ is invertible and

$$(M\mathbf{1} + E)^{-1} = M^{-1}\mathbf{1} - M^{-2}E + M^{-3}E^2(\mathbf{1} + E/M)^{-1}.$$

The remainder has norm at most $2\beta^2/M^3$. The Schur complement for the original-side compressed resolvent is

$$S_M(z) = A - z - C(D + M - z)^{-1}C^* = A - z + F_M,$$

where

$$F_M = -M^{-1}CC^* + M^{-2}C(D - z)C^* + \mathcal{R}_M, \quad \|\mathcal{R}_M\| \leq \frac{2c^2\beta^2}{M^3}.$$

The hypotheses give $\|R_A F_M\| \leq 1/2$, so

$$S_M(z)^{-1} = R_A - R_A F_M R_A + R_A F_M R_A F_M R_A + \mathcal{T}_M,$$

with

$$\|\mathcal{T}_M\| \leq 2\|R_A\|^4\|F_M\|^3 \leq \frac{16c^6}{\eta^4 M^3}.$$

Substituting the displayed expression for F_M , the first-order term is

$$M^{-1}R_A C C^* R_A.$$

The second-order terms are

$$-M^{-2}R_A C(D - z)C^* R_A \quad \text{and} \quad M^{-2}R_A C C^* R_A C C^* R_A,$$

coming respectively from $-R_A F_M R_A$ and $R_A F_M R_A F_M R_A$. The part containing $\mathcal{R}_M$ contributes at most

$$\frac{2c^2\beta^2}{\eta^2 M^3} \leq \frac{4c^2\beta^2}{\eta^2 M^3},$$

and the omitted cross terms in $F_M^2 - M^{-2}(C C^*)^2$ contribute at most

$$\frac{8c^4\beta}{\eta^3 M^3}$$

under $M \geq 2\beta$. Combining these estimates gives the stated bound for $E_M(z)$. $\square$

A.113 Proof of Result A.3

Proof of Proposition A.3. Since $V \geq v_0 \mathbf{1}$, every singular value of $D + MV - z$ is at least $Mv_0 - \beta$. Hence the copy block is invertible and

$$\|(D + MV - z)^{-1}\| \leq (Mv_0 - \beta)^{-1}.$$

The original-side Schur complement is

$$S_M^V(z) = A - z - C(D + MV - z)^{-1}C^*.$$

The perturbation of $A - z$ has norm at most $c^2/(Mv_0 - \beta) < \eta/2$. The resolvent identity therefore gives

$$\|(S_M^V(z))^{-1} - (A - z)^{-1}\| \leq \|(S_M^V(z))^{-1}\| \|C(D + MV - z)^{-1}C^*\| \|(A - z)^{-1}\| \leq \frac{2c^2}{\eta^2(Mv_0 - \beta)}.$$

By the Schur complement formula, $(S_M^V(z))^{-1}$ is exactly the original-side compressed resolvent.

For the eigenvector estimate, the copy equation is

$$(D + MV - \lambda)\psi_{\text{copy}} = -C^*\psi_{\text{orig}}.$$

The lower bound $V \geq v_0 \mathbf{1}$ gives

$$\|(D + MV - \lambda)\xi\| \geq (Mv_0 - \|D\| - \Lambda)\|\xi\|$$

for every copy-sector vector ξ . Applying this to $\xi = \psi_{\text{copy}}$ gives the displayed leakage bound. $\square$

B Numerical experiments and reproducibility

The numerical experiments record the finite matrices used to test the mechanisms in the paper: doubled bulk gaps on sampled meshes, the intrinsic Poincaré–Lefschetz/compression comparison, boundary localization after jamming, finite-size flux insertion, chirality reversal, collar support, and the M^{-1}/M^{-2} decoupling laws. They do not replace the uniform doubled-bulk gap hypothesis and they do not define separate K -theoretic indices for individual boundary components. The localization, transport, flux, disorder, and refinement-scaling figures are included to document finite-matrix behavior. They are checks of the mechanisms proved above, not inputs to the proofs.

Pipeline and examples. The companion code generates boundary triangulations from a subdivided icosphere, verifies the doubling identities, assembles $B, S, H_{\pm}(D(\mathcal{K}))$, forms $H_{\pm, M}$, and evaluates the localization, transport, and decoupling quantities below. The reported computations correspond to the linked repository snapshot 1c888c900cddbfff078f2b55b445ee01f2b4bd17, accessed 16 May 2026.

Table 5: Mesh sizes for the three boundary triangulations $\mathcal{K}$ and their doubles $D(\mathcal{K})$. The counts agree with the identities in Theorem 3.3.

ex	$\#\partial$	$ \partial\mathcal{K}_0 $	$ \partial\mathcal{K}_1 $	$ \mathcal{K}_0 $	$ \mathcal{K}_1 $	$ \mathcal{K}_2 $	$\dim \mathcal{H}(\mathcal{K})$	$ D_0 $	$ D_1 $	$ D_2 $	$\dim \mathcal{H}(D)$	g
ex1	1	22	22	73	194	122	389	124	366	244	734	0.5666
ex2	2	88	88	272	728	456	1456	456	1368	912	2736	0.3380
ex3	3	78	78	329	912	582	1823	580	1746	1164	3490	0.3272

Table 6: Selected numerical outputs: the measured doubled bulk half-gap g for $H_+(D(\mathcal{K}))$, the number of jammed eigenvalues selected by $|E| < g$ at $M = 50$, and fitted drift slopes when reported.

ex	g	# in-gap ($ E < g$)	v_+	v_-
ex1	0.5666	5	-0.890	0.893
ex2	0.3380	14	–	–
ex3	0.3272	12	–	–

Table 7: Finite- M checks. The measured slopes are fitted from the log-log data generated by the companion script.

Quantity	Theoretical scale	Observed check
Compressed resolvent error	M^{-1}	slope -0.943
First-order resolvent error	M^{-2}	slope -1.942
Mean squared copy leakage	M^{-2}	slope -1.965
First-order eigenvalue localization	M^{-2}	slope -1.901
Contour margin $\delta_H(\Gamma)$	positive	positive on sampled M -range
Window separation $g_M(E_{\max})$	positive for large M	linear growth after threshold
Original-copy coupling support	bounded collar	all nonzero entries inside measured collar

Meshes, doubled gaps, and jammed spectra. Figure 1 shows the boundary triangulations and Figure 2 shows the sorted spectra of $H_+(D(\mathcal{K}))$. In the displayed finite matrices all three doubled models have a positive measured gap at zero. Figure 3 then shows the spectra after adding $M\mathbf{P}_{\text{copy}}$ with $M = 50$. The high-energy cluster is copy-supported; the low-energy levels inside $|E| < g$ are the finite-size interface candidates.

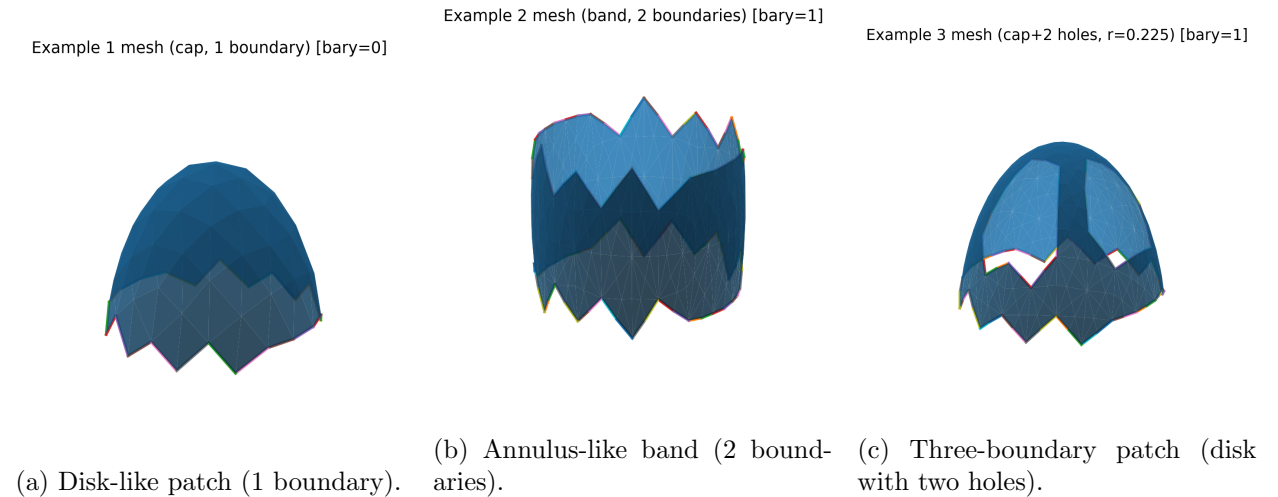
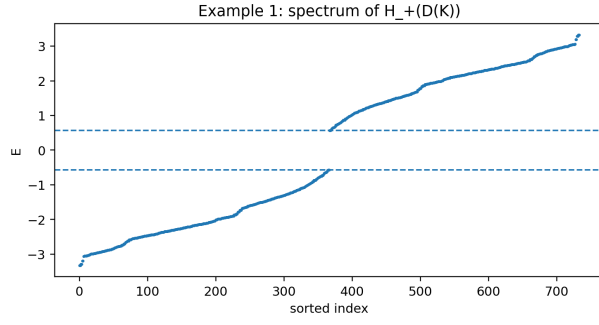
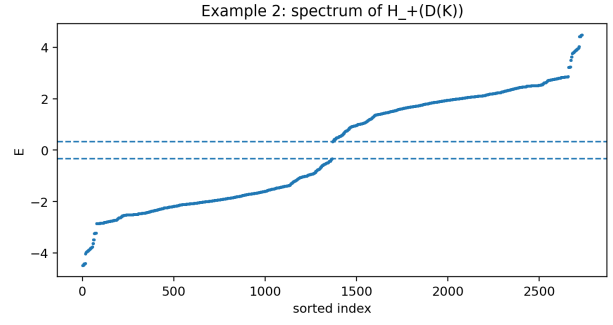


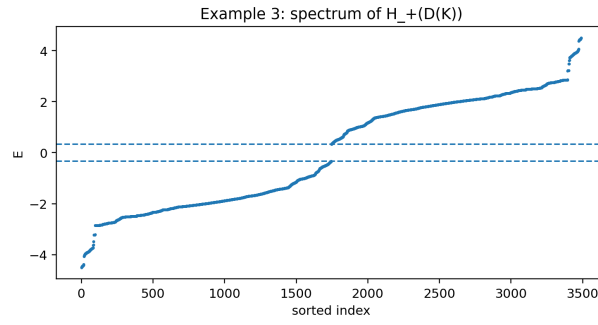
Figure 1: Boundary triangulations used in the experiments. Boundary edges are highlighted.



(a) Example 1.



(b) Example 2.



(c) Example 3.

Figure 2: Sorted spectra of $H_+(D(K))$. Dashed lines mark the measured half-gap $\pm g$, where $g = \min |\text{spec}(H_+(D(K)))|$.

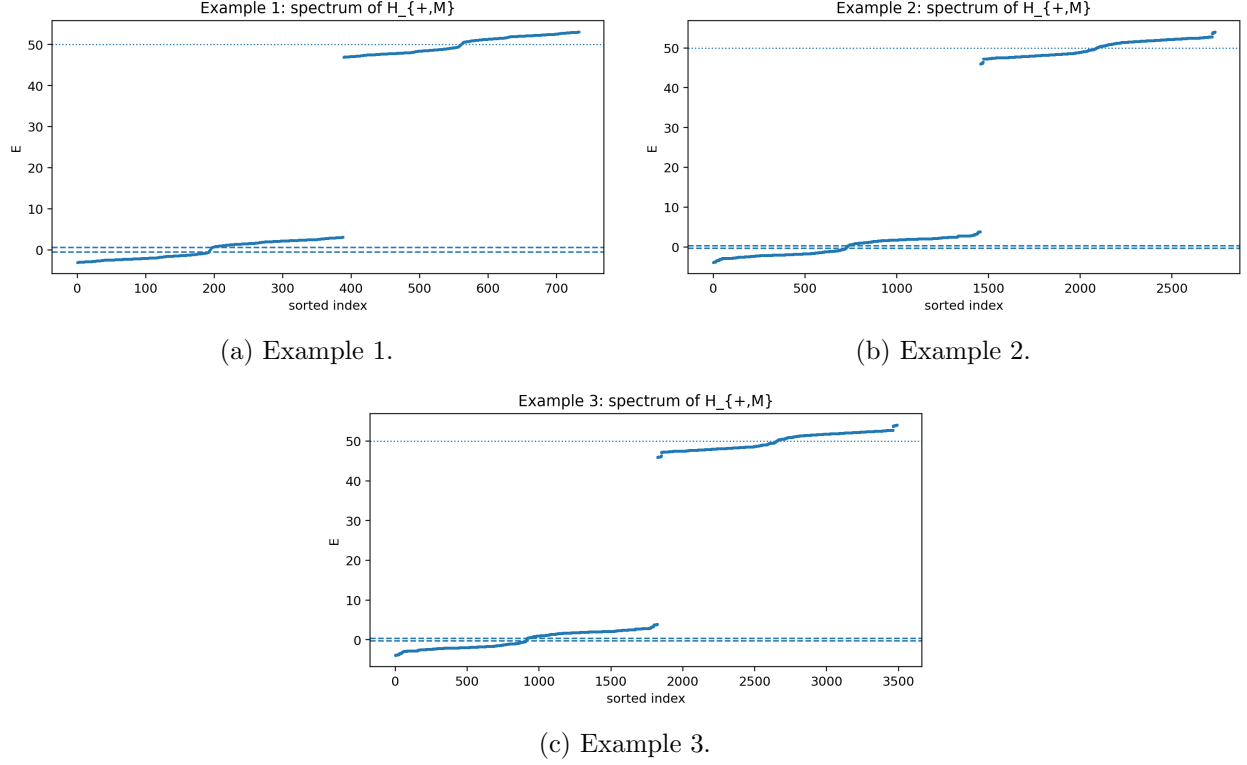


Figure 3: Spectra of $H_{+,M}$ at $M = 50$. Dashed lines indicate the measured doubled bulk gap scale inherited from $H_+(D(K))$.

Boundary localization. Figure 4 plots the boundary weight $w_{\partial}(\psi) = \|\mathbf{P}_{\partial}\psi\|^2$ for Example 1. Figure 5 repeats this check component by component for the two- and three-boundary examples. The in-gap states have enhanced boundary weight, while the high-energy copy cluster has negligible original-boundary weight.

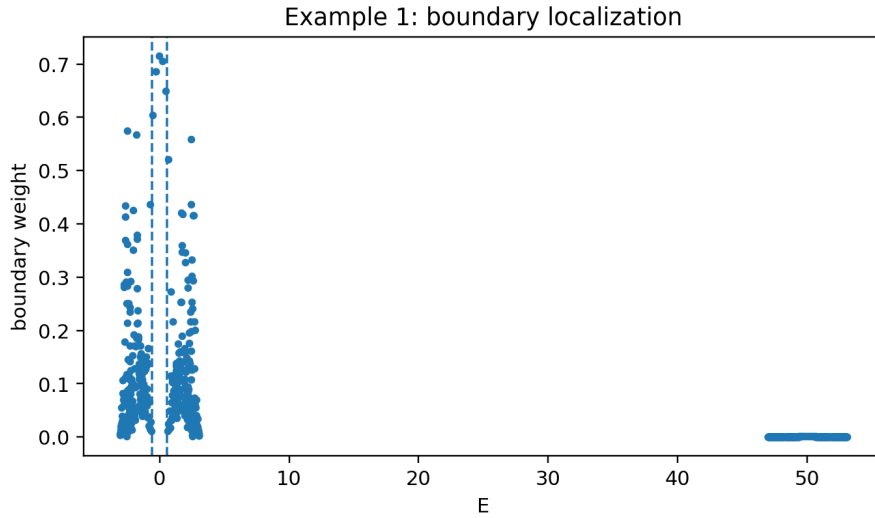


Figure 4: Example 1: boundary weight versus eigenvalue. States selected by $|E| < g$ have enhanced boundary weight.

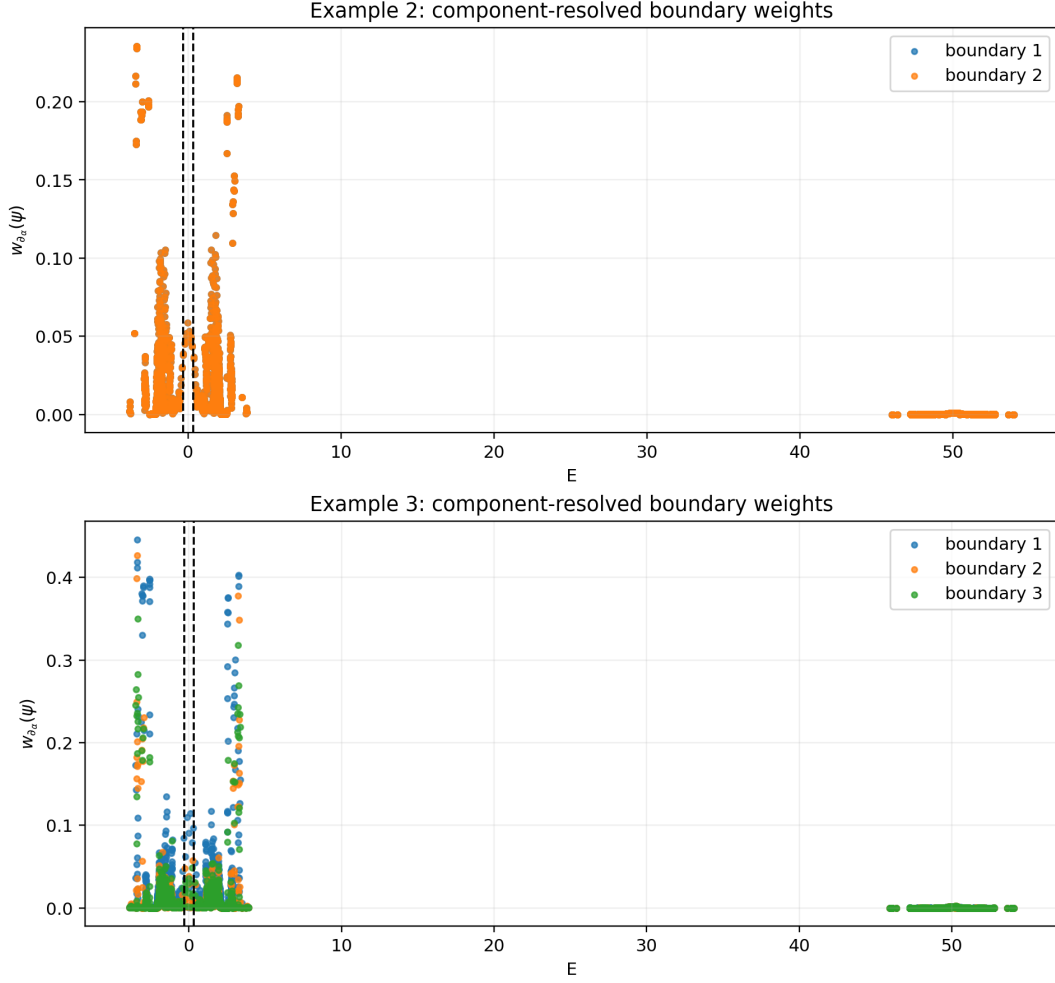


Figure 5: Examples 2 and 3: component-resolved boundary weights. The dashed vertical lines mark the finite-size gap window $|E| < g$.

Intrinsic boundary comparison. Figure 6 and table 8 compare the intrinsic Poincaré–Lefschetz boundary Hamiltonian $H_+^{\text{PL}}(\mathcal{K})$, the doubled original-side compression $H_+^{\partial}(\mathcal{K})$, and the finite- M jammed model. This is the numerical check most directly tied to Theorem 3.12: the discrepancy $R_{\partial,+} = H_+^{\partial}(\mathcal{K}) - H_+^{\text{PL}}(\mathcal{K})$ is visible but confined to the boundary collar, and the jammed low-energy spectrum tracks the compressed boundary model.

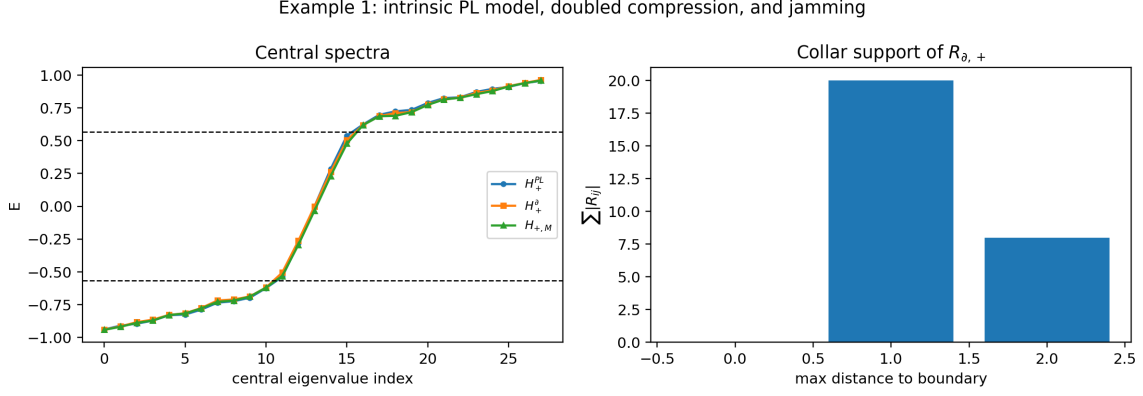


Figure 6: Example 1: central spectra of $H_+^{\text{PL}}(\mathcal{K})$, $H_+^{\partial}(\mathcal{K})$, and $H_{+,M}$, together with the measured support distribution of the collar correction $R_{\partial,+}$. The correction is not norm-small, but it is boundary-collar supported.

Table 8: Example 1: numerical size and support of $R_{\partial,+} = H_+^{\partial}(\mathcal{K}) - H_+^{\text{PL}}(\mathcal{K})$.

example	$\dim \mathcal{H}(\mathcal{K})$	rank R	support dim.	support radius	$\ R\ $	$\ R\ _F$
ex1	389	32	44	2	1.0000	3.7417

Boundary transport and chirality. We evolve a wavepacket projected to the finite in-gap subspace and order the boundary vertices cyclically along each component. Diagonal ridges in Figure 7 show one-way boundary motion. Figure 8 and Figure 9 show that the signed drift reverses under $H_{+,M} \leftrightarrow H_{-,M}$. These are finite-matrix transport checks; the global K_1 -class remains the one in Theorem 6.38.

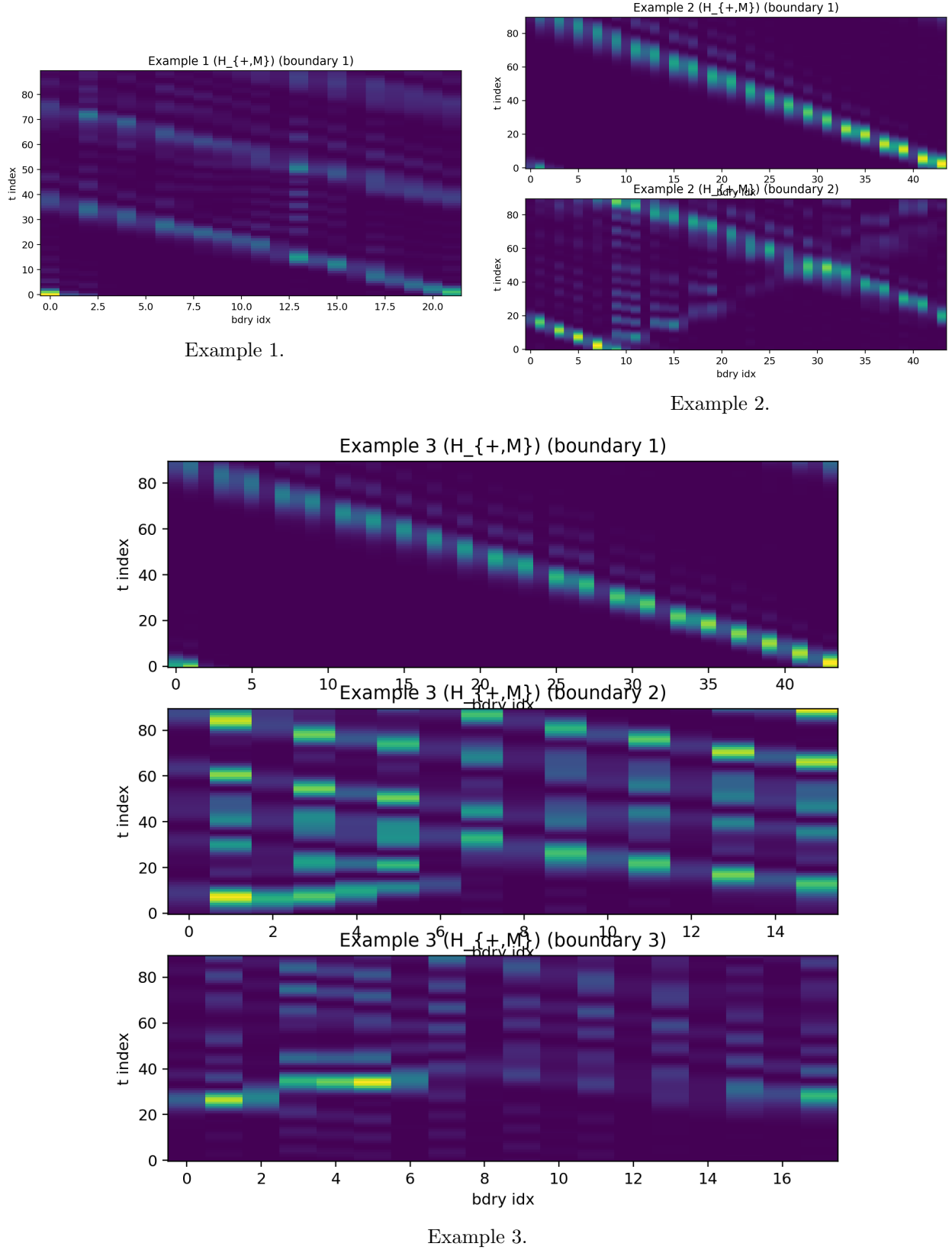


Figure 7: Boundary transport heatmaps for wavepackets projected to the finite-size in-gap subspace.

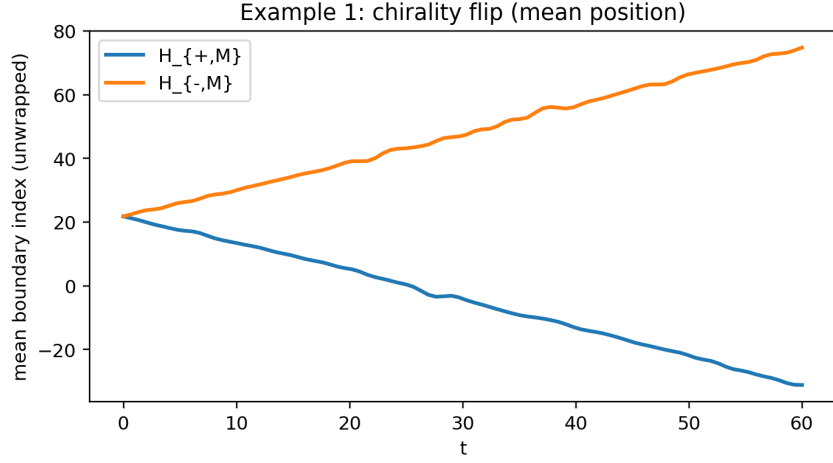


Figure 8: Example 1: opposite signed drift for $H_{+,M}$ and $H_{-,M}$.

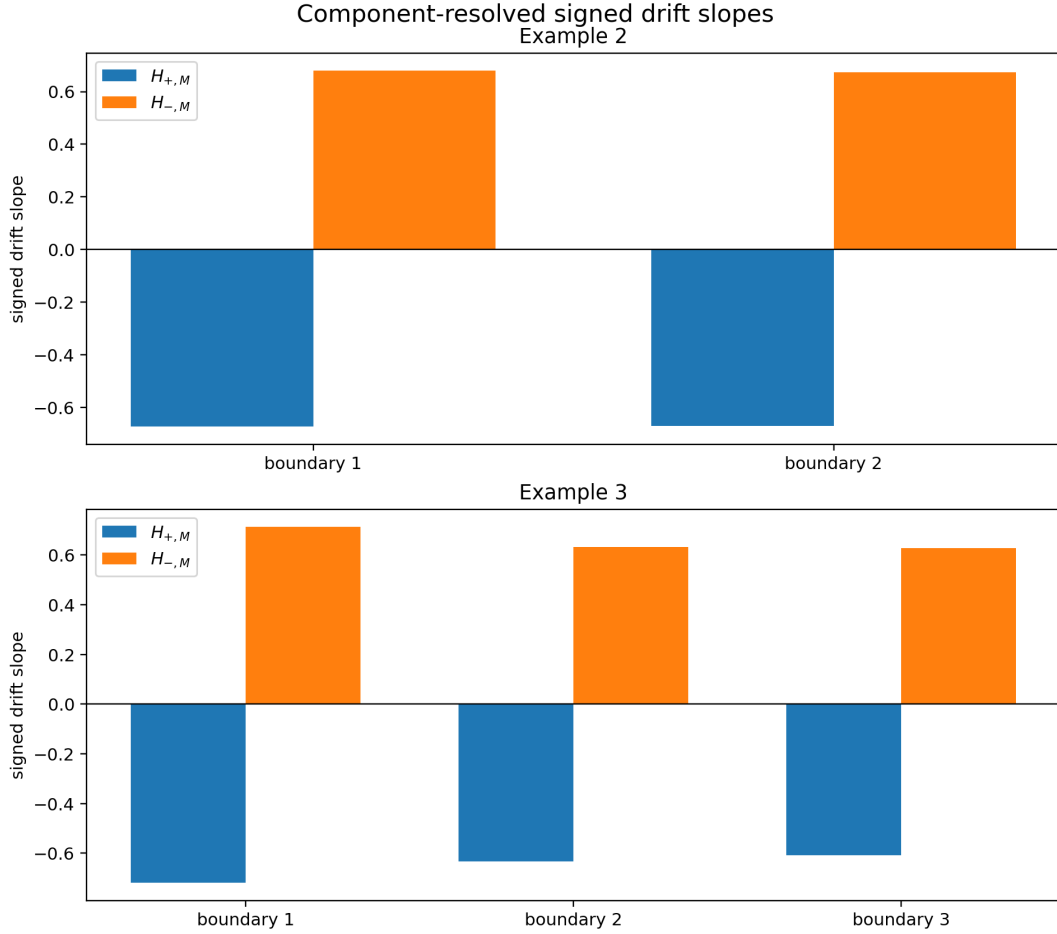


Figure 9: Component-resolved signed drift slopes for the two- and three-boundary examples. The sign reverses under $H_{+,M} \leftrightarrow H_{-,M}$ on every sampled boundary component.

Flux insertion. For the annular example we also insert a $U(1)$ twist across a fixed angular cut

and recompute near-zero eigenvalues for $\theta \in [0, 2\pi]$. Figure 10 is a finite-size spectral-flow check for the same interface class appearing in Theorem 6.38: edge branches move through the bulk gap under flux, with the sign-changing events summarized in Table 9. This figure is evidence for the interface mechanism, not a replacement for the coarse bulk–interface pairing theorem.

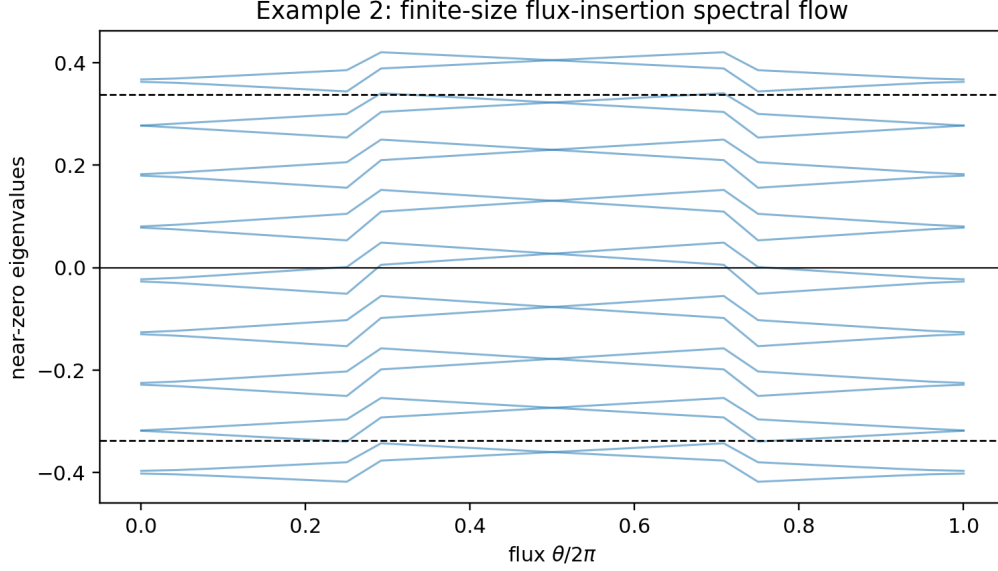
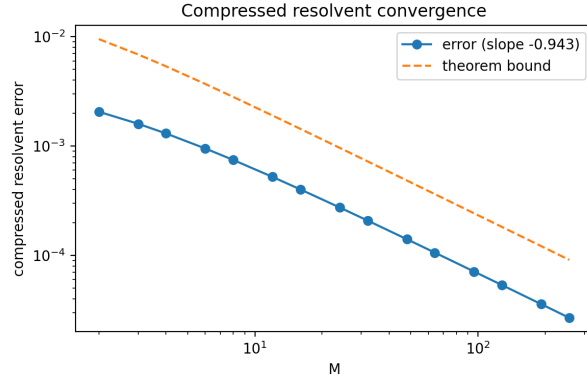


Figure 10: Example 2: finite-size flux-insertion spectral flow for $H_{+,M}$. The curves are the computed near-zero eigenvalues as the twist angle runs from 0 to 2π .

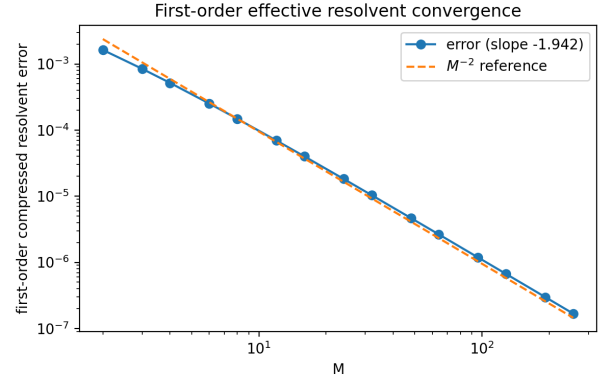
Table 9: Example 2: flux-insertion check. Sign changes count crossings between consecutive sampled flux values among the near-zero eigenvalues computed by sparse shift-invert solves.

example	flux samples	eigenvalues/sample	sign changes	$\min E $
ex2	25	18	4	0.0011

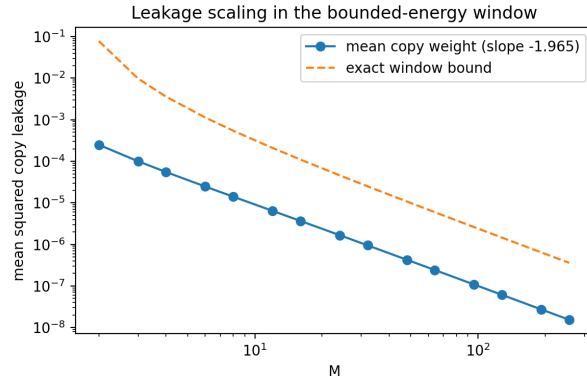
Finite- M decoupling checks. The next figures test the quantitative estimates in Section 4. The compressed resolvent error follows the predicted M^{-1} behavior from Theorem 4.2. The first-order resolvent correction, squared copy leakage, and first-order eigenvalue localization follow the predicted M^{-2} behavior from Theorems 4.5, 4.11, and 4.10.



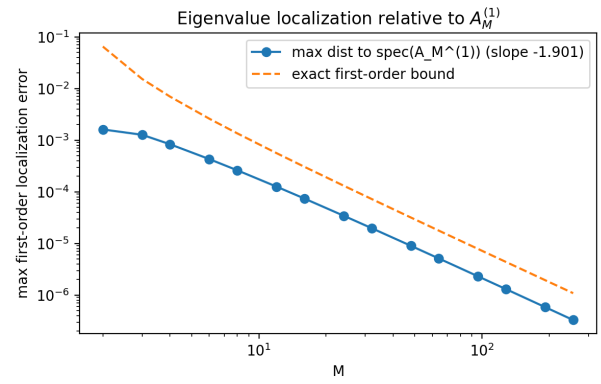
(a) Compressed resolvent.



(b) First-order resolvent.



(c) Copy leakage.



(d) First-order eigenvalue localization.

Figure 11: Log-log finite- M checks. The fitted slopes are reported in Table 7.

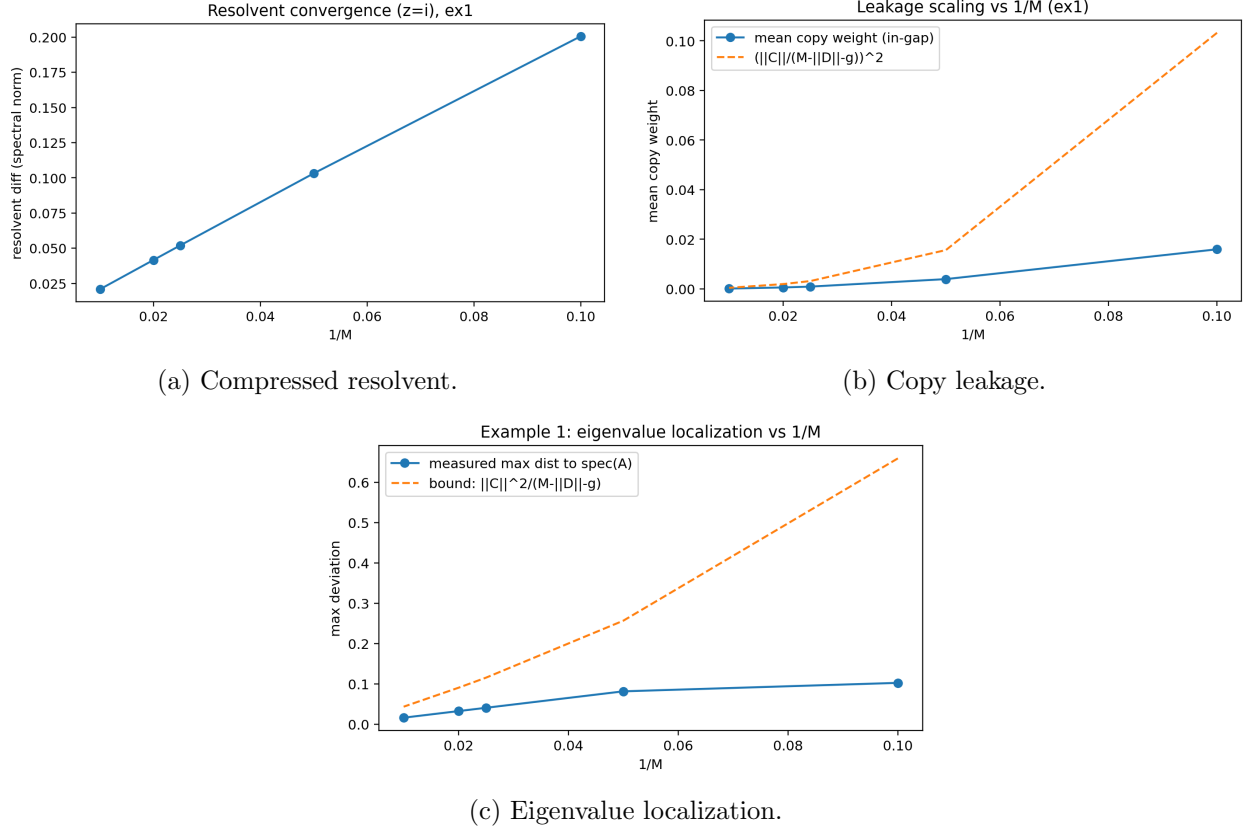


Figure 12: Example 1 finite- M decoupling quantities with explicit comparison bounds.

Table 10: Example 1: numerical eigenvalue-localization summary for several M values.

M	$1/M$	# in-gap	$\max_{ E <g} \text{dist}(E, \text{spec}(A))$	bound
10.0	0.1000	4	0.1027	0.6591
20.0	0.0500	4	0.0817	0.2568
40.0	0.0250	5	0.0409	0.1156
50.0	0.0200	5	0.0328	0.0907
100.0	0.0100	5	0.0164	0.0437

Collar support, contour separation, refinement, and robustness. The collar and contour checks in Figure 13 test hypotheses used by the contour projection and interface arguments. The refinement plot in Figure 14 records the measured bulk half-gap across the sampled subdivision levels; it is a finite-size check, not a proof of the uniform doubled-bulk gap. Figure 15 records a separate refinement check for the relative-boundary comparison: the support fraction of $R_{\partial,+}$ decreases with subdivision and remains in a radius-two boundary collar in the sampled family. Figure 16 adds weak onsite disorder and shows that the qualitative boundary transport ridge persists. Figure 17 tests the robustness of the low-energy boundary sector under a nonconstant positive copy-side confining potential.

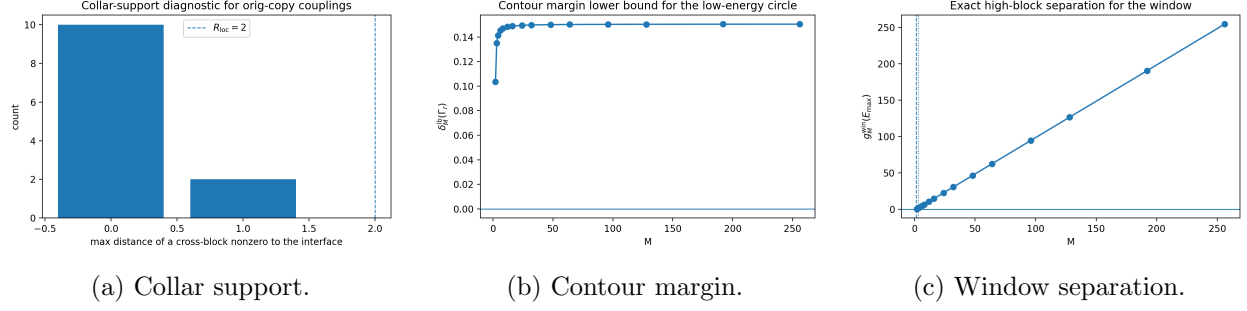


Figure 13: Numerical checks of collar support and separation quantities used in the finite- M projection estimates.

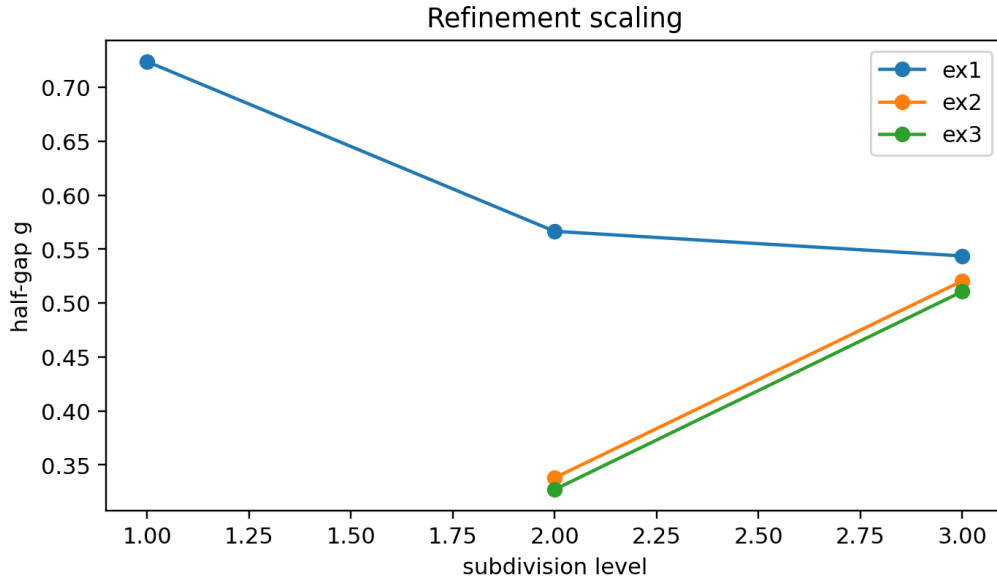


Figure 14: Measured bulk half-gap g_{gap} versus subdivision level for the three example families. This finite-mesh check does not replace the uniform doubled-bulk gap hypothesis.

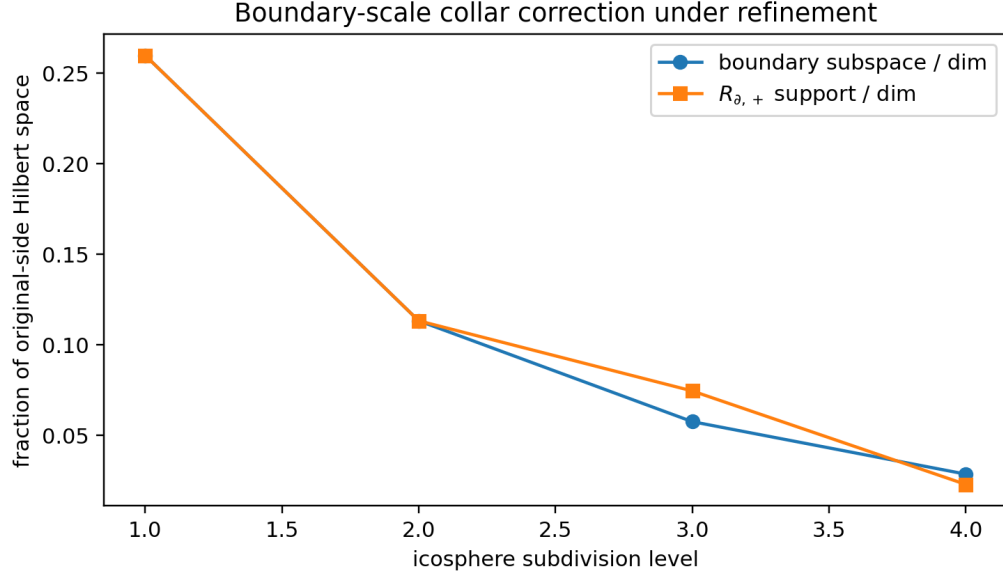


Figure 15: Example 1 refinement check for the relative-boundary comparison. The measured support fraction of the collar correction $R_{\partial,+}$ decreases with refinement, consistent with the boundary-rank interpretation in Lemma 3.15.

Table 11: Example 1: subdivision scaling of the support of $R_{\partial,+} = H_+^{\partial}(\mathcal{K}) - H_+^{\text{PL}}(\mathcal{K})$.

subdiv.	bary rounds	$\dim \mathcal{H}(\mathcal{K})$	boundary dim.	support dim.	radius	support/dim
1	0	77	20	20	2	0.2597
2	0	389	44	44	2	0.1131
3	0	1533	88	114	2	0.0744
4	0	6337	180	144	2	0.0227

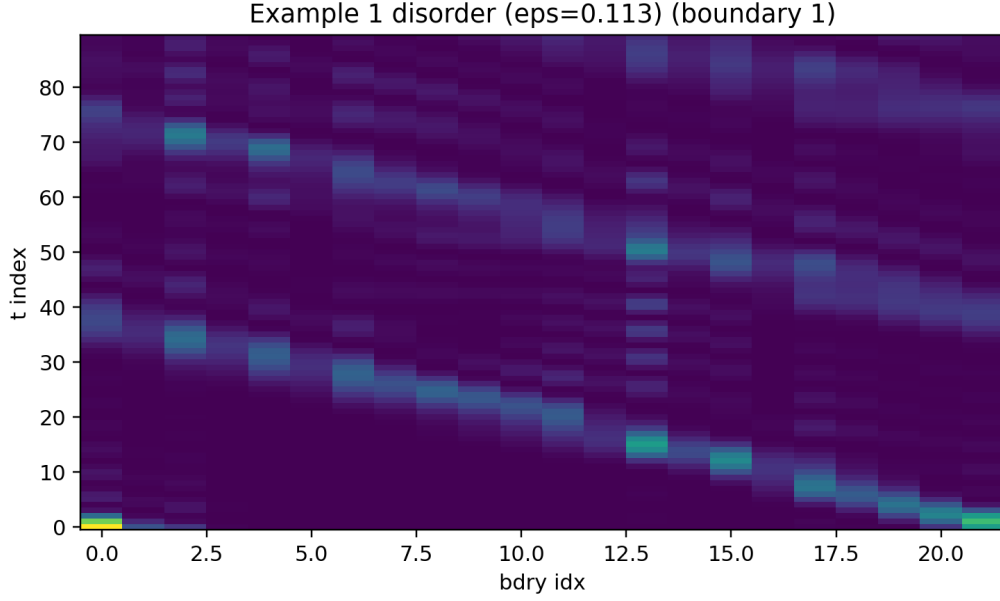


Figure 16: Example 1 with weak onsite disorder: the qualitative one-way boundary transport remains visible in the finite in-gap subspace.

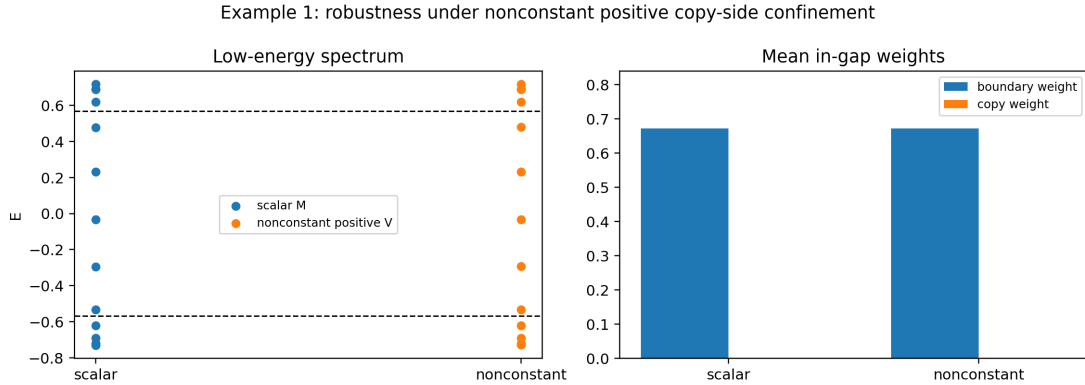


Figure 17: Example 1: scalar jamming compared with a nonconstant positive copy-side confining potential. The in-gap count and mean boundary weight are unchanged at the displayed precision, while the copy weight remains small.

Table 12: Example 1: robustness under nonconstant positive copy-side confinement. The nonconstant potential has entries in the displayed range on the copy sector.

copy potential	# in-gap	mean boundary weight	mean copy weight	potential range
scalar M	5	0.6721	5.982×10^{-4}	[50.0, 50.0]
nonconstant V	5	0.6722	5.942×10^{-4}	[32.7, 67.1]

B.1 Reproducibility appendix

The repository includes the rendered figures and summary tables. The companion code is available in the linked [project repository](#); the computations reported here correspond to main-branch snapshot 1c888c900cddbfff078f2b55b445ee01f2b4bd17, accessed 16 May 2026. The repository URL, snapshot hash, and included rendered artifacts are the availability record for this manuscript.

The script fixes the random seeds used for disorder and nonconstant copy-side confinement. It also exports the finite matrices, the barycentric adjacency graph, and the boundary-collar data used above. The dependency versions used for the local regeneration check were Python 3.14.3, numpy 2.4.3, scipy 1.17.1, and matplotlib 3.10.8. Matplotlib artifacts are PNG files.

B.2 Minimal reproduction script

After installing the packages listed in `requirements.txt`, the key periodic triangular certificate tables, including Tables 3 and 4, are regenerated with this command:

```
python periodic_triangular_bulk_certificate.py --check --write-table
```

The finite-matrix figure panels and data-backed tables in Appendix B are regenerated with this command:

```
python run_companion_numerics_with_robustness.py
```

The static theorem-map and notation tables are generated by this $\text{\LaTeX}$ source.

The local source tree expected in the GitHub repository contains the following numerical artifacts:

```
periodic_triangular_bulk_certificate.py
periodic_triangular_certificate_table.tex
periodic_triangular_extension_certificate_table.tex
run_companion_numerics_with_robustness.py
requirements.txt
mesh_stats_table.tex
numerics_summary.tex
pl_boundary_comparison_table.tex
pl_collar_scaling_table.tex
flux_spectral_flow_table.tex
eig_localization_summary.tex
copy_confinement_robustness_table.tex
fig_ex1_mesh.png
fig_ex2_mesh.png
fig_ex3_mesh.png
fig_ex1_bulk_spectrum.png
fig_ex2_bulk_spectrum.png
fig_ex3_bulk_spectrum.png
fig_ex1_jammed_spectrum.png
fig_ex2_jammed_spectrum.png
fig_ex3_jammed_spectrum.png
fig_ex1_boundary_localization.png
fig_component_localization_ex23.png
```

fig_pl_boundary_comparison.png
 fig_ex1_edge_propagation.png
 fig_ex2_edge_propagation.png
 fig_ex3_edge_propagation.png
 fig_ex1_chirality_flip.png
 fig_multiboundary_component_drifts.png
 fig_flux_spectral_flow_ex2.png
 fig_resolvent_loglog.png
 fig_first_order_resolvent_loglog.png
 fig_leakage_loglog.png
 fig_first_order_localization_loglog.png
 fig_resolvent_convergence.png
 fig_ex1_leakage_vs_M.png
 fig_ex1_eig_localization_vs_M.png
 fig_collar_support_histogram.png
 fig_contour_margin_lower_bound.png
 fig_window_separation_thresholds.png
 fig_refinement_scaling.png
 fig_pl_collar_scaling.png
 fig_ex1_edge_propagation_disorder.png
 fig_copy_confinement_robustness.png

References

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